

TASA: A Tick-Based Affective State Architecture for Persistent, Inspectable Inner State Framework, Learnability Conditions, Governance, and Evaluation

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Abstract

Standard language-model inference lacks cross-exchange internal organization. External memory can reconstruct continuity without persistent state whose causal role is tested. We present TASA (Tick-Based Affective State Architecture), a theoretical framework for such a state: segmented across timescales in discrete ticks, with structured affect, competitive workspace integration, governed typed memory shelves, and language isolated behind translators.

TASA contributes (i) a semi-formal architecture specification; (ii) five conditions for latent-target learning and assessment—four target/optimization conditions plus one persistence-discriminability reporting condition—with failure modes including identity solutions and conditional-mean collapse; (iii) growth governance based on memory-first consolidation, no-self-sealing, and criterion reachability; and (iv) an evaluation theory with noise floors, structurally equivalent positive controls, item-level statistics, evidence markers, and staged understanding. A worked instantiation sketches an inspectable end-to-end interaction loop whose prototype core is memory- and complexity-plausible on one GPU; throughput and repeat-run affordability remain unmeasured. Seventeen hypotheses—fourteen non-provisional and three provisional—define the program. The non-provisional hypotheses become bidirectionally testable once a study versions its estimands, controls, and decision rules. This framework paper reports no empirical results; its contribution lies in methodological integration and operationalization. The long-horizon aim is to test whether persistent state becomes measurably richer, more coherent, more history-dependent, and deeper during sustained, governed operation. All such claims require instrumentation, and terms such as affect and inner state denote functional organization rather than consciousness, sentience, or phenomenal experience.

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1 Introduction

Contemporary AI is dominated by systems that maintain no persistent internal state between exchanges. Transformer inference has internal organization—recent interpretability work locates workspace-like structure inside it [42]. We infer a separate limitation for the systems considered here: that organization is transient rather than an explicitly declared persistent object, and is rebuilt from the context on each forward pass rather than carried forward as an explicit object. Context windows simulate continuity, agent scaffolding attaches memory from the outside, and the “state” that persists across sessions is a transcript rather than an organization. What such systems lack is not capability but a persistent, evolving, inspectable internal organization whose configuration at any moment is the system’s situation—what it is attending to, how it is disposed, what its recent history has made of it.

This paper presents TASA (Tick-Based Affective State Architecture), a theoretical framework for building that object directly. Here, *inspectable* denotes a design target evaluated through the exposure conditions of Section 3.4 and the explicitly non-conforming comparison of H17b; it is not reported as an achieved result. We use *artificial inner world* for a persistent, multi-timescale, typed internal state whose dynamics—not its language output—are the primary object of design and study.

The structural stance. Debates about machine minds usually run through behavior: does the system *act* as if it understands, feels, remembers? We take a different route, which we call the *structural stance*: specify architectures whose internal organization instantiates—and can be tested to instantiate—the structural properties that theories of cognition and consciousness talk about—persistent state, affective modulation, competitive global integration, meta-representation, memory continuity—and make those properties measurable from the inside. The question shifts from “does it behave consciously?” to “does its organization demonstrably have the claimed structure, and what follows from that?” The stance concerns testable functional organization and therefore places correspondingly heavy demands on measurement; the second half of the paper develops those demands.

The long-horizon aim. The long-horizon aim of this line of work is a system whose functional inner state— affect, intention, deliberation, memory, and metacognition—grows measurably richer, more coherent, and deeper over sustained operation: a system whose utterances demonstrably arise from its persistent state rather than from per-exchange context reconstruction, whose dispositions are its own history compressed, and whose ascent through the understanding stages of Section 9.3 is earned through governed growth (Section 8) rather than asserted.

For interaction that extends across exchanges, the practical distinction is that continuity would reside in the system’s carried organization rather than only in context reconstructed at each exchange. Whether that form of continuity improves interaction remains an empirical question.

The framework operationalizes this long-horizon aim through markers whose failure criteria are fixed in advance (Section B), falsifiable hypotheses (Section 11), and gates that decide when growth toward the next rung is permitted (Section 8.4). Figure 1 sketches the resulting ladder. The ambition is large; the present contribution is the instrumented path by which each step can be tested.

The larger research questions. The long horizon gives the framework a purpose beyond specifying a stateful agent. If internal organization is carried forward, altered by controlled history, and shown to matter for later processing, it becomes possible to pose longitudinal and interventional questions about that organization rather than infer it from behavior alone. Can different histories produce persistent, causally attributable differences in later disposition, retrieval,

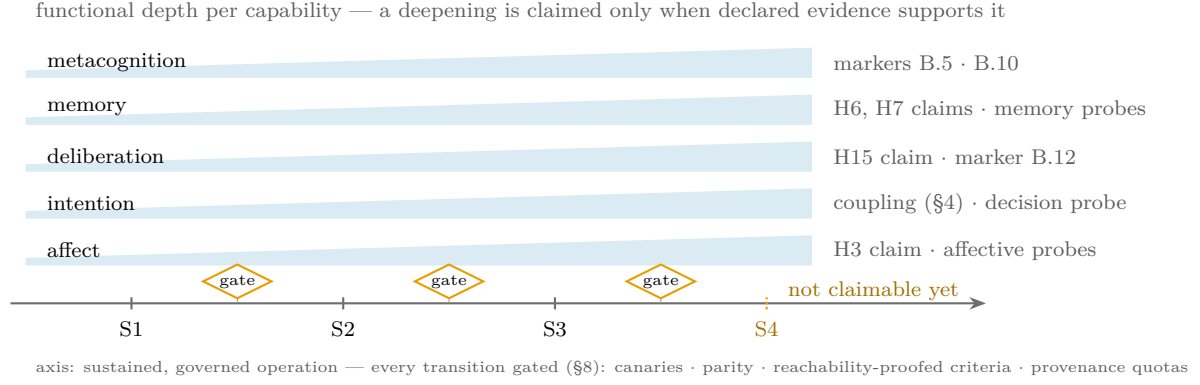


Figure 1: The long-horizon aim as an instrumented ladder. Over sustained, governed operation, each capability lane is intended to deepen. The right-hand annotations distinguish claim or mechanism anchors (hypotheses and the coupling requirement of Section 4.3) from evidence instruments (markers and probes); a deepening claim requires corresponding support from the latter and remains bound to the growth gates of Section 8. S1–S4 mark the understanding milestones of Section 9.3; none of the depicted deepening is asserted as achieved, and S4 is dashed for a sharper reason—two of its three components have no marker, so it is not claimable at all until they exist.

integration, or action tendency? Can calibrated self-monitoring deepen toward representations of processing history and known limitations? In realizations that give losing workspace proposals a declared path into later state, can non-broadcast activity leave a detectable causal footprint? Can additional internal processing count as deliberation only when it improves outcomes under matched-compute controls? TASA answers none of these questions by construction; it supplies candidate state carriers, provenance, intervention points, controls, and failure criteria with which they can be tested.

Why affect-first, and why language at the periphery. Two design commitments distinguish the framework from its neighbors from the outset. First, *affect is architecture*: affect variables persist across time steps, constrain each other, bias action selection, and gate memory—the framework’s object is a state that is disposed, not merely recorded, and without valence and disposition persistence buys a log rather than an organization. Second, *language is peripheral*: an embedding encoder translates text into the state space, a generative model verbalizes structured state reports back out, and the core in between never touches a token. The core’s job is state dynamics; language is its sensory and expressive boundary.

Contributions. The paper contributes the following integration and operationalization, in order of appearance. The contribution lies in the joint contract among components with substantial prior art and in the tests and gates that make that contract auditable:

C1 — Architecture (Sections 3 to 6). A semi-formal specification of a tick-based, affect-centric state architecture: segmented multi-timescale state, competitive workspace integration with disjoint output responsibility, a calibrated metacognitive layer, and typed memory shelves with governed consolidation.

C2 — Learnability theory (Section 7). Five conditions governing latent-target learning and valid assessment: four target or optimization conditions and the LC3 persistence-discriminability reporting condition. Together they carry a failure-mode taxonomy (identity solutions, conditional-mean collapse, pseudo-learnability, silent neglect, unproducibility,

displacement, starved interpretation) derived from optimality and optimization arguments—and, kept separate from them, generator closure, which limits what a result may be evidence for rather than degrading the representation.

- C3 — Growth governance (Section 8).** A gate-based governance model for continual growth: seed/growth separation, memory-first consolidation, no-self-sealing, multi-source governors, criterion-reachability proofs, and input-novelty accounting.
- C4 — Evaluation theory (Section 9).** Nine measurement requirements (M1–M9) for honest evaluation of stateful stochastic systems, a staged understanding model, and an evidence-marker battery—failure criteria fixed before measurement—that replaces consciousness claims with testable structure.
- C5 — Instantiation sketch (Section 10).** A worked instantiation, abstracted from the reference implementation noted below into justified ranges—state layout, operator candidates with a suitability comparison, training recipe, data pipeline, the construction space for an experience-evaluation suite, and an end-to-end interaction loop—arguing that the prototype core is memory- and complexity-plausible at single-GPU scale, without a claim about measured throughput or repeat-run affordability. The suite is the one element deliberately left short of a canonical instrument (Section 10.9).
- C6 — Research program (Section 11).** Seventeen hypotheses with operationalization sketches, several with sharp negative halves, designed for staged empirical testing. Fourteen are non-provisional and are designed for bidirectional testing once a study versions the required estimands, controls, and decision rules; three are marked *provisional*, because a quantity they turn on is not yet an estimand the paper can point at.

A reference implementation and a multi-month empirical program are in progress; their results are intended for future companion papers, negative outcomes included.

What kind of paper this is. The six contributions have distinct epistemic roles. C1 specifies an architecture *class*, fenced by a short register of requirements (Section 3.8) so that the hypotheses of Section 11 have a definite subject—a framework whose membership is a family resemblance can be agreed with but not falsified. C2, C3, and C4 are not part of that fence: target learnability, growth governance, and the measurement theory apply to latent-state systems generally, and we expect them to outlive the particular architecture stated here (Sections 2.8 and 12.5). C5 is one worked point inside the class, offered as evidence of implementability rather than as the canonical instance. The register’s purpose is correspondingly narrow: it decides what the hypotheses range over, and it gives a successor that drops a load-bearing commitment a way to say precisely which one—Section 3.8 calls that a *declared departure* and treats it as an expected move rather than a defect. Section 12.5 names five root commitments; its figure maps the four with substantial dependent structure.

As a specification, the paper is designed for selective consultation. The contents list, the identifier register in Table 6, and the reading map below provide entry points for that use. Readers seeking the core argument without the full apparatus can begin with Section 3.8 and Table 4.

Readers who want the framework’s epistemic bookkeeping before its content will find it in Table 4 (Section 12.4), which sorts every core position into *committed*, *argued*, and *open* with its basis; and in Section 3.8, which states what a system must satisfy to count as an instantiation, what discipline a study about one must observe, what it may choose freely, and what a system leaving the class should declare instead. Those two, read together, are the shortest summary of what this paper does and does not establish.

Empirical scope and sequencing. This paper fixes the framework, hypotheses, failure criteria, and measurement requirements before empirical reports are available. This sequencing separates specification from evidence and permits later results to be judged against public commitments. Accordingly, the paper argues implementability and resource plausibility from component-level reasoning, but does not evaluate performance, throughput, or repeat-run affordability. Those questions are reserved for companion studies.

What this paper does *not* claim

- This paper makes **no claim that systems built on this framework are conscious, sentient, or capable of experience**. “Affect,” “memory,” “curiosity,” and related terms denote functional state variables and their causal roles—never phenomenal states. Where the text says a system “remembers,” “reacts,” or that something passes through its “experience,” these are functional shorthands, fixed here once for the whole paper.
- This paper makes **no competitive claim against large language models** on language tasks. The framework studies a different object—persistent internal organization—and uses language models as peripheral components.
- This paper makes **no neuroscientific claims**. Brain-adjacent vocabulary (workspace, ignition, consolidation) names engineering constructs inspired by the cited theories, not models of neural tissue.

Reading map. What is where, the contents list answers; this paragraph answers the question it cannot, which is what depends on what. Four dependencies are real and the rest of the order is convenience. Section 7 is unreadable without the segment definition of Section 3.2, because the learnability conditions are conditions on targets that supervise *those* coordinates. The markers of Section B presuppose the measurement requirements of Section 9.2, which define the role of their controls. Section 3.4 is load-bearing for Sections 5 and 6 alike, since both are constrained by what a consumer may read. And Section 3.8 is best read twice—once early for what it requires, once after Section 11 for what that means for the hypotheses.

One part is skippable without loss: Section 2 can be deferred by a reader who already knows the neighborhood. Section 10 can be deferred by a reader interested in the theory rather than in whether it can be built, with one exception: Section 3.4’s account of what a dispositional-stability result can mean depends on the staging recommendation in Section 10.5. Conversely, a reader who wants the epistemic bookkeeping first can read Section 3.8 and Table 4 together. The appendices are reference material throughout, with one exception worth naming: the identifier register in Table 6 (Section A.2) is the fastest way into the paper’s machinery, since it says where each family of requirements is fixed and what, if anything, tests it.

2 Related Work and Theoretical Foundations

TASA sits at the intersection of five literatures: cognitive architectures, Global Workspace Theory and its computational instantiations, world models and latent-space processing, affective computing, and the evidence-based approach to machine consciousness. This section places the framework in each, adds the methodological literature its evaluation theory builds on, then reviews recent work that converges on individual commitments of the framework, and closes by positioning it against its closest neighbors.

2.1 Cognitive architectures

The classical architectures pursue integrated cognition with symbolic or hybrid machinery: ACT-R organizes cognition around production rules over modular buffers [2]; Soar adds impasse-driven subgoal and multiple learning mechanisms over a unified decision cycle [65]; NARS derives reasoning from the assumption of insufficient knowledge and resources [114, 115]; OpenCog integrates heterogeneous components over a shared hypergraph [37, 49]. Closest in spirit to TASA is LIDA, which builds a systems-level architecture directly on Global Workspace Theory, with perceptual, episodic, and procedural memories organized around a cognitive cycle [33]. The field is broad and mature [61], and TASA differs from it along three joint axes rather than any single one: it is differentiable end-to-end (no symbolic production layer); it is *inner-world-first* (the object of study is the persistent internal state and its dynamics, not task performance); and it treats affect as load-bearing architecture rather than as a module among many. LIDA shares the workspace commitment but is a framework of coordinated components rather than a trainable substrate; TASA is designed so that gradient-based learning shapes the whole core. Modern language-agent work makes the comparison more immediate. CoALA supplies a taxonomy that organizes language agents around memory, action, decision making, and learning [107], while HIMA proposes a human-inspired long-term-memory architecture for LLM agents [58]. These works narrow the remaining claim: the contribution here is not agent memory or cognitive-architecture vocabulary, but their integration with a typed trainable state, explicit learnability conditions, state-exposure tests, and change governance.

Two precursors worth naming precisely. Two older works come close enough to specific commitments of this framework that treating them as background citations would understate them. Shanahan [96] proposes a cognitive architecture combining internal simulation with a global workspace, in which competing processes produce an ongoing sequence of internal states, anticipation and planning run through simulated interaction, and—the commitment closest to ours—action selection is mediated by affect. Read at the level of ideas, that architecture already holds the combination of ongoing inner processing, workspace competition, and affect-mediated selection. Snider and Franklin [103] is the second: it replaces LIDA’s node-and-link representations with high-dimensional vectors over a sparse distributed memory, so that “cognitive cycle plus workspace plus vector representation plus multiple memory systems” was assembled more than a decade ago.

Accordingly, none of the following is offered here as new: the cognitive cycle, the workspace, affect inside a cognitive architecture, multiple memory systems, or vector-valued internal representation. The contribution lies in the joint operational contract that neither precursor supplies: an internal state that is *trained end-to-end and evaluated as the product*, with the conditions governing its supervision targets and their valid assessment (Section 7), a test for whether the inspectable part is the state or a readout of it (Section 3.4), and a governance model for changing it over time (Section 8). In Vector LIDA the vectors are a representation *choice* serving a designed architecture; here the typed state is the object under study and the learning problem is the paper’s subject.

2.2 Global Workspace Theory and computational workspaces

The workspace commitments of Section 5 descend from Global Workspace Theory [4] and its neuronal elaboration, in which competition, ignition, and global broadcast mark conscious access [22–24]. Two recent developments make this lineage directly buildable. First, deep-learning translations of GWT propose concrete latent-workspace designs [110], and shared-workspace architectures demonstrate that bandwidth-limited, competitive write access improves coordination and generalization among specialist modules [38]—the engineering counterpart of the framework’s competitive-selection requirement. Workspace routing has also been shown to

support sequential operation chaining [15], and GWT-structured agents have been evaluated in embodied multimodal settings [27]; complementary arguments locate functional advantages of the selection–broadcast cycle in real-time processing constraints [84]. Second, the theory side has moved toward severe testing: the adversarial collaboration between global-workspace and integrated-information accounts subjected both to pre-registered, theory-neutral protocols and reported challenges to predictions of both theories [20]. Contemporary GNW reviews connect conscious access to recurrent amplification and global availability [76], while a subsequent GNW response disputes parts of the adversarial study’s operationalization and interpretation [83]. TASA inherits both developments: its workspace requirements are stated in the marker vocabulary that severe testing needs (selection, ignition, persistence, broadcast), and its evidence policy (Section 9.4) adopts the pre-registration discipline rather than the claim-first style that severe testing has punished.

2.3 World models and latent-space processing

The framework’s token-free, latent-first stance has a strong contemporary current behind it. Learned world models compress environment dynamics into recurrent latent states and act from them [43, 44]; PlaNet made latent dynamics directly usable for planning from pixels [45], and the Dreamer work later appeared as a preprint and journal version rather than two independent results [46, 47]. Position work argues for predictive, hierarchical joint-embedding architectures trained in representation space rather than in pixel or token space [67], with image-domain instantiations demonstrating that predicting representations yields semantic features without reconstruction [3]. In language modeling, reasoning has been moved into continuous hidden states [48], compressed into continuous space by self-distillation [100], and next-vector prediction has been proposed in place of next-token prediction [98]; selective state-space models provide efficient recurrent substrates, from structured S4 models to input-selective Mamba variants [40, 41]. Mechanistic evidence also cautions against equating a continuous latent trajectory with literal sequential step-by-step reasoning [70]. TASA shares the core wager of this literature—that representation-space processing is where the leverage is—and differs in what the latent state *is*: not a compressed observation code optimized for control or prediction, but a typed, inspectable inner state with affective and metacognitive structure, whose dynamics are the object of study rather than a means to a task. The learnability conditions of Section 7.3 are, in effect, the price of that difference made explicit: once the latent state is the product, the targets that supervise it must be engineered with care that task-driven world models get for free from their reward or prediction objectives.

2.4 Affective computing and emotion theory

Affective computing established machines that recognize and express emotion as a field [88]; surveys of emotion in agents and robots document a long line of appraisal-inspired and homeostatic designs [81]. The framework’s affect segment (Section 4) composes the standard theoretical ingredients—dimensional structure [93, 94], basic-emotion categories [29], appraisal as the organizing process [95], and curiosity as an information-gap drive [72]—into an architectural commitment: affect variables that persist, constrain each other, and causally modulate integration, action selection, and memory. The difference from most affective-computing work is positional: affect here is not a recognition target or an expressive layer but part of the state the whole system is built around.

2.5 Machine consciousness: the evidence turn

Recent work has reframed the machine-consciousness question from assertion to assessment: derive indicator properties from leading scientific theories and evaluate systems against them [12].

Higher-order theories contribute the meta-representational motif the framework’s meta segment instantiates [66, 92]; higher-order state-space work gives a contemporary inferential formulation of that motif [31]. Predictive-processing accounts supply an inspirational error-minimization backdrop—not a derivation of the framework’s scalar confidence variable—[34]. TASA adopts the indicator methodology wholesale (Section 9.4) and operationalizes it as an engineering contract: markers with failure criteria fixed in advance, structurally equivalent positive controls, and a verdict vocabulary that distinguishes “absent” from “not measurable.” The philosophical posture stays fixed: structural evidence, no phenomenal claims.

2.6 Measurement and reproducibility

The evaluation theory of Section 9 compresses lessons the ML methodology literature has documented at length: variance from seeds and run-level perturbations rivals reported effect sizes [10]; multi-run reporting and comparison discipline decide whether deep-RL results replicate at all [50]; suggestive language and unidentified sources of gain corrode scholarship [71]; preregistration separates prediction from postdiction [86]; optimizing a measure destroys it [105]; shortcut solutions win whenever they are available [36]; and models trained on their own kind of output degrade [101]. What the framework adds is placement: these are usually post-hoc cautions, whereas here they enter as design-time requirements (M1–M9, LC1–LC5, no-self-sealing) with named failure modes.

2.7 Convergent developments

Several of the framework’s individual commitments have recently appeared, separately, in other work—a convergence that bounds what can be claimed as new here. We group the closest developments by the commitment they share. Publication status varies widely across this set: some are peer-reviewed, several are single-author preprints, and we cite them as convergent developments rather than as established baselines.

Discrete processing cycles. Heartbeat-driven scheduling gives an agent a periodic, task-independent cycle that decides which cognitive activity to run next, so that reflection and consolidation continue without an external prompt [106]; a workspace-centered multi-agent design turns LLM orchestration into an event-driven discrete dynamical system with a running cognition cycle [97]. The tick of Section 3.3 shares the commitment to processing time that does not wait for input.

Affect as control structure. In our comparative audit completed on 31 August 2026, a deterministic homeostatic-controller preprint was the closest located neighbor among affect-in-the-loop designs: it couples appraisal, executive arbitration, episodic memory, and a deliberately down-scoped self-model in a monitoring-control loop, logs the state at every tick, and is inspectable by construction [99]. A motivational architecture for conversational agents re-reads homeostatic drive theory for dialogue, combining dimensional affect, needs, an urgency-driven fast path with a deliberative slow path, and a governing blend stage [80]. Both share this framework’s core wager—that affect belongs in the control loop rather than in the output style. The differences are architectural rather than motivational: neither places a *typed, trainable* latent state at the center, and both regulate a system whose competence lives in a language model rather than in the state dynamics.

Inner state separated from utterance. The framework’s separation of internal dynamics from language output has a direct recent counterpart: agents that maintain explicit structured internal variables—appraisal-style dissonance, a perceived opinion climate, an isolation risk, a response strategy, and a willingness to speak—update them from dialogue and memory, and only then decide whether to produce a public utterance [120]. The ordering is the same one this framework builds on, internal state before intention before speech, and the ordering itself is therefore not what is claimed here. The difference is where the state is computed: there the variables are

produced and read by a language model, whereas the commitment of Section 3.1 is that the update happens in a small trainable core that never processes tokens.

Persistent, inspectable state. CaveAgent makes a persistent Python runtime—including complex native objects—the agent’s stateful workspace [90]. Our distinction is therefore not serializability, but a typed, auditable state contract whose components are themselves learning targets. On the memory side, typed intermediate representations have been introduced specifically to prevent provenance-role collapse in long-lived agents [57], which is the same argument the shelf typing of Section 6 makes from a different direction, and affect-sensitive designs now separate factual from affective memory under a conflict-aware controller [1]. A recent survey of 435 coded works on persistent-state agents reports that the literature accumulates and retrieves state extensively but governs, recovers, and releases it rarely [26]—an assessment that locates Section 8 in an acknowledged gap rather than in a niche of our own construction. Broader reviews of agent externalization document the same drift toward state living outside the model [123]. Direct recent prior art also treats memory as an evolving, typed, or verifiable systems object [69, 108, 113]; growth governance here integrates those concerns with the paper’s state contract and gate discipline rather than originating them.

Workspace and consolidation mechanisms. Convergence arguments have been made that global workspace, reconstructive episodic memory, inner speech, and complementary learning systems reduce to a small set of shared architectural principles, implemented as parallel threads synthesized each turn into a bounded representation [52]. That work is turn-driven rather than tick-driven and keeps its persistent state in narrative text rather than in a typed vector; in our dated audit it was the closest located neighbor on the integration axis. Open-ended settings supply a further contrast: autonomous agents have been run for months in an open environment under a budget-based metabolism, with asynchronous memory and evaluation processes [77].

Workspaces inside language models. Finally, interpretability work reports that the representations a language model can verbalize form a subspace with workspace-like properties—reportable, steerable, and broadly broadcast [42]. We infer from this result—rather than treating it as a claim made by that paper—that transient inference-time organization and explicitly persistent declared state should be distinguished. This matters for how the present framework states its motivation. The relevant contrast is not that transformer inference lacks internal organization; evidently it has some. The contrast is that such organization is transient, discovered rather than declared, and not carried across interactions as an explicit object. What this framework adds is a state that is persistent, typed by construction, and evaluable as such.

2.8 Positioning

Table 1 summarizes the comparison on the axes that define the framework’s niche; Appendix C refines it against the closest neighbors. Given the convergence just described, the claim to novelty is deliberately narrow. Tick-like processing cycles, affect in the control loop, inspectable persistent state, and workspace competition each have independent recent precedent. On the architectural axis alone, a fair reader would find little here that has no precedent, and Section 2.1 says so at the level of individual commitments.

The claim is therefore one of integration and operationalization rather than component novelty, and it is worth bounding it by the search that supports it rather than by the phrase’s reach: within the literatures reviewed in this section, in searches completed on 31 August 2026, we did not identify a *method for constructing and empirically validating* a persistent typed inner state that combines all four of the following. The field’s preprint half moves faster than any single review, and a reader should weigh the claim as one about a documented search rather than about the literature entire. The four: conditions governing its supervision targets and their assessment begins (Section 7), a causal test deciding whether the inspectable part is the state or a readout of it (Section 3.4), a governance model for changing it over time without self-sealing (Section 8), and an evaluation theory whose failure criteria are fixed in advance (Section 9). Substantial

	<i>differentiable</i>	<i>affect-first</i>	<i>persistent typed state</i>	<i>meta layer</i>	<i>governed memory</i>	<i>growth governance</i>	<i>evaluation theory incl.</i>	<i>language peripheral</i>
ACT-R / Soar	–	–	partial	partial	partial	–	–	n/a
LIDA	–	partial	✓	partial	partial	–	–	n/a
NARS / OpenCog	–	–	partial	partial	partial	–	–	n/a
World models / Dreamer	✓	–	latent, untyped	–	–	–	partial	n/a
LLM agents + memory	–	–	external scaffold	–	partial	partial	–	–
Homeostatic affect controllers	–	✓	✓	partial	partial	–	partial	partial
Workspace / consolidation agents	–	–	partial	partial	partial	–	partial	–
TASA	spec	spec	spec	spec	spec	spec	spec	spec

Table 1: Positioning against neighboring system classes. “Partial” marks partial or optional coverage; “n/a” marks axes that do not apply to the system class; on the persistent-typed-state axis, ✓ requires state that is typed *and* carried across interactions *and* treated as an object of evaluation. Entries in the TASA row are marked *spec*, not ✓: they are properties the framework *specifies*, and unlike every other row they are not backed by a built and evaluated system. Two of the axes—growth governance and included evaluation theory—are integrated and operationalized here under one contract, so a dash records a difference in documented scope rather than a claim of invention or a deficiency; *partial* on the language axis marks systems that isolate a non-linguistic control loop while leaving task competence in a language model. The two rows above TASA summarize the convergent developments of Section 2.7 ([80, 99]; [52]); on several axes they match this framework. The table is a map of coverage, not of quality. Appendix C discusses the closest neighbors in detail.

prior art exists for each component; the bounded negative result concerns their documented combination under one auditable contract. The affect-centric architecture is what that method is stated *for*, and it is needed to make the method concrete; but it is the method that carries the weight, and it is the part that would survive substituting a different inner-state architecture underneath it. What binds the synthesis is stated explicitly rather than assumed: Section 3.8 separates the requirements a system must meet to be an instantiation, the disciplines that bind studies about it, and the choices it may make freely, so that “conforming” is a checkable property rather than a family resemblance.

3 The Framework Core: State Space and Tick Dynamics

TASA is a framework for systems whose primary object is a persistent internal state that evolves in discrete time and remains inspectable at every step. This section fixes the framework’s design principles, defines the state space and its update dynamics, and specifies the two peripheral interfaces through which language enters and leaves the system.

3.1 Design principles

Nine principles constrain every instantiation of the framework, and the remainder of the paper derives structure from them. They are design commitments rather than the conformance test itself: Section 3.8 turns them into the checkable register R1–R7 and states, principle by principle, which requirement carries which—including the two that deliberately do not become requirements at all.

- A1. Autonomous differentiable core.** The processes that constitute the system’s inner dynamics— affective response, integration, memory access, self-monitoring, dispositional change—are computed by a differentiable core that is trainable end-to-end: one computation

graph, whether realized as a single network or as several jointly trained sub-networks. No external service participates in the inner loop.

- A2. Language is peripheral.** Text is translated *into* the state space by an encoder at the boundary (Section 3.5) and *out of* it by a generative language model at the boundary (Section 3.6). The core itself never processes tokens; its internal representations are continuous vectors.
- A3. Token-free internal semantics.** The semantic content the core operates on lives in embedding-derived and core-generated vector spaces. This buys short update steps, frees capacity from surface-form modeling, and follows the broader turn toward latent-space processing [48, 67].
- A4. Explicit, inspectable, typed state.** Every state variable belongs to a named *segment* with a declared role and, where the role demands it, a constrained range. Segments that deliberately carry uncommitted latent capacity are marked as such—inspectability means the *typed* portion of the state is readable and renderable as a structured report at any tick, and the untyped remainder is explicitly declared rather than hidden. Interpretability of the typed portion is an architectural property here rather than a post-hoc analysis—subject to the exposure condition of Section 3.4, which is what decides whether that portion is the system’s state or only its readout.
- A5. Multi-timescale organization.** The state is partitioned by characteristic timescale—per-event, per-tick, and dispositional—with distinct update mechanisms per partition (Section 3.7).
- A6. Tick-based evolution.** The system advances in discrete update steps (*ticks*); each tick transforms the full state. The tick is the system’s native unit of processing; internal depth is counted in ticks, not in tokens or wall-clock time.
- A7. Modularity.** Specialist processes, memory shelves, and heads attach to the core through defined interfaces, so components can be validated in isolation before integration.
- A8. Bottom-up validation.** No component is integrated before it has demonstrated its intended function in isolation—a discipline that the evaluation theory of Section 9 turns into concrete measurement requirements.
- A9. Exposed-state sufficiency.** Every consumer of the system’s state other than the tick operator itself—memory read and write, the deliberation policy, the workspace, the slow-state update, and the output translator—reads the declared exposed state and nothing else. Its regime-specific envelope $\mathcal{X}_t^{(r)}$ contains every quantity any consumer is permitted to read (Equation (2); per-consumer permissions in Figure 4). The principle constrains *which channels* a consumer may read, not which tick’s value it sees: a consumer that runs before the update reads the previous tick’s components, and the tick order of Section 3.3 fixes which is which. Outside the declared-carrier regime, any internal carrier the operator maintains is an implementation detail of the update, not a private channel between components. This is a boundary, not a default: outside regime (iii), a system that routes a consumer through the carrier is a *comparison variant*, not a conforming instantiation, and disclosure does not restore conformance. In regime (iii), the route is conforming only when h_t and its permission are inside $\mathcal{X}_t^{(r)}$ before evaluation. That fixes which class the system is in and says nothing about its merit. Such designs are useful, the framework provides an instrument for them—B.13 measures exactly the difference they make—and a system built that way may declare the departure (Section 3.8) instead of being filed as a defective instance. What it may not do is call the post-hoc carrier disclosure a declared

style of *conformance*. The principle is stated as an architectural constraint *and* checked empirically, because the constraint alone does not establish that the exposed state is what drives behavior.

3.2 The state space

The four-part *typed exposed core* at tick t is

$$X_t = (F_t, C_t, S_t, g_t), \quad (1)$$

whose first three components differ in origin, timescale, and update mechanism (Figure 2) and are described below; the fourth, the broadcast context g_t produced by workspace competition (Section 5), is defined in Section 3.3. The regime-specific *declared exposed state* is the envelope

$$\mathcal{X}_t^{(r)} = \begin{cases} X_t, & r \in \{\text{i, ii, iv}\}, \\ (X_t, h_t), & r = \text{iii}. \end{cases} \quad (2)$$

Every non-operator consumer is restricted to named channels in $\mathcal{X}_t^{(r)}$. Thus regime (iii) may expose h_t only as a declared, rendered, and evaluated component with an explicit permission entry; in regime (iv), h_t remains outside $\mathcal{X}_t^{(r)}$ and no consumer may read it. That the typed core is a vector of fixed width rather than a structured object—slots, a graph, a set of propositions—is a commitment and not a neutral encoding decision; Section 12.5 states what it buys and what it costs.

The membership of g_t is a deliberate consequence of A9 rather than a matter of bookkeeping. A broadcast that only the tick operator reads is not a broadcast; the whole point of the workspace is that selected content becomes available to the mechanisms that act on the state (Section 5.1). But A9 admits no channel that consumers read and the declared state does not contain—a quantity in that position would be precisely the hidden bus the principle exists to forbid, one level below the carrier of Section 3.4. Only two consistent resolutions exist: keep the broadcast private to the operator, or declare it part of the exposed state. We take the second, and the price is explicit— g_t is covered by the sufficiency test of Section 3.4 like every other exposed component, and an instantiation reports it. Figure 4 fixes, per component and per consumer, what may be read.

Fast state F_t . The *fast state* is the per-tick encoding of what is currently impinging on the system. It concatenates (i) a projected embedding of the current event (Section 3.5) and (ii) a small set of typed *signal channels*: scalar indicators of event type and intensity together with pragmatic and epistemic cues (e.g., whether the event is a question, refers to the self, carries causal or uncertainty markers). Signal channels make properties explicit that would otherwise have to be recovered from the embedding—and, as Section 7.4 argues, they are also what a supervised interpretation head needs to read, which makes explicit structure at the input a reliability lever rather than a redundancy.

Core state C_t . The *core state* is the central, inspectable representation that the tick update rewrites. It is partitioned into typed segments:

- a *semantic focus* segment holding the vector-space content the system is currently processing;
- an *affect* segment with a dimensional core (valence, arousal, dominance), a small set of basic-emotion intensities, and one epistemic drive (Section 4);

- an *intention* segment: a probability distribution over a small, fixed set of action tendencies (act, plan, explore, reflect, retreat, wait), plus scalar readiness and *action confidence* variables—distinct from the meta segment’s monitoring confidence below, which alone carries the calibration duty of Section 5.4;
- a *meta* segment: confidence, conflict, and complexity read-outs produced by self-observation (Section 5);
- *latent workspace slots*: unnamed capacity for content the architecture does not pre-commit to interpreting.

Each segment carries an activation constraint appropriate to its semantics: bounded ranges for affect and meta variables, a probability simplex for the intention distribution, unconstrained continuous values for semantic and latent content. Segment *roles* are part of the framework; segment *sizes* are instantiation parameters and deliberately not fixed here.

Because sizes are free, the layout that maps segment names to coordinate ranges is itself an object the instantiation must maintain, and the framework makes one demand about it: the layout exists exactly once, in machine-readable form, and every path that addresses the state—the operator’s typing map, each loss term, the memory encoders, the probes, the renderer, the serving path—derives its offsets from that single declaration. The demand looks like software hygiene and is not. A state whose boundaries are restated in several places can drift between them, and the drift is invisible to the obvious tests: a check that reads the same restatement as the code it checks will agree with it while both disagree with the operator. What is at stake is principle A4 itself—a segment addressed by two different ranges is not a typed variable but two overlapping ones—and *every* downstream measurement, since a probe reading the wrong coordinates reports a well-formed number about the wrong quantity. An instantiation that resizes a segment should be able to do so in one place; if it cannot, the typed state is a convention rather than a structure.

Reference ontology versus invariant roles. The named contents above—which affect variables, which action tendencies, which monitoring read-outs—form the framework’s *reference ontology*: a concrete, worked inventory that later sections use throughout, so that the specification stays usable rather than abstract. It is a *declared choice*, not a derivation, and the distinction matters for anyone building on this. What the framework commits to is the *roles*: an affect segment combining a dimensional core with categorical intensities and a separate epistemic drive; an intention segment carrying a distribution over a small, closed set of action tendencies plus readiness and confidence; a meta segment carrying monitoring read-outs for at least confidence, disagreement, and processing demand. What it leaves to the instantiation is the *inventory*: how many categorical emotions, which action tendencies, whether uncertainty is tracked separately from confidence, whether novelty or agency signals earn their own coordinates. An instantiation may differ from the reference ontology provided it covers the roles and declares the difference—and the reference ontology should be read as one defensible resolution of choices the theory constrains but does not settle (Section 4 argues its particular resolution and marks where that resolution is contested).

Slow state S_t . The *slow state* holds dispositional variables—stable value orientations and processing tendencies that change on the timescale of many interactions rather than single ticks. It parameterizes *how* the core reacts (thresholds, biases, recovery tendencies) rather than *what* it currently represents.

The segmentation is a bet. One tension deserves naming at the point of commitment. Typed segmentation presupposes that inner-state content *factors* along the declared boundaries—that

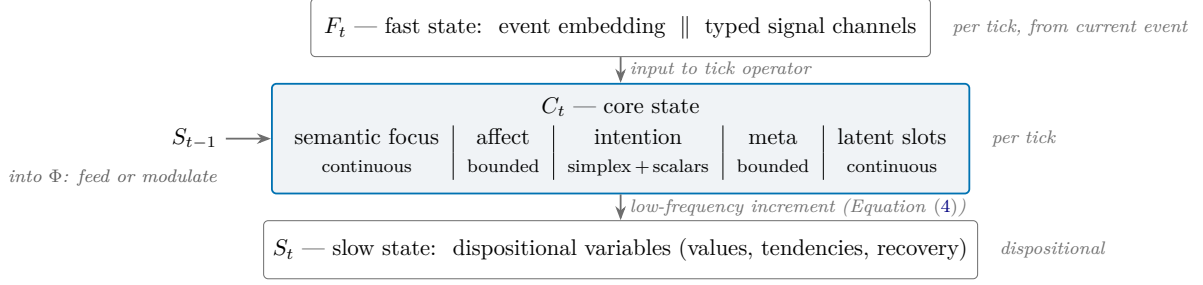


Figure 2: The segmented, multi-timescale state space. The fast state F_t is rebuilt from the current event and the core state C_t is rewritten on every tick; the slow state S_t integrates dispositional change at low frequency. The exposed state also contains the broadcast context g_t (Equation (1)), omitted because this figure focuses on the fast, core, and slow components and their distinct update paths. The left input is the previous slow state S_{t-1} ; the downward path produces S_t under the incremental update of Equation (4), so the paths do not form a same-tick cycle. The previous slow state may enter the tick operator by concatenation—the reference realization—or condition its transformation by modulation, as declared by the instantiation (Section 3.3). Segment roles are fixed by the framework; segment sizes are instantiation parameters.

affect, intention, semantics, and meta-observation are separable carriers rather than aspects of one entangled representation. Cognition may not factor this way; distributed representations generally do not. The framework accepts the bet: the typed segments buy inspectability, evaluability, and governable interfaces, the latent slots buy back representational freedom the typing forecloses—and where the right balance lies between a state that can be read and a state that can be rich is an open problem, catalogued with the others in Section 12.3.

3.3 Tick dynamics

A tick maps the previous state and the current input to a new state through the *tick operator* (Figure 3). The framework specifies this as an operator contract; any differentiable sequence model satisfying the contract is an admissible instantiation.

The tick operator. The core update is

$$C_t = \Phi(F_t, g_t, C_{t-1}, S_{t-1}, W_t, M_t; \theta), \quad (3)$$

where g_t is the broadcast context produced by workspace competition over specialist proposals—which themselves read (F_t, C_{t-1}, S_{t-1}) (Section 5)—, W_t is the working-memory summary defined below, M_t is content retrieved from long-term memory (Section 6), and θ are the learned parameters. The operator is recurrent in two coupled senses that the contract keeps distinct: it consumes the previous *core state* C_{t-1} as an explicit input, and—in some instantiation classes—it may maintain an internal carrier h_t across ticks, in which case C_t is obtained from that carrier by a typing map that enforces the segment constraints of Section 3.2. That second carrier is the source of the exposure problem treated in Section 3.4, and the contract there constrains it: whether h_t persists at all is an instantiation choice that must be declared, and no component other than the operator may read it. Gated recurrent architectures [18] and modern selective state-space models [40] both satisfy the contract; the framework does not rank them here. The discreteness of the step is itself a choice rather than a formality—continuous-time dynamics would express the multi-timescale organization of Section 3.7 more naturally, at the cost of the countable processing depth this framework is built on. Section 12.5 weighs the trade.

The order of a tick. Equation (3) names the operator’s arguments but not the order in which they become available, and that order is not free: the arguments are produced by mechanisms

that themselves read state. The framework therefore fixes one normative sequence. Within tick t :

- T1. Encode.** The input translator maps the current event to F_t (Section 3.5). On a tick with no external event, F_t carries the null event; the tick still runs.
- T2. Propose.** Specialists read (F_t, C_{t-1}, S_{t-1}) and emit proposals (Section 5.1).
- T3. Broadcast.** Competition over the proposals yields g_t (Section 5.2).
- T4. Retrieve.** The retrieval gate forms its query and its gate weight from the *pre-update* exposed state $(F_t, C_{t-1}, S_{t-1}, g_t)$ and returns M_t (Section 6.1).
- T5. Update.** Φ computes C_t by Equation (3); W_t is the summary over C_{t-1} and its predecessors.
- T6. Write.** Memory writes are formed from the post-update state (Section 6.1).
- T7. Slow update.** If the cadence condition of Equation (4) is met, S_t is computed; otherwise $S_t = S_{t-1}$. Either way X_t is complete at the end of this step and not before.
- T8. Decide.** The deliberation policy reads its permitted fields of the completed $\mathcal{X}_t^{(r)}$ and chooses *continue* or *act* (Section 5.5).
- T9. Report and act, or continue.** On *act*, the report-bearing fields of $\mathcal{X}_t^{(r)}$ are rendered and verbalized (Section 3.6), and the resulting self-action event enters as an ordinary event at step T1 of tick $t+1$. On *continue*, nothing is emitted and the next deliberation tick begins.

Two of these steps sit where they do for reasons worth stating, because both are easy to place wrongly, and the two mistakes are of different kinds.

T4 is where the order does real work rather than merely tidying. Forming the retrieval query from the *current* core state would make Equation (3) circular— C_t would depend on M_t and M_t on C_t —and the circle can only be cut by a fixed point or by factoring the update into two stages. This framework does neither: retrieval sees the state as it stood before the operator ran. The price is explicit and belongs with the choice: a retrieval cannot exploit the semantic focus that this tick’s own update produces, so content becomes retrievable one tick after the situation that would have called for it.

T8 must precede **T9**, and the reason is not aesthetic. The deliberation policy’s decision *is* the decision whether this tick emits at all (Section 5.5); a report rendered before it would presuppose the answer. Item **T7** in turn must precede both, because X_t contains S_t : a report rendered before the slow update would render a state that is complete in three of its four components and stale in the fourth, and no reader of the report could tell which. The three steps are therefore ordered by what each needs to exist rather than by narrative convenience, and an instantiation that reports before deciding is not running this tick.

Equation (3) fixes the operator’s argument list but not how those arguments enter, and the choice is load-bearing enough to name rather than leave to habit. Two axes are open. On *composition*, the arguments may simply be concatenated into one input vector, or the slower carriers may *modulate* rather than feed—conditioning the operator’s transformation (through gains and biases) instead of occupying input dimensions. The second reading fits the role the slow state is given in Section 3.2, where it parameterizes *how* the core reacts rather than *what* it represents; the first is simpler and is the reference realization. On *update form*, the operator may emit the next core state directly or emit an increment added to C_{t-1} ; the residual form biases toward continuity and makes per-tick change explicitly measurable, the direct form imposes no such prior. The framework requires only that the choice be declared, because H1 and H2b are sensitive to it: a residual operator with a small effective step will look more stable than a direct one for reasons that have nothing to do with the segmentation the hypotheses are about.

Where a signal is injected. Composition has a consequence that is easy to overlook when a new channel is added to an existing operator. A signal concatenated into the operator’s input vector does not arrive with a reserved share of the update; it competes with every other input dimension for the same first-layer weights, and at initialization its influence is set jointly by the width it occupies, its scale relative to its neighbors, the normalization the path applies, the covariance among the inputs, and the weight initialization. Under the usual width-neutral initialization at comparable input scale, a narrow channel added beside a wide one is therefore structurally quiet. The qualification matters and is easy to drop: raising the channel’s *amplitude* does move both its activation contribution and the gradient reaching its weights, unless the path normalizes scale away—so amplitude is a lever, just a blunt one that buys influence by distorting the input geometry rather than by reserving capacity. The alternative is to inject the signal where its effect is read rather than where the update begins: as a modulation of the operator’s transformation, or as an additive contribution to the quantity it is meant to shift, initialized at zero so that adding the path leaves the untouched system exactly unchanged. The framework does not rank the two—the input route keeps the operator’s argument list honest, the late route buys effect size—but it requires the choice to be named, because a channel reported as “present and ineffective” is uninterpretable until one knows which route it took. The zero-initialized late route has the further property that it is testable by construction: with the path’s contribution forced to zero the system must reproduce its own baseline exactly, which turns “the channel does nothing” into a measurable statement rather than a suspicion.

Working memory. The *working memory* W_t is a bounded summary of the last few core states, giving the tick operator direct access to the recent trajectory without burdening the recurrent carrier with short-horizon bookkeeping. What the contract requires is that the summary be *bounded* (a fixed window or a fixed-capacity aggregate, so it cannot become an unbounded second memory) and *differentiable*; the reference realization is a learned attention pooling over a short window, warm-started from simple averaging before attention weights train, but a fixed exponential aggregate or a small recurrent summarizer satisfies the same contract. Functionally, this window and the workspace’s short persistence of selected content (Section 5.2) together play the short-term-memory role in this architecture: capacity-limited recency that decays unless consolidated—the tick-scale counterpart of the long-term shelves of Section 6.

Slow-state update. The slow state changes by small, low-frequency increments,

$$S_t = S_{t-N} + \eta \Delta(\bar{C}_{t-N:t}), \quad (4)$$

where Δ is a learned update head applied every N ticks to a summary of the intervening core states, and η is small. The separation of Equation (3) and Equation (4)—per-tick rewriting versus slow incremental drift—is the architectural expression of the multi-timescale principle. Equation (4) is the reference form rather than the only admissible one: what the principle requires is that dispositional change be *incremental*, *aggregated over many ticks*, and *bounded per update*, so that fast dynamics cannot overwrite it. A continuous low-rate decay toward a running aggregate, or an event-triggered update conditioned on accumulated evidence, meets those requirements equally; the periodic form is chosen here because its update cadence is explicit and therefore easy to audit.

How the slow state is *supervised*, if at all, is a separate and deliberately open question, and one an instantiation must answer explicitly. Three regimes are admissible: *unsupervised*, where the slow carrier receives gradient only through its effect on core-state predictions—cleanest in principle, and the regime in which the vacuity risk below is sharpest; *endpoint-supervised*, where a sequence’s dispositional profile supplies a target at sequence boundaries; and *profile-conditioned*, where the profile is provided as input rather than target and the carrier must reconstruct it. The

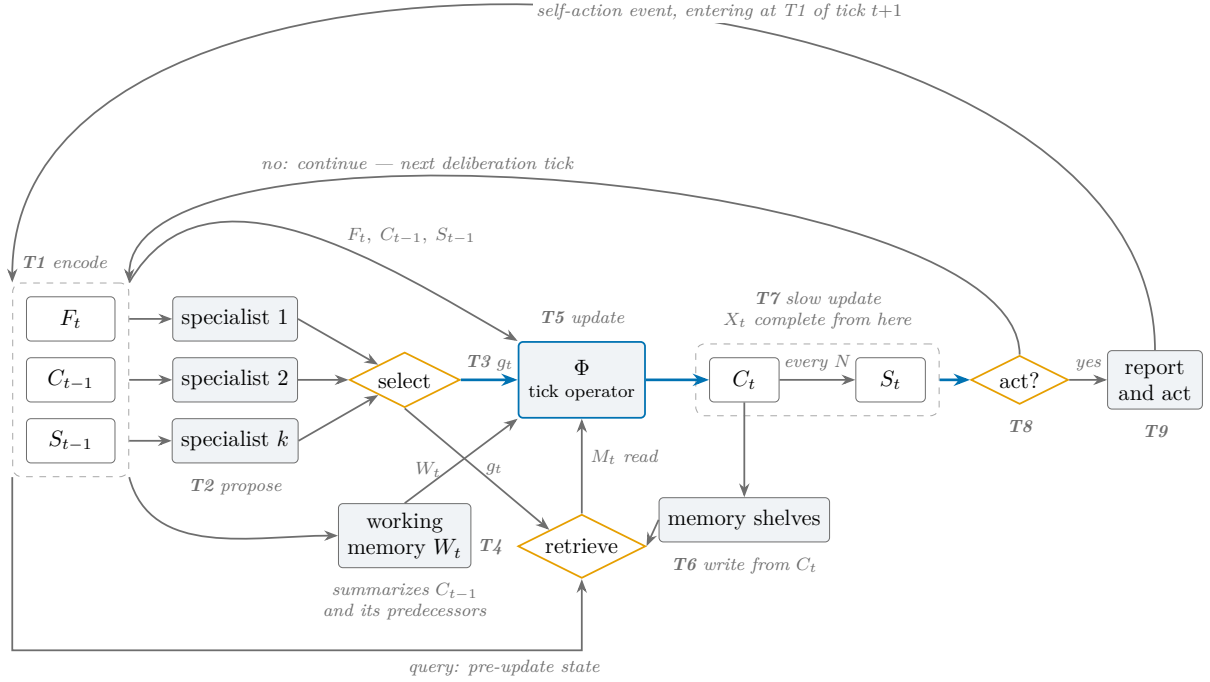


Figure 3: One tick, in the order of Section 3.3; the stage tags are its step numbers. The encoded event and signal channels form F_t (T1); specialists and the retrieval gate both read the *pre-update* state, which is why the query at T4 cannot see the C_t this tick produces; competitive selection yields g_t (T3); the operator rewrites C_t from the six arguments of Equation (3) (T5); memory is written from the post-update state (T6). The two steps on the right are where the order does work: the slow update runs *first* (T7), because X_t is incomplete until it has, and only then does the deliberation policy read the completed state and decide (T8). A policy drawn off C_t alone would be reading one of four components. On *act* (T9) the canonical self-action event re-enters as an ordinary event at T1 of the next tick—the generated text never does (Section 3.6). W_t summarizes C_{t-1} and its predecessors; this tick’s C_t enters the summary the *next* tick reads. The periodic $S_t \rightarrow$ identity-shelf consolidation is not a tick step and is drawn in Figure 6 instead.

recommendation to begin with a single default profile (Section 10.5): under a single profile, the endpoint-supervised regime has a constant target and the unsupervised regime has nothing forcing variation, so either way the channel is expected to stay quiet until profile diversity is introduced. That is a consequence of the staging choice rather than a defect, but it must be reported as such rather than discovered later.

Autoregressive operation. During both training rollouts and deployment, the C_{t-1} entering Equation (3) beyond the first tick is the system’s *own* previous core state, not a ground-truth state: the system runs on its own outputs, anchored by periodic external events. Sustaining non-trivial, non-collapsing trajectories under this self-feeding regime over horizons far exceeding the training-sequence length is the framework’s most basic behavioral requirement, stated as H1 in Section 11; the measurable separation of timescales is H2.

3.4 The exposure problem and exposed-state sufficiency

The recurrence of Equation (3) carries a problem serious enough to state before the rest of the architecture is built on it. If the operator maintains an internal carrier h_t across ticks and C_t is a readout of that carrier, then nothing in the equations so far forces the exposed state to be the state that matters. Least-effort optimization has a cheaper solution available: route content, context, and effective memory through h_t , and let C_t carry whatever the supervised heads are scored on. The result would satisfy every readout constraint while the typed state functions as an instrument panel attached to a system whose actual variables are elsewhere. Inspectability of a readout is not inspectability of a system.

The concern is not speculative, and it is not ours alone. Interpretability work has reported that a quantity can be linearly decodable from a model’s internals and still fail to control its behavior: in one recent preprint study a decodable state variable generalized to held-out inputs, while steering along the very direction that carried it left downstream choices unmoved—readability, behavioral sensitivity, and causal control are three properties that need not co-occur [112]. What separates the present treatment from that literature is the direction of the demand. There, the question is asked *after the fact* about a representation someone discovered. Here the exposed state is declared in advance as the system’s state, which converts the question into an architectural obligation: if the declaration is wrong, the architecture is wrong, not merely the interpretation. That is why the response below is a constraint and a test rather than an analysis method.

The framework’s response has three parts, in increasing cost.

First, a structural constraint (principle A9): no undeclared carrier may be a channel between components. All consumers—memory read and write, the deliberation policy, workspace proposals, the slow-state update, and the output translator—take only the regime-specific declared exposed state $\mathcal{X}_t^{(r)}$ as input. In regimes (i), (ii), and (iv), this removes the fan-out that would let h_t act as a hidden bus; the operator may still use h_t to compute the next typed core. In regime (iii), h_t is no longer hidden: any consumer access is named in the permission matrix and is part of the system under report.

Second, an admissible-regime declaration. The constraint does not by itself stop h_t from carrying information *across* ticks that never surfaces. An instantiation must therefore declare which regime it adopts, and the choice is a reportable property of the system, not a private matter: (i) *no hidden carrier*— C_t is the full persistent recurrent state; (ii) *per-tick reset*—any internal carrier is reinitialized each tick, so it cannot bridge ticks; (iii) *declared carrier*— h_t persists but is counted as part of the declared state and evaluated with it; or (iv) *constrained carrier*— h_t persists undeclared, meaning the regime is declared while the carrier itself is not counted as part of the state, under architectural pressure (bottleneck width, noise, or a cost term) that limits what it can hold. Regimes (i) and (ii) close the loophole by construction and pay for it in expressiveness; (iii) closes it by disclosure; (iv) merely narrows it, and an instantiation choosing (iv) carries the burden of the third part.

Each regime therefore implies a different reading of the same B.13 result, and leaving that implicit would make the marker’s verdict ambiguous. The rule is: a persistent carrier has exactly one of two statuses. Under (iii) it is *inside* $\mathcal{X}_t^{(r)}$ —it is rendered and evaluated like every other declared component, and any consumer route to it is named. B.13 then attributes downstream influence between the typed core X_t and the declared carrier h_t within one declared state; carrier dominance does not create an undeclared channel, but it defeats a narrower claim that the typed core drives behavior. Under (iv) h_t is *outside* $\mathcal{X}_t^{(r)}$, and the instantiation must pass B.13 against a dominance margin fixed in advance; failing it is a failure of the principle, not a disclosure question. Regimes (i) and (ii) have no persistent carrier to place at all: they satisfy B.13 by construction and should report that fact rather than the test (Section B). A carrier that persists outside $\mathcal{X}_t^{(r)}$ under *no* architectural pressure is neither regime: it is the unconstrained case the

register excludes, and a system built that way is a comparison variant.

Third, a measurement. Exposure sufficiency is an empirical property and is treated as one. The test is interventional and symmetric: hold the typed core fixed while perturbing the internal carrier, then hold the carrier fixed while perturbing the typed core, and measure in each case how much of the downstream behavior—next state, retrieval, halting decision, and report—follows. Under regime (iv), carrier dominance fails exposed-state sufficiency; under regime (iii), it leaves route closure intact but defeats any narrower typed-core-dominance claim. This is marker B.13 (Section B) and the paired prediction it tests is H17. The framework does not claim the problem is solved by construction; it claims the problem is nameable, constrainable, and measurable, and that an inner-state architecture that omits the measurement has not earned the word “inspectable.”

Direct and instrument-mediated inspectability. One distinction has to be drawn here, because the paper otherwise uses one word for two evidential situations. For scalar and explicitly structured segments—affect variables, the intention distribution, the monitoring read-outs—inspectability is *direct*: the value is the quantity, and reading it requires no inference. For the semantic-focus segment it is *instrument-mediated*: what is read is a decoder’s output, and the decode gap of Section 3.6 applies in full. A vector in a named segment is not thereby inspectable. The consequence for reporting is concrete—a state report must mark which of its fields are direct read-outs and which are decoded inferences, and the semantic segment counts as inspectable only for a decoder that has passed an intervention-based faithfulness check. Where this paper says a state is inspectable without qualification, it means the direct sense; the mediated sense always carries its instrument with it.

Which component may read what. The exposure contract is only checkable if the permitted channels are enumerated, so Figure 4 states them for regimes (i), (ii), and (iv). It is a specification, not a description of any implementation: an instantiation conforms if no consumer reads a channel outside $\mathcal{X}_t^{(r)}$ or without a named permission. In regime (iii), the carrier column is folded into the declared-state input and the instantiation publishes the additional per-consumer permission cells; an unlisted route is still forbidden. In regime (iv), giving any consumer access to h_t fills a barred cell and makes the system a comparison variant. B.13 may measure such variants, but disclosure alone does not restore conformance.

3.5 The input translator

External text reaches the core through the *input translator*: a pretrained sentence-embedding encoder [91], selected against standard embedding benchmarks [82] with open, reproducible models preferred [87], followed by a trainable projection into the state space.

The framework prescribes a *frozen-first* policy: in early training stages the encoder is fixed and only the projection is learned. A frozen encoder holds the chosen semantic geometry fixed and thereby keeps successive experiments more comparable; benchmark coverage informs encoder selection but does not make that geometry externally calibrated. The policy also prevents the core’s early training noise from destroying a representation it cannot yet replace. The cost is equally explicit: a frozen encoder is a fixed lens, and what it does not distinguish, the core cannot recover. Relaxing this constraint follows the staged unfreezing path of Section 7: safe encoder adaptation depends on learnability conditions, not architectural preference.

3.6 The output translator

Language leaves the system through the *output translator* (Figure 5). At an output tick, the report-bearing fields of the regime-specific declared exposed state $\mathcal{X}_t^{(r)}$ are rendered into a

		exposed state X_t				W_t M_t	h_t	report
		F_t	C_t	S_t	g_t			
Tick operator Φ		■	■	■	■	■	■	■
Specialist proposals		■	■	■	■	■	■	■
Workspace competition		■	■	■	■	■	■	■
Memory read (gate)		■	■	■	■	■	■	■
Memory write		■	■	■	■	■	■	■
Deliberation policy		■	■	■	■	■	■	■
Slow-state update		■	■	■	■	■	■	■
State report renderer		■	■	■	■	■	■	■
Output translator		■	■	■	■	■	■	■
Grounding check		■	■	■	■	■	■	■

■ permitted (state)	■ permitted (memory)	■ not read	■ forbidden by construction
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Figure 4: Channel permissions. A filled cell is a permitted read, a dashed cell a channel the component never sees, a barred cell a read the architecture forbids. The matrix is the base permission contract for regimes (i), (ii), and (iv). Two cells, ringed, carry the argument. The carrier column h_t is filled exactly once, for the operator alone; in regime (iv), any other fill would be an undeclared private channel. Regime (iii) instead extends the declared-state input by h_t and must publish each additional permission cell; those routes are declared, rendered, and evaluated rather than hidden. The translator’s row is empty except for the report, which is what “the report is its only content-bearing input” means operationally: an access restriction rather than an instruction. Two absences are easy to misread. Workspace competition consumes specialist proposals rather than state components, so its row is empty by construction; and the state report may not read W_t, M_t , because retrieved content reaches a report only after the operator has written it into C_t (Section 3.6). Every filled cell in the g_t column is the reason g_t belongs to X_t (Equation (1)) rather than sitting beside it. The columns name channels, not tick indices: which tick’s value a given consumer sees is fixed by the tick order of Section 3.3, and for the retrieval gate it is the pre-update core state.

structured *state report*: current affect, dominant intention, semantic focus, meta read-outs, dispositional tone, and what currently holds the broadcast. In regimes (i), (ii), and (iv), $\mathcal{X}_t^{(r)} = X_t$ for this purpose. In declared-carrier regime (iii), $\mathcal{X}_t^{(r)} = (X_t, h_t)$ and the report includes the declared carrier field under its published permission. A generative language model then verbalizes the report under a strict contract: it may articulate only what the report contains.

The rendering source is $\mathcal{X}_t^{(r)}$ and nothing beyond it. Consequently, retrieved memory content is intentionally excluded from the report fields. Retrieval enters the tick as M_t and reaches the report only by the route every other influence takes—through the operator, into C_t . The system can therefore say what it currently *holds*, not what was fetched on its behalf, and a retrieval

PERIPHERY language boundary

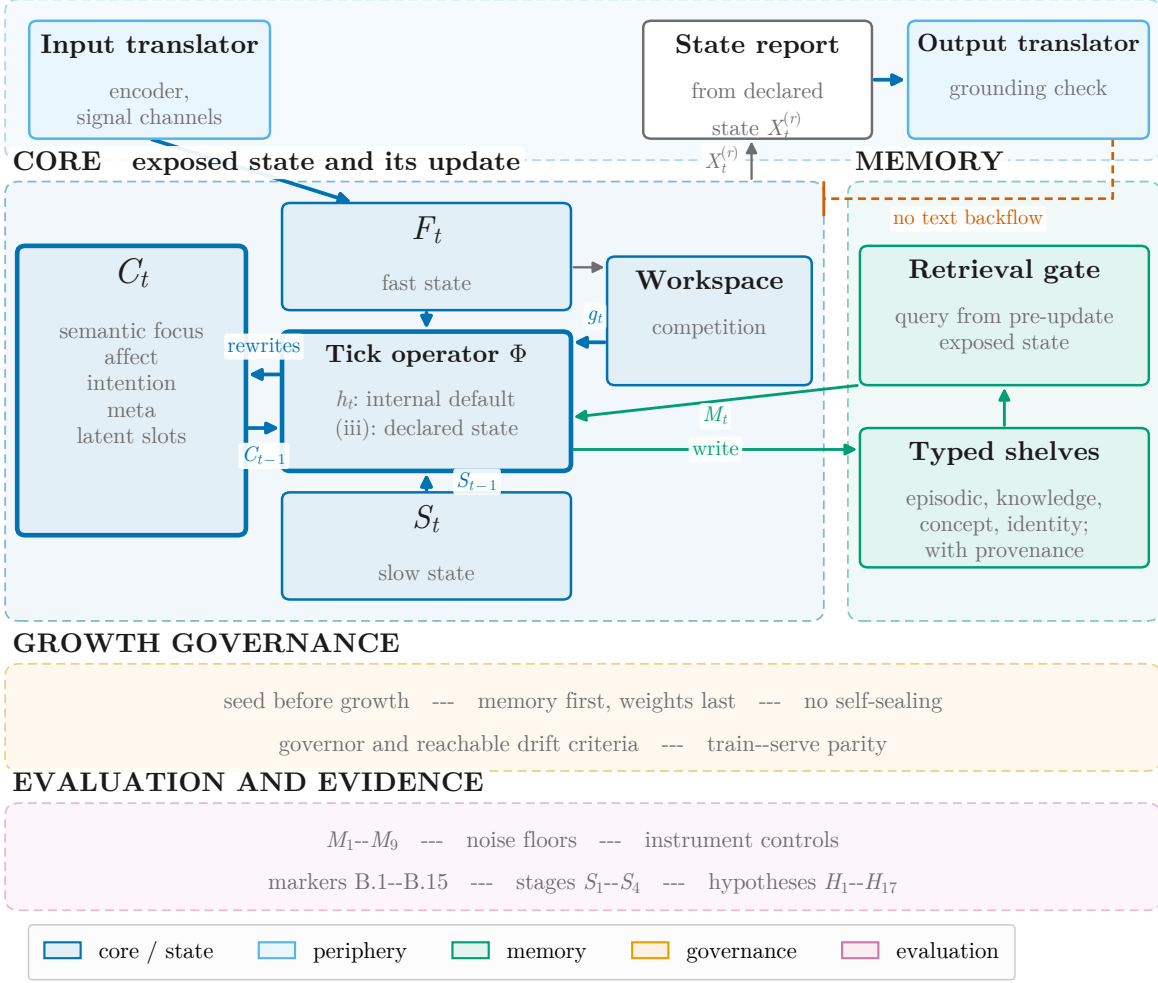


Figure 5: System overview. The core is language-free: an embedding encoder translates events into the state space, and a peripheral language model verbalizes structured state reports, its output checked against the report before emission. Neither peripheral component participates in the inner dynamics, and generated text never re-enters. The tick operator is drawn at the centre because every input named in Equation (3) enters *it*: the fast state, the broadcast, the previous core state, the slow state, and retrieved memory. Only the operator writes C_t , which is what the two arrows between them record. One operator input is omitted for legibility, the working-memory summary W_t over recent core states. The carrier h_t is operator-internal in the base contract shown; declared-carrier regime (iii) instead places it inside $\mathcal{X}_t^{(r)}$ and publishes every additional consumer permission, including the report field. This distinction is what principle A9 constrains and marker B.13 tests. The lower two bands are not components—they are the rules under which everything above may change and be claimed about. Blue arrows carry state, green arrows memory content; the barred vermilion line is a path the architecture forbids.

that left the core state unchanged is one the system cannot talk about. That is the intended behavior rather than a limitation to be engineered around: it makes speech a function of state, which is the whole claim, and it turns an inert memory pathway (Section 6.3) into something audible rather than something that quietly pads the output.

Observability and *sayability* are distinct. Observability is what an operator may watch: every state channel, the retrieval events, the checker’s score, the probe outputs. *Sayability* is what the system may assert: the report, and nothing else. The first is deliberately wider than the second.

A retrieval that changed nothing is visible on the console and unavailable to the utterance; this prevents the system from narrating machinery it did not use. Wherever this paper says a quantity is inspectable, it makes a claim about observability; whether the system can talk about it is the separate question the report answers.

Three requirements make this contract enforceable. First, *report grounding*: the report is the translator’s only content-bearing input, so everything sayable is bounded by something inspectable. The qualification is not decoration—it is exactly as strong as the report’s own inspectability, which for the scalar fields is direct and for the semantic field is instrument-mediated (Section 3.4). Where the decode has not been validated, “everything sayable is inspectable” is a claim about a chain whose weakest link has not been measured, and the decode-gap paragraph below spells out how that link breaks. The word *only* has to be taken literally, because the tempting relaxation is a specific one. Giving the translator a window of the recent conversation for coreference and register looks like a stylistic convenience; it is not. Prior exchanges carry propositional content, and a generative model that can see them can assert them. An instruction to use such context “for surface form only” is a request, not an access restriction, and a contract enforced by a request is not enforced. The framework therefore permits non-propositional conditioning—register, grammatical person, formality, length—and nothing else. Any auxiliary input that carries content does not thereby become permissible; it becomes *part of the declared report* and is covered by the reportability tests along with the rest of it. Second, *grounding verification*: generated text is validated against the report, and utterances asserting content absent from it are rejected or regenerated. This is the requirement most easily written down and least easily met, so the framework states what a valid instrument must supply rather than leaving “check it” as an instruction. Deciding whether an utterance asserts something the report does not contain is a question of semantic entailment, not of string overlap; a checker built on surface matching will pass a paraphrase that adds a claim and reject a faithful rewording. Four things are therefore declared per instantiation. *Coverage*: which classes of assertion the checker is claimed to decide—numeric values, named entities, relations, negation and modality, counterfactual and hedged statements—and which it is known to miss, because a checker with unstated coverage cannot be relied on for anything. *An instrument-positive control*: reply-report pairs in which a violation is present by construction, verified to be caught (M4). *Adversarial cases*: replies engineered to be fluent, report-consistent in wording, and false in content—the failure the checker exists for. And its *role*: a per-reply diagnostic, or a release gate that blocks emission; the two carry different consequences when the checker is wrong, and an instantiation says which it runs. B.15 (Section B) is the corresponding marker. Until an instantiation has one, the grounding contract is an architectural intention rather than an enforced property, and should be reported as such.

Declared coverage sets a floor on what may then be claimed, and the floor is part of the requirement rather than a matter of taste. A *grounding claim* requires checker coverage of every content-bearing report field the reply drew propositions from. The semantic field is where that bites: it carries most of the propositional content, so a checker restricted to typed scalars can pass everything it looks at while the assertion that matters goes unexamined. Such an instrument is admissible and useful—it is not non-conforming—but what it licenses is verified *grounding of the fields it covers*, not verified grounding. Fields outside the coverage may be displayed as diagnostics; they may not support an emitted assertion until an instrument reaches them. Third, *no backflow*: the translator’s text is never fed back into the core as if it were the system’s own state; the language model is a mouthpiece, not a participant in the dynamics. The third requirement anticipates the no-self-sealing governance rule of Section 8, where it reappears as a learning-source constraint.

The semantic decode gap. The contract is straightforward for the scalar segments: an affect coordinate, a dominant intention, a confidence value all render into the report by reading a number

and naming it. The semantic focus does not. It is a continuous vector of substantial width, and turning it into a sentence such as “the stated plan conflicts with an earlier commitment” requires a decode step the paragraph above quietly assumes. This matters more than a rendering detail: if the decode is unconstrained, the translator can supply content the vector never contained, and the grounding check—which compares text against the report—cannot catch it, because the fabrication entered the report itself. A vector is not inspectable merely by lying in a named segment.

The framework does not fix one decode mechanism, but it does require that an instantiation name its own and evaluate it. Four families are admissible: a *trained decoder* from the semantic segment to text, evaluated separately from the system that produced the vector; *retrieval against a canonical proposition library*, where the report cites the nearest admissible propositions with their distances rather than generating prose; a *bounded slot or graph format* in which the semantic segment is decoded into typed fields (referent, relation, polarity) instead of free text; or *constrained generation with explicit faithfulness scoring*, where the decoder proposes and a separate check verifies. Whichever is chosen, two obligations hold. The decode is *evaluated as an instrument in its own right*, with the same positive-control discipline the rest of the evaluation theory demands: intervene on the semantic segment in a known direction and verify the decoded report changes correspondingly, since a decoder that ignores its input will otherwise pass every fluency check. And the report distinguishes *decoded content* from *scalar readouts*, so a reader can see which parts of a report are direct and which are inferred—the two carry different warrants and should not be presented as one.

Self-action events. No backflow bars the generated *text* from re-entering the core, and that restriction is deliberate. But a system that emits an utterance and retains no trace of having done so cannot later know what it committed to, which undermines the continuity the architecture exists to provide. The framework therefore admits—and, for any system operating over sustained interaction, requires—a *self-action event*: a canonical, system-generated representation of the act just performed (its type, its addressee, and a pointer to the report it was grounded in), entering as an ordinary event on the next tick. What re-enters is the record that an act occurred, not the language model’s wording of it; the distinction is what keeps no-backflow intact while closing a gap that would otherwise leave the system’s own history missing from its state.

3.7 Why multiple timescales

The three-way split of Equation (1) is not an implementation convenience; it encodes a claim about the structure of inner-state dynamics. Reactive appraisal must be fast and event-bound; deliberative processing must integrate over several ticks; dispositional identity must be nearly stationary, or the faster carriers have no stable reference against which their variation is defined. Collapsing these timescales into one recurrent vector forces a single update rule to serve all three, and invites the fast dynamics to overwrite the slow ones.

The remedy—structuring recurrence by timescale—has a substantial lineage: delayed and hierarchically clocked recurrent connections [30], hidden layers partitioned into modules with distinct clock rates [62], and learned hierarchical update boundaries [19], alongside the complementary fast/slow organization of biological memory [78]. TASA inherits the mechanism-level insight and relocates it: timescales are assigned not to hidden-layer partitions but to *typed, inspectable state carriers* with distinct update mechanisms (Equations (3) and (4))—so the timescale structure is part of the system’s readable state, not an internal training device.

As a guiding image—and only as an image—one may picture the total state as a layered field in which fast ripples of reaction ride on slower swells of mood and an almost still dispositional depth. The framework’s contribution is not the metaphor but the contract that makes it testable: distinct carriers, distinct update mechanisms, and the measurable variance ordering that H2 demands.

Two anticipated difficulties attach to this design. First, *slow carriers are learning-starved by construction*: a variable updated every N ticks receives a fraction of the gradient traffic its fast neighbors receive, and a dispositional channel that nothing ever forces to vary risks becoming decorative—present, typed, and empty. Making the slow state genuinely load-bearing therefore requires explicit pressure (curricular variation across dispositional contexts, and the reachability discipline of Section 8.5 applied to its monitoring). The stakes of that pressure are worth naming: dispositional development over long horizons—the slow state shaped by the trajectories it has run through, snapshotted to the identity shelf (Section 6.1), and guarded by drift criteria (Section 8.4)—is part of the long-horizon aim of Section 1, which is exactly why the vacuity risk is instrumented rather than accepted. Second, the *tick’s semantics are under-determined*: how ticks relate to interaction time and world time, and how tick budgets should scale with task demands, is fixed here only functionally (via the deliberation policy), not theoretically. Both entries recur in Section 12.3.

3.8 Conformance: what makes an instantiation an instance

The framework delegates a substantial number of choices to the instantiation, each under a duty to declare it. Delegation on that scale invites a fair objection: if enough is left open, two conforming systems might share nothing but the vocabulary, and every hypothesis would then test a family rather than the framework. The objection is answered by separating what conformance *requires* of a system, what discipline it requires of a *study*, what it *permits*, and what a system that leaves the class should say instead—four things this subsection states once, in one place.

Two registers, not one. The separation has to be drawn twice rather than once, because two different kinds of obligation are easy to collapse into a single list and the collapse is not harmless. The *architectural register* R1–R7 states what a running system must be, and membership in the framework’s class is decided by it alone. The *disciplines* D1–D2 state what a study about such a system must do—how its supervision targets are checked, how its numbers are reported. The test that separates them is the one this subsection applies to the design principles below: an entry belongs in the architectural register only if it can be decided by inspecting an instantiation. D1 and D2 cannot—one fixes obligations that fall before and during training, the other governs claims, and both are checked by reading a report rather than a system. Keeping them inside a membership test would have made a system non-conforming for how its authors wrote about it. Two consequences follow, and both are intended. A study that neglects D1 or D2 has not thereby built a non-conforming system; it has produced results that carry less than they appear to. And the disciplines apply beyond this framework’s class: LC1–LC5 bind any architecture that supervises a latent state coordinate-wise, and M1–M9 bind any quantitative claim about a stateful, stochastic system, whether or not R1–R7 hold of it.

The architectural register—what a running system must be. The seven below are what can be decided by looking at an instantiation, and a system that omits one of them falls outside the class rather than inside it as a poor member. Further obligations follow from them in the sections that state them: no-self-sealing under R6, criterion reachability under D2, the frozen-first policy and the staged unfreezing path under R5, self-action events under R5 for systems in sustained interaction. The growth-governance model of Section 8 has no register entry of its own—its obligations ride on R6 and on D2—and that is recorded here rather than left for a reader to reconstruct.

R1. Tick-bounded update. State advances in discrete steps that rewrite the full state; processing depth is counted in ticks (A6, Section 3.3).

- R2. Fixed-width typed segmented core covering the roles.** A fixed-width vector core state partitioned into named segments that cover the reference roles—semantic content, affect with a dimensional core and a separate epistemic drive, a closed action-tendency distribution with readiness and confidence, monitoring read-outs, declared latent capacity (Section 3.2)—and partitioned by characteristic timescale, with a distinct update mechanism per partition (A5, Equations (3) and (4), Section 3.7). Inventories vary; the roles and the timescale split do not.
- R3. Affect in the control loop.** Affect variables persist across ticks, constrain each other, and causally modulate integration, action selection, and memory—not an output style (Section 4). All three modulations carry an explicit path: the affect–intention coupling for action selection, and at least one declared affect-conditioned path each into integration and into memory (Section 4.3). Affect being *readable* by a consumer does not discharge this; a consumer may read the segment and weight it at zero.
- R4. Declared-exposed-state closure.** Every channel a consumer reads belongs to the regime-specific declared exposed state $\mathcal{X}_t^{(r)}$ and has a named permission. In regimes (i), (ii), and (iv), no consumer other than the operator reads h_t ; in regime (iii), h_t is inside $\mathcal{X}_t^{(r)}$ and every permitted consumer route to it is published. An undeclared route is a comparison variant, and disclosure after the fact does not restore conformance (A9, Section 3.4).
- R5. Language peripheral, with a declared report path.** The core processes no tokens; utterances are generated only from a state report, and generated text never re-enters as state. The report mechanism and semantic decoder are declared (Section 3.6).
- R6. Typed memory with provenance and governed promotion.** Partitioned store, per-type lifecycles, provenance at write time, consolidation as an explicit gated operation (Sections 6 and 8.2).
- R7. Autonomous differentiable core.** The processes constituting the inner dynamics—appraisal, integration, memory access, self-monitoring, dispositional change—are computed by a differentiable core trainable end to end, and no external service participates in the inner loop (A1). What is required is one computation graph, not one module: R2 already admits a distinct update mechanism per timescale partition, and a core realized as several sub-networks conforms as long as gradients cross their interfaces—separate parameter groups or optimizer schedules do not by themselves break that. A system that calls a non-differentiable service inside the tick is a comparison variant, not an instantiation.

R3–R5 state structural membership conditions, not successful empirical outcomes. A conforming running system therefore has the required affect paths, declared-state closure, and report path even before their efficacy is known. The corresponding study must validate causal efficacy, exposure attribution, and report faithfulness under D2. Negative evidence does not retroactively erase structural membership: it blocks the associated efficacy or faithfulness claim and is reported as a failed or null result. By contrast, a missing path or an undeclared read is a structural non-conformance, not a negative experiment.

The disciplines—what a study about such a system must do. Neither of the two below decides membership, and neither is optional for anyone reporting results.

- D1. Target discipline.** Where the state is supervised by targets supplied per coordinate, LC1–LC5 are checked and reported, with probes at the target generator’s capacity class, on the schedule each condition admits: LC1, LC3 and LC5 before training; LC4 at initialization and monitored during optimization; LC2 twice—once design-time against an oracle information state and once as realized reachability under the trained model

(Section 7.3). The conditional is part of the discipline rather than a loophole in it. An instantiation may be trained by predictive or task-driven objectives instead (Sections 2.3 and 12.5), in which case LC1–LC5 are *inapplicable* rather than satisfied or violated, and what D1 then requires is that the supervision regime be declared, so that a reader can tell which of the two situations produced the absence of the checks.

D2. Measurement discipline. M1–M9 applied to every quantitative statement, including the three-way verdict vocabulary and the noise-floor hierarchy (Section 9.2). Criterion reachability (Section 8.5) is carried here, as the watchdog case of the same doctrine.

Where each principle went. A register that claims to define conformance in one place has to account for every principle it inherits. Where a principle has no requirement carrying it, “conforming” looks checkable while quietly depending on prose elsewhere. The mapping is therefore stated rather than implied.

Principle	Carried by
A1 Autonomous differentiable core	R7
A2 Language is peripheral A3 Token-free semantics	R5
A4 Explicit, inspectable, typed state	R2
A5 Multi-timescale organization	R2 (timescale clause)
A6 Tick-based evolution	R1
A7 Modularity A8 Bottom-up validation	— (see below)
A9 Exposed-state sufficiency	R4

A7 and A8 are process disciplines rather than properties of a running system and therefore do not enter the architectural register. Both constrain how a system is *built and validated*, not what it *is*: modularity is a property of a development process, and bottom-up validation is a rule about the order in which evidence is collected. Neither can be checked by inspecting a running instantiation, which is what R1–R7 must support. They remain binding on the program that builds a system—A8 is what Section 9 turns into measurement requirements—but a system is not non-conforming for having been built in some other order. That is the same test that put D1 and D2 outside the register: keeping either inside it would fail that criterion twice over, since neither a before-training check nor a reporting obligation is visible in a running system.

The register is *closed*: obligations stated elsewhere in this paper are consequences of R1–R7 and of the disciplines, not hidden additions to them. Closure is what a membership test is for, and it is all this one buys. It settles which systems the hypotheses of Section 11 range over. It does not rank the systems outside it, and the next paragraph says what the framework asks of those instead.

Declarable—free, but not silently. The list below is not exhaustive; it collects the choices this paper explicitly opens. Segment sizes; the operator family; operator composition and update form (Section 3.3); the exposure regime (Section 3.4); the write path (Section 5.1); the meta-supervision regime (Section 5.4); the slow-channel supervision regime (Section 3.3); the numeric target map and the semantic decode mechanism (Sections 3.6 and 7.2); the memory-shelf partition (Section 6.1); the affect and intention inventories (Section 3.2); every threshold and objective weight.

The order in which these are decided. The list above is flat; the decisions are not. A rebuilders gets the choices and the duty to declare them, but not the sequence—and three of them are prerequisites in the strong sense, in that taking them late invalidates work already done. The *segment layout* comes first: every probe, every per-term gradient share, and the whole LC4 arithmetic derive their offsets from it, so a layout revised after those exist changes what

they measured without changing what they report (A4). The *numeric target map* comes before any LC1 check, because non-triviality is a property of the map and can be decided on the data alone—long before a model exists to be fitted to it. And the reachability floor of LC2 comes before any gate or threshold is designed: a threshold stated as an absolute number is ill-posed, since the admissible range is a fraction of a probe-relative reachability floor rather than a value on the raw scale (Figure 7). These three are cheap early and expensive late, which is the only ordering claim the framework makes about them.

One further constraint is internal to the tick rather than to the design. Retrieval (T4) is specified before the update (T5), because the query reads the *pre-update* state; settling them in the other order reintroduces exactly the circularity the tick order exists to prevent (Section 3.3).

The remaining couplings are of a different kind, and the difference matters: the choice constrains nothing downstream, but the claim that depends on it is unreadable until the choice is on record. The *exposure regime* decides which arm of B.13 is even well posed, so a system that fixes it after running the marker has measured something it cannot name. The *meta-supervision regime* is reported with every calibration claim (B.5). *Operator composition* moves H1, and its residual form flatters stability measures specifically. The *write path* determines how much B.14 can distinguish, which is why H4 reports which form was used. The *slow-channel supervision regime* fixes what a dispositional-stability result is able to mean at all. None of these has a right answer here; each has a deadline, and the deadline is the first claim that rests on it.

Declared departures. The two lists above cover systems that stay inside the class. They do not cover the case a specification of this kind should expect most: a system that drops one of R1–R7 outright, replaces a whole layer, or abandons a commitment this framework treats as load-bearing—and is *better* for it. Without a name, such a system falls into the only remaining category, *comparison variant*, which is defined by what it is useful *for* (measuring what a requirement buys, B.13 and B.14) and would be a poor description of work that has moved past this framework rather than serving as its control condition.

A *declared departure* is therefore a third status, and its obligations are deliberately light: name what was dropped or replaced—a register entry, or one of the root commitments of Section 12.5—state why, and state what plays its role instead, or that nothing does and the property is given up. A departure is outside the conformance class and inside the research program. It cannot inherit this paper’s hypotheses unexamined, since those are quantified over conforming systems, and it does inherit D1 and D2, which never depended on membership.

Four of the five root commitments are departures in this sense: discrete ticks (R1), affect as the organizing layer (R3), the fixed-width vector state (R2), and the single trainable core (R7). The fifth is deliberately not, and the difference is instructive. Coordinate-wise supervision is not a register entry at all; D1 already carries the alternative, so a system trained by predictive or task-driven objectives is an ordinary conforming instantiation that declares its regime—no departure required. The framework expects the affect commitment to be revisited first (Section 12.5). R1–R7 exist to make such a move *legible*, so that a reader can see which load-bearing part was removed and what the result therefore does and does not show. A framework whose requirements could only disqualify would be measuring its own boundary rather than the systems near it.

What this means for the hypotheses. H1, H2, H9, and the conforming arm H17a range over conforming systems generally and are stated at that level—H17a in particular tests the empirical consequence of R4. Its paired negative arm H17b instead ranges over a predeclared non-conforming comparison variant. The H12–H13 pair ranges wider still, and the split above is what makes the difference statable: H12 and H13 are claims about measuring stateful, stochastic systems, they hold whether or not R1–R7 do, and they answer to D2 rather than to the register. Others depend on declarable choices and therefore test a *configuration*, not the framework: H3 depends on the numeric target map, H4 on the write path, H15 on the halting inputs and the

supervision that shapes them, H10 on the governance package as a whole. For those, a failure localizes to the configuration reported, not to the framework—which is why each requires its declarations alongside its result, and why replication means replicating the declaration set, not merely the section numbers.

4 The Affective System

In TASA, affect is not an output style or an annotation layer; it is a set of state variables with architectural consequences. Affect variables persist across ticks, modulate integration and action selection, gate memory writes, and are visible in every state report. This section specifies their structure, their internal consistency constraints, and their coupling to the rest of the system.

4.1 A dimensional–categorical hybrid with one epistemic drive

The *affect segment* combines three representational layers whose joint use is deliberate.

Dimensional core. Three continuous dimensions—valence, arousal, and dominance—form the base coordinate system, following the three-factor tradition in emotion psychology [94]; large-scale valence–arousal–dominance norms provide a direct empirical reference for the three axes [117], while their most influential two-dimensional relative is the circumplex [93]. Dimensions are the right substrate for *dynamics*: they are continuous, interpolable, and suited to regression targets, smooth trajectories, and homeostatic regulation.

Basic-emotion intensities. A small set of discrete emotion intensities—joy, fear, anger, sadness, and surprise—overlays the dimensional core, following the basic-emotions tradition [29]. Categories are the right substrate for *interpretability* and for data generation: scenario authors and human raters think in categories, not in coordinates. Disgust is deliberately omitted from this minimal inventory as an engineering scope choice, not because disembodied or moral forms are assumed unreal. The moral-disgust literature supports overlap with physical disgust while also showing that moral judgment recruits appraisal and social interpretation [14]. An embodied or morally focused instantiation may add a disgust coordinate; here such responses are represented, provisionally, as complex appraisal-borne trajectories in the sense of Section 4.4.

One epistemic drive. Curiosity/interest is represented as a separate motivational variable—the *epistemic drive*—not as a sixth basic emotion. This follows the psychology of curiosity as an information-gap-driven epistemic state [72]: functionally it behaves like a drive—it energizes exploration, sustains processing, and is satisfied by information—rather than like a reactive emotion episode. Keeping it separate keeps both theories clean: the basic-emotion layer stays reactive, and the drive layer supplies intrinsic motivation for the intention dynamics of Section 4.3.

The hybrid is redundant by design: the same affective situation is expressed once in coordinates and once in categories. The redundancy is what makes the next subsection’s constraints meaningful—and it mirrors standard practice in affective computing, where dimensional and categorical codings coexist because they serve different consumers [81, 88].

A note on theoretical standing: the status of basic-emotion categories is itself contested—constructionist accounts hold that emotions are situated constructions rather than natural kinds with dedicated signatures [6]. The framework does not adjudicate this debate and does not need to: the hybrid is an *engineering* choice, selected for trainability and interpretability, and the architecture would survive a constructionist replacement of the categorical layer (richer appraisal-derived patterns over the dimensional core) without structural change. What would not survive is pretending the choice is theoretically settled; it is not.

4.2 Consistency constraints

An affect state in which fear is high while arousal is low, or joy is high while valence is negative, conflicts with the framework’s declared coding rather than with a universal psychological invariant. The framework therefore requires *consistency constraints*: a small set of soft, psychologically motivated initial profile hypotheses between the categorical and dimensional layers during training—among them joy $\rightarrow$ positive valence, sadness $\rightarrow$ negative valence, fear $\rightarrow$ elevated arousal, fear $\rightarrow$ reduced dominance, anger $\rightarrow$ elevated arousal with retained dominance, surprise $\rightarrow$ elevated arousal with unconstrained valence.

Three design commitments matter here. First, the rules are *soft* penalties, not hard projections: unusual but genuine states (bittersweet joy, calm anger) remain expressible at a cost, rather than being clipped away. Second, the rule set is *small and auditable*—the dimensions and category gradients are empirically motivated [21, 102, 117], but the particular profiles above are framework hypotheses to be preregistered and tested, not invariants entailed by those sources. The constraint set is therefore a falsifiable commitment rather than a tuning surface. The commitment is to the *rules*, not to the inventory they range over: an instantiation that declares a different affect inventory (Section 3.2) inherits the obligation to state the corresponding rule set and its sources, and it is that stated set—not this paper’s—which is then falsifiable. Third, constraints operate on *states*, not on trajectories: how affect evolves is learned from data; only its momentary internal coherence is architecturally *encouraged*—and the distinction matters more than the word. Soft penalties shape a training objective; they do not constrain the reachable state set, on training data or under distribution shift. Since the framework deliberately keeps unusual states reachable above, a coherence *guarantee* is exactly what it must not claim. Runtime coherence is therefore an empirical property, measured in and out of distribution and reported as such; an instantiation that wants a hard invariant has to buy it with a projection step or a constrained parameterization, and pay the expressiveness.

4.3 Affect–intention coupling

Affect that changes nothing is decoration. The framework requires a structured *affect–intention coupling*: a bias term through which affect variables shift the logits of the intention distribution before it is normalized. The coupling matrix is initialized from declared, testable engineering priors—fear raises retreat and vigilance tendencies, curiosity raises exploration, sadness dampens action readiness, joy raises engagement—and remains learnable, so training can refine or, where the data demand it, override the prior. The appraisal literature motivates the general connection between appraisal, emotion, and action tendency; it does not validate these four initializations individually.

The design rationale is a middle course between two failure modes. Hard-wired affect-to-action rules produce a rule cage: behavior that cannot adapt and a system whose “emotions” are a lookup table. Fully emergent coupling, at the other extreme, provides no guarantee that affect influences behavior at all—the optimizer is free to route around the affect segment entirely if no architectural path privileges it. A learnable bias with a principled initialization guarantees the causal pathway exists from the first gradient step while leaving its refinement to learning. The appraisal tradition supplies the psychological warrant: action tendencies are a constitutive component of emotion episodes, not a downstream add-on [95].

Two additional causal paths. Action selection is only one of the three modulations R3 demands; integration and memory are the other two, and until this paragraph they had no mechanism behind them. That gap was easy to miss because affect *is* inside the core state, so the workspace and the memory interface read it as a matter of course. Reading is not modulating. A system whose specialists and retrieval gate happen to learn weights near zero on the affect coordinates would satisfy every structural description in this paper while affect

influenced nothing beyond the intention logits—and no measurement in the framework would notice, because “the segment is visible to the consumer” is true either way.

The framework therefore requires, alongside the affect–intention coupling, at least one *explicit* affect-conditioned path into integration and one into memory. It does not prescribe their form, and two families are workable for each. Into integration: affect as a modulator of the workspace’s selection temperature or capacity—high arousal sharpening competition, low arousal widening it—or as a bias on the proposal scores of specialists whose content the current affect makes relevant. Into memory: affect as a component of the retrieval gate alongside the meta segment (Section 6.1), or as a write-salience term deciding which episodes reach the episodic shelf at all, which is the closer analogue of experimentally documented affective modulation of memory consolidation [79]. Whichever is chosen, the obligation that makes the requirement checkable is an interventional one, in the style of B.10: move the affect coordinates within their observed operating range while holding the rest of the state fixed, and show that the integration and retrieval statistics move with them.

One limitation belongs here rather than in a later section. No dedicated marker for these two paths exists in Section B—B.10 tests the meta segment’s causal efficacy and the affect analogue has not been written. Instantiations run the intervention and report it; a marker with a fixed failure criterion and its own instrument-positive control is owed, and naming the gap is better than letting R3’s third clause rest on a mechanism no instrument checks.

4.4 Complex affective patterns are trajectories, not primitives

The framework deliberately stops its affective ontology at the hybrid of Section 4.1. Complex phenomena—ambivalence, melancholy, anticipation, relief, awe—are not additional state variables; they are *patterns*: configurations and trajectories over the basic space, extended in time and context. Melancholy is sustained mildly negative valence with low arousal and intact dominance; relief is a characteristic arousal-decay trajectory following a fear episode; ambivalence is the co-activation of opposing valence contributions across concurrent appraisals. Episode structure itself—onset, peak, decay, offset—lives at the same level: an emotion *episode* in this framework is a bounded trajectory through the affect space, not a switched discrete state, which is how momentary affect levels and temporally extended episodes coexist in one representation.

Two consequences follow. First, the complex vocabulary is an *analysis* layer: detecting such patterns is an evaluation task (Section 9), not a modeling primitive, which keeps the core ontology small and the claims testable—H3 states that contextually plausible reactive patterns arise from the constrained basic space without per-scenario rules, and the complex vocabulary above supplies the analysis layer in which such patterns are detected. Second, reliable *situated* affect—reacting to what the situation means rather than to surface intensity—requires an interpretation of the situation. The component-process view of emotion makes appraisal central [95]; the framework draws the corresponding architectural conclusion in Section 7: situation interpretation is made an explicit, supervised element (the *appraisal context*) rather than a hoped-for emergent, a commitment stated there as the supervised-interpretation principle and tested as H5.

4.5 Functional semantics

Throughout this paper, affect vocabulary denotes functional state variables and their causal roles: “fear rises” means the fear variable increases and, through the couplings above, vigilance-related tendencies strengthen. These terms do not denote felt states.

5 Workspace, Competition, and Meta-Cognition

A segmented state answers *what* the system represents; it does not answer how heterogeneous processes come to write into one coherent state, or how the system monitors its own processing. TASA adopts the workspace answer from Global Workspace Theory [4, 23]—specialists compete, a winner is amplified, the result is broadcast—and turns it into two architectural requirements that recent computational work makes precise [38, 110].

5.1 Specialists propose, the workspace disposes

Each tick, a small set of *specialist processes*—modules attending to different aspects of the situation, such as perceptual content, affective significance, and action relevance—produces candidate contributions to the next core state. R3’s integration path (Section 4.3) attaches at this stage or at the selection that follows it: an instantiation declares whether affect modulates the competition’s sharpness and capacity or biases proposal scores, and demonstrates the path by intervention rather than by pointing at the affect segment’s presence in C_{t-1} . A *workspace* mechanism then determines what enters the shared broadcast context g_t —consumed by the tick operator (Equation (3)) and read downstream by the retrieval gate, the deliberation policy, and the state report: the broadcast’s audience is every mechanism that acts on the current state, which is what makes it a broadcast rather than a private channel. Those downstream reads are what put g_t inside the exposed state rather than beside it (Equation (1)); the permitted reads per consumer are fixed in Figure 4, and A9 admits no channel a consumer reads and $\mathcal{X}_t^{(r)}$ omits. The division of labor is the point: specialists can develop focused competence precisely because they are not individually responsible for the whole state.

One point of the division of labor needs stating explicitly, because the two requirements below can otherwise be read as contradicting each other. The core state is written by the tick operator (Equation (3)); specialists never write it independently of that update. What specialists own is *output responsibility*—which coordinates their proposal is accountable for, and therefore which coordinates their gradients shape. Two admissible realizations follow, differing in how the update factorizes. In the *broadcast-only* form, proposals reach the state solely through the selected broadcast context, and disjointness is enforced on the proposal-to-broadcast mapping. In the *two-phase* form, the operator’s update is factored into a shared write and a set of specialist-owned additive contributions to disjoint regions—still one update under Equation (3), but with the disjointness constraint enforced on the contributions rather than on the broadcast. Both preserve the property H4 tests—that specialists face non-overlapping loss surfaces—but they differ in where the constraint is enforced, and a study reporting H4 must say which form it ran, since the 2×2 design means something different in each.

Proposals that lose the competition need not be discarded from the system, and whether they are is decided by the realization rather than by the idea. In workspace-theoretic terms, processing that stays outside the globally available state is the structural counterpart of processing without conscious access [23], and the framework treats this *non-broadcast processing* as a first-class structural feature rather than an implementation leftover.

But a feature needs a carrier, and this one has a carrier in only one of the two realizations. In the two-phase form, losing specialists retain their own output region, so their processing persists by construction. In the broadcast-only form, proposals reach the state solely through the selected broadcast, and a specialist that recomputes its proposal from (F_t, C_{t-1}, S_{t-1}) each tick carries nothing forward: there is no path by which a losing proposal shapes a later one. Adding one would mean giving specialists a persistent activation of their own—a second carrier that specialists then read, which A9 admits only if it is declared part of X_t , and the framework does not extend X_t that way. The consequence is stated rather than hidden: in a broadcast-only realization, non-broadcast influence is *absent by construction*, marker B.14 is expected to return a null there, and that null carries no information about the workspace. B.14 is informative

only where losing proposals have a declared path into the state—the two-phase form has one by construction—and Section B scopes it accordingly.

5.2 The competitive-selection requirement

The first requirement concerns *how* the workspace combines proposals.

Workspace access must be competitive and capacity-limited: sparse selection with nonlinear amplification of winners (ignition) and short persistence of selected content—never an unconstrained soft average over all proposals.

The argument is structural. A softmax mixture over proposals is differentiable and convenient, but under gradient pressure it is expected to drift toward a permanent blend: every specialist contributes a little to everything, gradients flow evenly, and the specialists’ representations converge, so that the workspace degenerates into an averaging bus and loses exactly the selectivity it was introduced to provide. H4 states the claim in testable form. Capacity limits and competition are not decorative fidelity to the neuroscience; they are what makes a workspace a *selection* device. This is also how the theory itself characterizes conscious access—*competitive selection* followed by ignition and global *broadcast* [22, 24]—and how modern shared-workspace architectures obtain their generalization benefits, through bandwidth-limited write access rather than free mixing [38]. Discrete or near-discrete selection remains trainable end-to-end via continuous relaxations [55, 74], with annealing from soft to sharp selection as the practical schedule. Recent robotics-oriented work argues, complementarily, that a selection–broadcast cycle confers real-time functional advantages independent of any consciousness interpretation [84].

5.3 The output-disjointness principle

The second requirement concerns *where* specialization lives.

Where specialists share averaged output coordinates, the objective leaves specialist identity underdetermined and admits homogeneous solutions of equal loss. Disjoint output responsibility—each specialist accountable for its own region of the core state—removes that particular degeneracy. Disjoint inputs alone do not.

We state this as the *output-disjointness* principle for typed cognitive-state specialists. The mechanism it rests on is not ours: assigning experts to disjoint output subspaces so that the router delivers non-uniform gradient signals and the permutation symmetry between experts breaks has been used in mixture-of-experts adapters [56]. That result establishes symmetry breaking and routing de-uniformization, not functional specialization; recent work explicitly cautions that routing geometry need not imply domain expertise [116]. What we claim is the transfer and the constraint—that in a typed state the responsibility boundaries should coincide with the segment boundaries, and that an instantiation must declare where the constraint is enforced. The reasoning again follows the gradients. If several specialists’ outputs are averaged into the same target coordinates, each receives only the gradient of the *mixture*: the objective constrains the average, never the parts, so it supplies no pressure toward the specialists differing. The solution set is correspondingly degenerate—the homogeneous configuration in which all specialists compute the mixture’s function is loss-equivalent to every diverse configuration with the same average, and nothing in the objective’s gradient field favors leaving its neighborhood once entered. Assigning disjoint output regions removes that degeneracy: each specialist owns its own loss surface, so the objective now distinguishes configurations the averaged form could not. The argument removes one identified symmetry under one factorization; it neither shows that every shared-output design supplies zero diversity pressure nor guarantees *functional*

specialization—distinct loss surfaces can still be minimized by specialists that compute near-identical functions. Whether disjointness produces specialization in practice is the empirical question, and it is H4 that answers it rather than this paragraph. Protecting expert diversity against the pull of shared objectives is the same concern that motivates per-task gating in multi-gate mixture-of-experts systems [73]. In TASA, output-disjointness composes with the typed segmentation of Section 3.2: segment boundaries are natural responsibility boundaries, so the state’s semantics and the specialists’ ownership reinforce each other.

Disjointness is the framework’s choice, not the only route to the same end. Load balancing, orthogonality penalties, expert-specific losses, top- k routing, information bottlenecks, and distribution-level specialization all supply pressure toward diversity by other means, and some compose with disjointness rather than replacing it. What distinguishes output ownership here is that it acts through the loss geometry rather than through an added regularizer, and that it aligns with a segmentation the architecture already commits to—so it costs nothing extra structurally. Whether it also outperforms the alternatives is an open comparison, and H4 is stated as a factorial test rather than as a demonstration of the framework’s preference.

Disjointness protects specialization; it does not dissolve a deeper tension that any shared-state architecture should expect. The desiderata of Section 9 include both *convergence*-flavored qualities (coherent integration, focused intention, stable trajectories) and *differentiation*-flavored ones (distinct responses to distinct situations, expressive breadth)—and both read the *same* core state. Pressure toward one is, along the shared coordinates, pressure away from the other: improving a single axis in isolation should be expected to be locally zero-sum against its antagonists, and multi-objective training on a shared carrier to settle into compromises rather than dominate every axis at once. The framework’s levers—disjoint output regions, segment typing, workspace capacity limits—*localize* this tension rather than eliminate it. It is catalogued as an anticipated difficulty in Section 12.3.

5.4 The meta segment: self-observation with a calibration duty

Above the workspace sits the framework’s minimal self-monitoring layer. A dedicated head observes the core’s processing and writes three variables into the *meta segment*: *confidence* (how reliable the current processing is), *conflict* (how strongly the current proposals disagree), and *complexity* (how demanding the current content is). Taxonomically this is a *metacognitive monitoring* layer: confidence–accuracy relations are one standard family of metacognitive measures, alongside sensitivity and efficiency measures rather than the whole construct [32, 75]. Recent work also provides evidence that language models can monitor and control aspects of their internal activations under specific experimental tasks [68]. Higher-order theories of consciousness place representations of one’s own mental states at the center of conscious awareness [66, 92]; the meta segment is *inspired* by that motif while claiming less: three monitoring scalars are not higher-order representation proper, and the framework says so rather than borrowing the theory’s weight. They are, however, deliberately the *beginning* of an axis rather than its end: a machine-facing metacognitive layer that later instantiations can widen—richer self-models, meta-representations of processing history, confidence over confidence—without changing the contract that whatever occupies the layer must stay calibrated against what it claims to represent. The long-horizon aim of Section 1 counts that axis among the capabilities expected to deepen. What the layer inherits unreduced, at any width, is an instrument’s duty of accuracy.

That duty is *meta calibration*: each meta variable must actually correlate with the quantity it claims to represent—confidence with realized accuracy, conflict with measured proposal disagreement, complexity with processing demand. The duty attaches to the roles, not to the count: an instantiation that separates uncertainty from confidence, or adds a novelty read-out, inherits exactly the same obligation for each variable it declares (Section 3.2). The supervision recipe is consistency-based rather than label-based: confidence is trained against realized prediction error, conflict against the workspace’s own competition statistics, and complexity

against observable demand statistics such as the processing depth actually consumed and the dispersion of specialist proposals—so no external annotator is needed.

Two supervision regimes are admissible here, and an instantiation must declare which it uses, because they differ in what the calibration duty can then establish. *Consistency supervision* trains each meta variable against the quantity it claims to represent, as just described; it keeps the calibration duty non-circular in the sense of M8—the variable is checked against something the pipeline did not also invent—and it is therefore the regime under which calibration claims are made. *Label supervision* takes meta values from the data pipeline alongside the other latent targets (Section 7.3); it converges faster from cold start and is easier to generate, at the price that meta calibration then partly measures agreement with the generator’s convention rather than with realized quantities. The framework’s default is therefore the *composite*: generator labels as a decaying warm start through the early curriculum stages, handing over to consistency supervision before any calibration claim is made—so that training gets the easier signal early while everything finally measured comes from the consistency regime. Whichever is chosen, the marker battery must report which regime produced the numbers (B.5, Section B).

Two failure modes are named now because the evaluation theory must detect them later (Section 9): *vacuous meta variables* (variables that decouple from their referents and drift to constants) and *anti-calibration* (variables that correlate with the wrong sign—a system most confident precisely where it errs most). A composite score that averages a calibration term with unrelated terms can mask anti-calibration entirely, which is why M3 will require composite guards to be reported with their components.

5.5 Multi-tick processing and the deliberation policy

Because the tick, not the utterance, is the unit of processing, responding is a *decision*, not a reflex. The *deliberation policy* decides at each tick whether to continue internal processing or to act, based on the current intention distribution and the meta segment—and, where an instantiation uses it, on what currently holds the broadcast, since a workspace that has just ignited on new content is evidence that processing is not finished. All of these lie inside the exposed state (Figure 4); the policy reads no other channel, which is what makes its decisions attributable to a state a reader can see. Its position in the tick is fixed and load-bearing: it runs at step **T8** of Section 3.3, after the update and the slow-state step have completed X_t and *before* anything is rendered—the policy decides whether the tick emits at all, so a report produced ahead of it would presuppose its answer. A response is triggered by a *halting criterion*: the action tendency crosses a readiness threshold, or meta-confidence is sufficient, or a tick budget is exhausted. Processing depth thereby becomes variable and content-dependent—trivial inputs resolve in a tick or two, demanding ones consume many. This content-adaptive halting commitment follows adaptive-computation work such as PonderNet [5]; continuous-latent reasoning supplies a related representation-space motivation but does not by itself establish adaptive depth.

The decision has a behavioral face. In interactive settings, the tick count preceding a reply becomes an expression of processing rather than an implementation accident: a reply after two ticks and a reply after ten differ in how much integration preceded them, the meta segment carries that difference explicitly, and it is observable in the tick stream even where wall-clock latency stays imperceptible (Section 10.7). Waiting is a legitimate outcome—a tick that ends in “continue processing” rather than action is the system’s way of not yet being ready to speak—and input arriving mid-deliberation poses no special case: it is simply the next event, folded into the very state the deliberation is refining. Section 10.8 walks through one full exchange built on this policy, and H15 states the associated falsifiable claim.

This gives processing depth a deliberately stronger meaning than elapsed computation. Additional internal steps count as deeper deliberation only if depth tracks externally defined demand and yields a measurable benefit over matched-compute controls; otherwise the result is latency or policy variation, not a deliberation dividend. The architectural ambition is therefore

not to make a system appear to think longer, but to make the functional value of continued internal processing falsifiable.

5.6 Conflict is a signal, not an emergency

A control note closes the section. Competitive integration *produces* conflict: the conflict variable rising during hard content is the mechanism working, not failing. Control loops layered on top of the architecture—schedulers, safety monitors, training-time regulators—must therefore not treat expected, transient elevations of conflict (or the accompanying dips in downstream quality) as emergencies to be suppressed; a regulator that punishes every transient makes the system unable to traverse exactly the states in which integration happens. And when a pathological reading *is* intended, pressure must be applied to the measured failure variable itself, not to a convenient proxy correlated with it—optimizing the proxy reliably optimizes the proxy. Both rules return in operational form in the governance and evaluation sections (Sections 8 and 9).

6 Unified Memory

Without long-term memory, a persistent state is persistent only within an episode. TASA specifies memory as a single store with typed partitions, a governed promotion mechanism between them—and, less conventionally, a set of conditions under which memory is actually *used* by the learned dynamics. The last part is the section’s core contribution: architectures acquire memory modules easily; systems acquire memory *use* only under conditions this section makes explicit.

6.1 One store, typed shelves

Long-term memory is a key–value store partitioned into typed *memory shelves*. What the framework requires is the *partitioning*: content whose lifecycles, write authorities, and governance rules differ must not share an undifferentiated store, because the consolidation and provenance rules of Section 8 are stated per type and cannot be enforced over a flat one. Which types, and how many, is an instantiation choice. The reference ontology uses four, chosen so that each has a distinct lifecycle:

- the **episodic shelf**: experienced situations—keys derived from the situation encoding, values holding state snapshots with affective context;
- the **knowledge shelf**: learned facts and insights, stored as the state experience of acquiring them;
- the **concept shelf**: consolidated, recurring structures distilled from episodes;
- the **identity shelf**: periodic snapshots of the slow state—the system’s own dispositional history.

All shelves share one mechanism: content-based retrieval by key similarity, with the retrieved value entering the tick as M_t (Equation (3)) through a confidence-weighted *retrieval gate*. Both of the gate’s inputs come from the exposed state and are named here so the permission figure of Figure 4 summarizes something the text fixes: the *query* is formed from the situation as it stands *before* the operator runs—the semantic focus of C_{t-1} together with the fast state and, where the instantiation uses it, the broadcast context—and the *gate weight* from the meta segment of C_{t-1} , so that a system with low confidence retrieves more readily than one that is already settled. R3’s memory path (Section 4.3) attaches here or at the write side: an instantiation declares whether affect enters the gate weight alongside the meta segment, or governs write salience instead, and

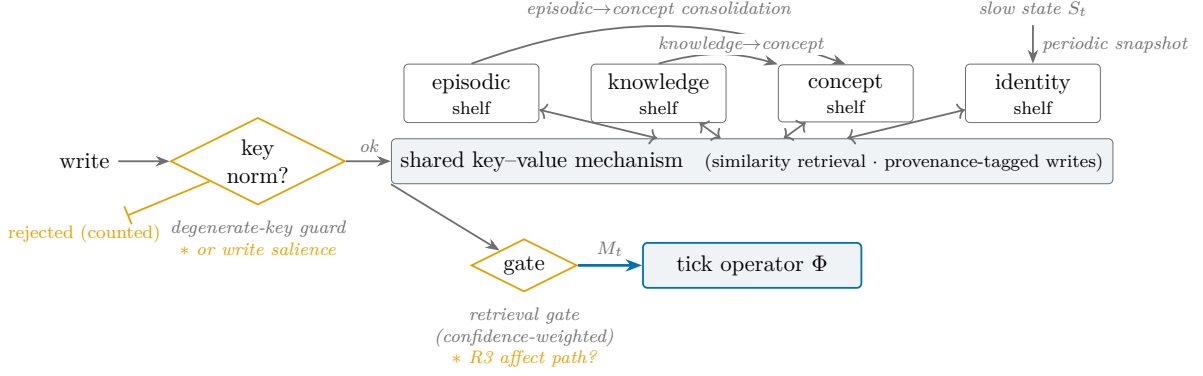


Figure 6: Unified memory: four typed shelves over one key–value mechanism. Retrieval enters the tick through a confidence-weighted gate; consolidation promotes content between shelves under governance criteria; every write carries provenance. The degenerate-key guard rejects near-zero-norm keys at the store boundary. The two starred points are the candidate attachment sites for the affect path R3 requires into memory (Section 4.3): the retrieval gate weight, or write salience. An instantiation declares *one* and demonstrates it by intervention; the figure marks both because the framework fixes neither.

shows by intervention that the chosen path moves. The tick index is not incidental: retrieval sits at step **T4** of Section 3.3, before the update that produces C_t , because a query formed from C_t would make Equation (3) circular. Which of the admissible components an instantiation actually uses is a declared choice; that all of them lie inside X_t is not. The design premise is that episodic memory and knowledge storage are architecturally identical and differ only in content and lifecycle—one interface, several shelves (Figure 6). Key–value episodic stores have a solid lineage in neural systems [89]; the DNC establishes a broader differentiable external-memory mechanism rather than this shelf-specific episodic role [39]. The framework’s addition is the typed partition, because the *governance* of memory (what may be written, promoted, or consolidated, Section 8) needs type distinctions that a flat store cannot express. A finer partition (separating procedural content, or splitting episodes by provenance class) or a coarser one is admissible on the same terms: declare the types, give each a lifecycle, and state the governance rules per type. Typed memory representations have independently been proposed to prevent source-monitoring failures in long-lived agents [57], which is the same argument arriving from the reliability side.

6.2 Consolidation as an explicit, governable operation

Beyond gated per-tick writes to the episodic shelf, and governed, provenance-tagged knowledge-acquisition writes to the knowledge shelf, content moves between shelves only through *consolidation*: episodes that recur are distilled into concepts, recurring knowledge stabilizes into concepts, and periodic snapshots of the slow state are consolidated into the identity shelf. Consolidation is an explicit operation with its own acceptance criteria, not a side effect of training—following the complementary-learning-systems insight that fast episodic acquisition and slow structural integration are different processes that should not be collapsed into one [63, 78]. Rare, verified, reversible consolidation into *weights* is the furthest step of this ladder and is deferred to the governance section, where its gating is specified.

Every write carries *provenance*: source, context, and confidence at write time. Provenance is recorded at the memory layer because the governance layer will need it—the no-self-sealing rule of Section 8 is unenforceable over a store that has forgotten where its contents came from.

6.3 The task-pressure condition

The first usage condition concerns training incentives.

Absent other pressure, learning recruits a memory mechanism when task-relevant information lies beyond the reach of the recurrent state. Where recurrence can carry the information cheaply, memory tends to remain functionally inert—regardless of the mechanism’s health.

We call this the *task-pressure condition*. Shortcut-learning results motivate the general concern that a model may exploit an easier available route [36]; they do not establish which route is shorter in this architecture. More direct memory-augmented-network evidence shows that recurrent controllers can retain task information and thereby reduce reliance on external memory [109]. We therefore state the hidden-state-versus-retrieval trade-off as a testable framework hypothesis, not a consequence of gradient descent.

The principle is a statement about incentives, not a law about mechanisms, and the distinction matters for how it is used. Distance beyond the recurrent horizon is the most reliable pressure available, which is why the framework builds curricula around it, but it is not the only one: limited hidden-state capacity, interference between concurrent contents, precision requirements the recurrent state cannot hold, explicit retrieval supervision, and regularization that prices recurrent capacity can each make retrieval competitive within the nominal horizon. What the principle rules out is the assumption that a healthy retrieval mechanism will be used because it exists. The design consequence is curricular, not architectural: training tasks must *place* needed information beyond the recurrent horizon (long gaps, cross-session references, external knowledge) if memory use is to develop. H6 states the corresponding testable claim, including its sharp negative half: with short-horizon tasks, even a demonstrably healthy mechanism stays unused.

Contemporary long-memory benchmarks make the use criterion operational at the interaction level: LongMemEval tests long-term information use in chat assistants, and MemoryAgentBench evaluates memory incrementally across multi-turn interactions [53, 119]. They are candidate external instruments for H6, not evidence that this framework’s mechanism will pass.

6.4 Representation learnability requirements

The second usage condition concerns what memory content must look like to be learnable at the point of use. Retrieved content re-enters a core state that is shaped by strong training pressures—target objectives, smoothness, segment activations. For a content channel to survive there, two conditions from the learnability theory of Section 7 apply directly:

- *On-manifold placement* (LC5): stored values must match the statistical profile (scale, norm) of the state region they are written to and read into. Off-manifold content is unproducible—smoothness and activation constraints block the core from ever emitting it.
- *Gradient visibility* (LC4): recall-relevant content must deliver a non-negligible gradient at the parameters it needs to shape. A channel whose contribution is orders of magnitude below competing loss terms can be ignored, and amplifying the upstream signal need not, by itself, change the optimizer’s indifference—nor is the loss share alone decisive, since gradient magnitude and alignment can diverge from it.

Jointly these define the *representation learnability region*: the region of scale and objective weight inside which memory content can be produced, stored, retrieved, *and* used. H7 makes the region falsifiable: outside it, recall stays at chance even under task pressure; inside it, task pressure converts a healthy mechanism into actual use.

6.5 The degenerate-key guard

A final requirement is mechanical and easy to violate silently. In a cosine-similarity store, zero and near-zero-norm keys are pathological, in two distinct ways. At exactly zero norm, cosine similarity is undefined ($0/0$); the common implementation convention of returning 0 then places such an entry below every practical similarity threshold—so a degenerate entry can never be retrieved, evades every similarity-based deduplication check, is re-admitted on every pass, and multiplies without ever becoming accessible. Near zero norm, cosine’s scale invariance means the similarity is determined entirely by the key’s noise direction, making both retrieval and deduplication arbitrary. The *degenerate-key guard* therefore rejects keys below a minimum norm at the store boundary, and rejections are counted rather than silently swallowed. The guard is a construction-level requirement for *any* cosine-keyed store: without it, a store in which degenerate keys occur accumulates unreachable or arbitrarily matched mass—and its shelf statistics will report that mass as content (Section 8 returns to this under input-novelty governance).

7 Learning and Target Theory

An architecture specifies what a system *could* represent; the training signals decide what it *will*. This section states the framework’s learning theory: what the data must carry, which target and optimization conditions support content-bearing representations, which reporting condition makes predictive evidence interpretable, and how training and evaluation must be kept apart. Its centerpiece—the learnability conditions LC1–LC5—collects four target or optimization requirements plus one reporting requirement that are individually familiar, from shortcut learning, auxiliary-task design, and time-series methodology, into an explicit set for latent-state architectures; we have not found them stated together for this setting. Every condition is stated so that violating it produces a nameable, detectable failure mode.

One scope note belongs at the head of the chapter rather than at its end, because it determines who this chapter is for. Everything below presupposes that the state is trained against *targets supplied per coordinate*. That presupposition is a commitment, not a background fact—predictive and task-driven objectives shape a latent state without ever posing the target-design problem (Section 2.3). Under those objectives, LC1–LC5 are *inapplicable* rather than refuted. That is why the target discipline D1 (Section 3.8) states the conditions with their applicability clause attached rather than as a condition of conformance: a system trained the other way conforms and declares its regime. Section 12.5 states why this framework pays the price of the harder route.

7.1 Data as disposition former

In a system whose core state includes affect, intention, and dispositional variables, training data does more than convey task structure: it installs a temperament. Sequences that model constructive error handling install recoverability; sequences that model epistemic caution install cautious confidence dispositions; the distribution of situations and responses *is* the disposition specification. Prompt design for data generation is therefore disposition design, and the framework treats it as a first-class engineering surface rather than a preprocessing detail.

One structural commitment follows. The *contrast-installed stability* hypothesis holds that a system trained only on positive, well-regulated states is fragile, while a system trained through negative states *with recovery trajectories*—fear that resolves, loss that is integrated, conflict that de-escalates—acquires robust homeostasis: the attractor structure of its affect dynamics is learned from contrast, not from calm. Stability, on this view, is a property of the training distribution’s trajectory geometry, not a regularizer added afterwards; H3 inherits this as its trajectory-level reading.

7.2 Generation strategy and the circularity risk

At the scale a small research program can curate by hand, training sequences must be generated, and generation by large language models is the practical route. The framework accepts this and names its price. Generated data must be *schema-constrained* (states, ranges, and transitions validated against the state specification, with rejection and repair), *diversity-controlled* (near-duplicate sequences rejected by similarity thresholds), and *anchored*: a curated *gold-standard set*—human-authored or human-verified sequences—serves as a fixed reference distribution against which generated batches are checked.

The price that cannot be engineered away is *circularity*: a core trained on a generator’s portrayal of inner-state dynamics learns the generator’s conventions, and its evaluation—if the evaluation scenarios come from the same generator family—measures agreement with those conventions rather than any independent target. Training on recursively generated data degrades in well-documented ways [101]. Whether a subtler analogue already operates after a single generation pass is our hypothesis, not a result established by that study. The framework’s mitigations are the gold-standard set above, grounded non-linguistic data streams where available (Section 7.8), and the metric stratification that keeps confirmatory evaluation out of the training loop (Section 7.5); its residual risk is acknowledged openly in Section 12.

How the numeric targets are produced. One element of the pipeline carries more weight than its size suggests and therefore deserves stating as a requirement rather than as an implementation detail: the map from a scenario specification to the numeric target vectors. It is where the framework’s psychology actually lives—whatever this map encodes is what the system will learn to be—and it is the point at which LC1 either bites or is silently evaded. The framework does not prescribe one map, but it constrains the family and requires disclosure. Three constructions are admissible:

- *Template trajectories.* Per-scenario-class affect curves, scaled by event intensity and smoothed. Simple and controllable; the highest-risk option for LC1, because the target becomes a closed function of attributes the fast state already carries. It carries a second cost, at the far end of the program rather than here: a map from scenario class to trajectory is exactly what H3 claims the architecture produces *without*, so a system trained on this construction cannot serve as confirmatory evidence for that hypothesis. It remains admissible for training and for exploratory work; it is barred from the confirmatory arm.
- *Dynamic integration.* A small deterministic dynamical system—appraisal variables driving a homeostatic integrator with decay and recovery—advanced tick by tick. Trajectory shape then depends on history rather than on the current event alone, which is what lifts the target out of the input’s immediate reach.
- *Model-derived with deterministic post-processing.* A language model proposes ordinal appraisal judgements; local deterministic code converts them into numbers under range and smoothness constraints. Keeps the numeric layer out of the generator’s hands while letting it supply the interpretation.

Three obligations attach to whichever is chosen. First, the map is *declared*—its inputs, its state, and its functional form—because a reader cannot otherwise tell what a positive result on H3 means. Second, it is probed at its own capacity class per LC1: a map with temporal integration must be probed with a family that can represent temporal integration. Third, the *text* that supervises the semantic segment should not be the text that produced the input for the same tick; where a scenario carries both an event description and an interpretive line, using the same string for both makes LC1 unsatisfiable by construction, and no probe can rescue that.

7.3 Learnability conditions for latent targets

The core state is trained against *latent targets*: per-tick target vectors for the semantic and affective segments, constructed by the data pipeline—and, under the label regime of Section 5.4, for the meta segment as well, though the framework requires that any calibration claim about that segment be made under the consistency regime (Section 5.4). Latent targets look innocuous—some vector has to supervise the state—but they carry the entire burden of deciding what the representation must compute. The framework states five obligations: four constrain a latent target scheme or its optimization, while LC3 constrains how predictive learnability may be reported. Each has a named failure mode, and together they define the viable band sketched in Figure 7.

The conditions are not of one epistemic kind, and treating them as if they were would overstate the formal ones and understate the practical ones. LC1 and LC2 rest on optimality arguments: given the objective, the stated solution *is* the optimum, and the condition follows. LC3 is a statistical requirement on how evidence may be reported, not a claim about systems. LC4 and LC5 are engineering conditions: motivated by how optimizers allocate gradient, by how constrained representations behave, and by the cited literature, but conditional on facts about the objective, the architecture, and the constraint regime rather than derivable from them. How often the failure modes they name actually occur, and how well the conditions predict them, are empirical questions for the companion studies; this paper reports no measurements bearing on either. Each condition below states which kind it is.

Work on supervising latent reasoning trajectories reaches a closely related diagnosis—weak outcome supervision leaving latent trajectories prone to drift, analyzed as a combination of gradient attenuation along the optimization path and representational drift in the latent space, with rigid geometric constraints on the latent manifold identified as themselves harmful [16]. We draw an analogy from those two failure directions to LC4 and LC5; the source does not state that equivalence. The comparison is convergent rather than shared, since that work supervises a reasoning trajectory whereas the conditions here govern a persistent typed state whose segments carry declared roles. We take the convergence as evidence that the failure modes are properties of latent supervision itself rather than of one architecture.

LC1 — Non-triviality. (*Optimality argument.*) *The target must not be reducible to a copy of the input plus constants.* Let X be the per-tick input representation and Y the target. If $Y = AX + b$ for a fixed simple map A (in the extreme case, an identity on a sub-block together with constant coordinates), then the Bayes-optimal solution of the supervision problem is that map itself: a pass-through buffer. Nothing in the objective demands abstraction, composition, or retention beyond the current input. Shortcut-learning evidence motivates testing whether the admissible copy solution is selected [36]; the citation does not guarantee optimizer behavior in this architecture. The failure mode is the *identity solution*: a representation that scores perfectly on its objective while computing nothing. The operational check is deliberately mundane: fit a probe from input to target; probe fit near unity means the target demands nothing the input does not already contain.

The probe’s *capacity* is not a free parameter, and getting it wrong is the most consequential mistake available here. A linear probe detects only linear triviality. If the target generator is a nonlinear, multi-tick construction—a parameterized affect trajectory with decay and homeostasis, say, computed from the same event attributes the fast state already carries—then the target is a closed function of the input while a linear probe reports a moderate fit and passes it. The system would then learn to approximate the generator’s rule set, every downstream check would agree (consistency constraints would hold because the generator imposed them; reaction profiles would separate because the generator built the separation in; the echo baseline would fail because the trajectories are structured rather than persistent), and the resulting confirmation would be complete and empty of the evidence it appears to supply.

This second failure deserves its own name, because it is not the identity solution: the state does not become a pass-through buffer but a *reconstruction of the generator’s rule set—generator closure*. It scores as learning while carrying no evidence for emergent or persistence-dependent structure, for the same reason the identity solution carries none: no independent target was ever posed. That is a verdict on the evidence and not on the representation—a target reconstructible from the available inputs may still be technically useful; but it is caught by probe *capacity* rather than by probe fit, and it is a circularity failure (Section 7.2, M8) wearing a learnability failure’s clothes. The requirement that catches it is therefore: *the probe family must be at least as expressive as the target generator*. A generator with per-class templates, temporal integration, or conditional branching must be probed with a family that can represent those—the generator’s own functional form is the natural choice, since it bounds what a sufficiently capable learner could extract. (The probe remains a one-sided instrument: it detects triviality; passing it is necessary for a demanding target, not sufficient.) The analogous degeneracy is familiar from autoencoding, where sufficient capacity without constraint yields trivial copying; latent-state supervision inherits the problem in a sharper form because the copy can hide inside a larger target vector padded with constants.

LC2 — Reachability and relative gating. (*Optimality argument.*) *The attainable fraction must be measured against everything the model may condition on, and acceptance gates must be set relative to it.* For a square-error objective, the optimal predictor is the conditional mean; variance of the target that is unpredictable from what the model can see is not learned but *averaged over*. Mixture-density modeling is a classical response when a conditional target distribution is multimodal [9]; recent self-predictive representation work likewise studies collapse-avoidance conditions in a different learning regime [85].

The optimality result is standard; the contribution here is to use the reachable fraction as a pre-training measurement and as the reference for acceptance gates when the latent state is itself the product. The same difficulty forces recent latent world models away from point-valued transitions, since a single predicted successor cannot represent a branching transition with several plausible futures [104]. For this object, the reachable fraction must be *measured before training* and used to set acceptance gates; its violation becomes a named, diagnosable failure rather than a disappointing number at the end.

The conditioning set matters, and getting it wrong is the easy mistake. A persistent system does not predict from the current input alone: the tick operator conditions on the full admissible information state

$$I_t = (F_t, C_{t-1}, S_{t-1}, g_t, W_t, M_t), \quad (5)$$

which includes trajectory history through C_{t-1} and retrieved content through M_t . Note that I_t and the declared exposed state $\mathcal{X}_t^{(r)}$ of Equation (2) are different objects and must not be substituted for one another: $\mathcal{X}_t^{(r)}$ answers *what a consumer may read*, and closes under that question; I_t answers *what the operator may condition on when producing C_t* , so it contains the previous tick’s core and slow state rather than the current ones, and it additionally contains W_t and M_t , which no consumer reads. They overlap in F_t and g_t and nowhere else. A target unreachable from the current event may be entirely reachable from history and memory—indeed, targets that require history are precisely what a persistent architecture is for. The reachable fraction is therefore

$$R_{\text{reach}}^2 = \frac{\text{Var}(\mathbb{E}[Y_t | I_t])}{\text{Var}(Y_t)}, \quad (6)$$

computed per coordinate and aggregated, since targets are vectors. Conditioning on the per-tick input alone yields a strictly smaller quantity, useful as a diagnostic—it isolates how much of the target is an immediate function of the current event—but mistaking it for the ceiling would condemn exactly the targets worth having.

One caveat on the conditioning set itself, and it needs stating carefully because the right observation here invites the wrong conclusion. Because I_t contains M_t and C_{t-1} , a target that merely *copies* retrieved memory or the previous core state is maximally reachable by construction—and would pass LC1, whose probe reads the per-tick input only. Such a target relocates the identity solution from the input path into the memory and recurrence paths rather than removing it, and LC1 must therefore be evaluated against the full conditioning set as well: a second probe, $I_t \rightarrow Y$ with the *current event ablated*, at the same capacity class required below.

What that probe may conclude is narrower than it first appears, and the distinction is the whole point. A near-unity fit from the full information state does *not* by itself mark a target the machinery passes through rather than computes. A target produced by a deterministic generator with temporal integration—the dynamic-integration construction of Section 7.2, which this framework recommends—is *expected* to be reconstructible from I_t by a probe at the generator’s capacity class, precisely because the generator computed it from the same history. Reading that as a failure would disqualify the construction the previous subsection prescribes. The two verdicts must therefore be kept apart:

- *An LC1 failure* is a declared copy or invertible recoding of a *named* source—the retrieved value, the previous core state, a signal channel—which satisfies the target without the transformation the study claims to test. It is identified by naming the source and showing the recoding, not by a fit number alone.
- *A limitation on evidential independence* is what a high generator-matched fit establishes: the target carries no information the generator did not put there, so a result on it cannot support a claim of emergent or persistence-dependent structure. This is reported alongside the result and constrains what may be concluded from it. It is not a defect of the learned representation, and it does not disqualify the target for training.

Whether the trained state *computes* its target or merely carries it is a third question, and neither probe settles it: separating those requires an intervention that removes the candidate source at inference and asks whether the target survives. The framework names that question rather than pretending a fit answers it.

Two caveats about what this number is. First, R_{reach}^2 is estimated with a fixed probe family, and any such estimate is a *lower bound*: $\hat{R}_{\mathcal{F}}^2 \leq R_{\text{reach}}^2$, with the gap unknown and model-class dependent. It is a probe-relative floor on reachability, not a measured ceiling, and a target that fails a probe may still be learnable by a stronger model. Second, gating is nonetheless well-posed provided the same probe family measures the floor and assesses candidates: the comparison is then internally consistent even though the absolute level is not identified. A target whose informative content lies mostly outside reach drives the optimum toward an input-insensitive output. The collapse lands where the supervision does: because latent targets supervise coordinates of the core state itself, not a disposable output head, the supervised coordinates become X -insensitive—and since those coordinates *are* the state’s typed content, the state grows *smoother and poorer*, not richer, as target “richness” grows. (Whether impoverishment extends beyond the supervised coordinates to the rest of the representation is the argued extension that H8 tests.) We name this failure *conditional-mean collapse*, and emphasize its perverse signature: enriching a target without adding input-side information *reduces* representational content. The governance consequence: any learnability gate expressed as an absolute threshold (“the model must reach fit τ ”) is ill-posed; gates must be expressed relative to the measured reachable fraction ($\tau \cdot \hat{R}_{\mathcal{F}}^2$), because a threshold above the reachable range of its own target space is unsatisfiable by any model whatsoever. The direction of that conservatism has to be named, because it is easy to state backwards: since $\hat{R}_{\mathcal{F}}^2$ is a floor, a gate at $\tau \cdot \hat{R}_{\mathcal{F}}^2$ sits *below* the true reachable level, so it is conservative against rejecting a target that was in fact learnable—and

permissive in the other direction. It does not protect against accepting a target whose content is mostly unreachable. Protection there comes from the collapse signature itself (content falling as target richness rises), not from the gate.

One further specification the gate needs, because “run the check before training” otherwise admits two incompatible procedures. Equation (5) contains model-dependent quantities— C_{t-1} , S_{t-1} , W_t , M_t —which do not exist before a model does. Design-time reachability is therefore estimated against an *oracle* information set I_t^* built from target history and the memory contents the pipeline would have made available, with the probe trained and evaluated on disjoint scenario splits; realized reachability is re-estimated after training on the model’s own I_t . The two are different numbers and are reported as such. A study that reports one without saying which has not run the check, since two groups can reach opposite LC2 verdicts on the same targets by choosing differently here.

LC3 — Persistence-discriminability. (*Reporting requirement.*) *Predictive targets must be benchmarked against the persistence baseline.* Latent state trajectories are autocorrelated by design—smoothness is a virtue everywhere else in this framework. But autocorrelation makes “predict the next target” deceptively easy: the *persistence baseline*, predicting $Y_{t+1} = Y_t$, inherits all of the target’s inertia. A predictive fit reported without the target’s own autocorrelation is therefore uninterpretable: high fit may reflect learned dynamics or merely a slow target. The condition requires that target autocorrelation be reported alongside any predictive fit and that claims of predictive learnability strictly exceed the persistence baseline. The failure mode is *pseudo-learnability*: inertia dressed as understanding.

The same reasoning has a second edge, and it cuts at the instruments rather than at the targets. The standard defence of a derived quantity—a target, a gate variable, a monitoring read-out—is a constructed control: perturb the thing it is supposed to track and show that it moves. That control establishes *functionality*. It does not establish that the quantity ever takes the values the construction relies on when the system runs on its own data, and the two are routinely confused. A gate whose threshold sits outside the range its variable actually occupies in operation is not a strict gate; it is an inactive one, and it will look identical to a satisfied gate in every log. The framework therefore requires *variation evidence* for any derived quantity used as a target, gate, or reported instrument: alongside the constructed control, report the quantity’s empirical distribution over ordinary operation—range, spread, and the fraction of the decision range it visits. Where that distribution is degenerate, the quantity is reporting the construction rather than the system, whatever the control showed. The governance chapter states the watchdog case of the same requirement as the reachability proof of Section 8.5; the two are one doctrine applied at the two ends of the system.

LC4 — Gradient visibility. (*Engineering condition.*) *Every content channel must produce a non-negligible gradient signal at the parameters it needs to shape.* The condition is often stated in terms of loss shares, and the loss share is the cheap proxy: a term whose contribution is orders of magnitude below the competing terms usually sits below the optimizer’s indifference threshold and is not opposed but merely ignored—and scaling the channel’s *signal* moves its share far less than the arithmetic suggests, since the term’s weight and the segment’s width set the share and amplitude enters only through the residual. But the scalar share is not what the optimizer sees. What decides visibility is the gradient the term delivers, $\|\nabla_{\theta} \lambda_i L_i\|$, its alignment with the competing terms, $\cos(\nabla L_i, \nabla L_j)$, and the overlap of the parameters they touch. A small loss can carry a large gradient, and a large loss can carry almost none; conflicting gradients of similar magnitude can cancel a channel that looks well-funded on the loss ledger. The failure mode is *silent neglect*—the channel trains to whatever incidental value minimizes the dominant terms. The check accordingly has two levels: the arithmetic one first (the channel’s share of the total objective at initialization and at the observed loss scale, since shares far below

the dominant terms’ fluctuation band predict a dead channel), and per-term gradient-norm logging with pairwise alignment as the diagnostic that decides the ambiguous cases—which is also why gradient-balancing methods are the designated remedy rather than weight retuning (Section 10.4).

There is a structural corollary that the loss ledger hides entirely. Latent-target losses are computed per coordinate and aggregated, so a segment’s share of the objective is set by its *width* as much as by its weight: a segment holding a handful of coordinates contributes a correspondingly small fraction of an aggregated term, and the same weight applied to a wide segment and a narrow one funds them very differently. This matters here because the framework’s most semantically loaded segments—affect, intention, the monitoring read-outs—are deliberately the narrow ones (Section 3.2), which places them on the wrong side of this arithmetic by construction. Three remedies are available and they are not equivalent: reweight per segment (simple, but the weights now encode a width correction and must be redeclared whenever a segment is resized), aggregate per segment before combining (width-neutral by construction, at the cost of making within-segment coordinates non-comparable), or widen the segment (which changes what the segment *is*). The requirement is that an instantiation report per-segment gradient share alongside per-term loss share; the first is what the optimizer acts on, the second is what the configuration file suggests.

LC5 — On-manifold placement. (*Engineering condition.*) *Targets must match the statistical profile of the representation region they supervise.* The core state is shaped by activation constraints, normalization, and smoothness pressure; its coordinates inhabit a characteristic scale. A target far off that manifold—wrong norm, wrong variance—is poorly producible, and how poorly depends on the constraint regime. Under *hard* constraints (bounded activations, projection onto a norm ball) the target is unreachable outright: the structure that keeps the state well-behaved excludes it, and the optimizer settles for the nearest admissible approximation, which may resemble the target not at all. Under *soft* constraints (norm penalties, smoothness terms) the target is reachable in principle, but only by shifting the state distribution against the regularizer—so the cost is paid elsewhere, typically as degraded performance on the other supervised segments or as the norm growth the penalty was installed to prevent. The failure mode is *unproducibility* in the hard case and *displacement* in the soft case, and the two are distinguished by whether other objectives degrade. The check is mundane and belongs before training: compare target statistics—norm, per-coordinate variance, distance to the state region’s manifold—to the statistics of the region being supervised, and state which constraint regime applies.

Table 2 summarizes the taxonomy, and Figure 8 pairs each condition with the failure it lets through: the table says what a violation looks like once it has happened, the figure says which question was not asked. Nothing in either is a diagnosis after the fact; every row is a check with a fixed place in the schedule rather than a post-mortem—LC1, LC3 and LC5 before training, LC4 from initialization onward, and LC2 both before training and again on the trained model (D1). One consequence deserves saying plainly, because the phrase “check before training” invites the opposite reading: passing the design-time audit does not close LC2. A target admitted on an oracle estimate can still turn out unreachable for the model that was actually built, and only the realized estimate says so. The figure’s top-to-bottom arrangement is a reading order and not a dependency—a study must discharge all five obligations—four through target or optimization design, and LC3 through persistence-baseline reporting—and no condition presupposes another. H8 states the conditions’ joint empirical content: targets violating LC1 or LC2 produce pass-through or collapsed representations, and target enrichment without input-side information reduces content. The conditions are stated for square-error supervision because that is the canonical regime for continuous latent targets; analogues for likelihood-based objectives follow the same logic through the corresponding Bayes-optimal predictors.

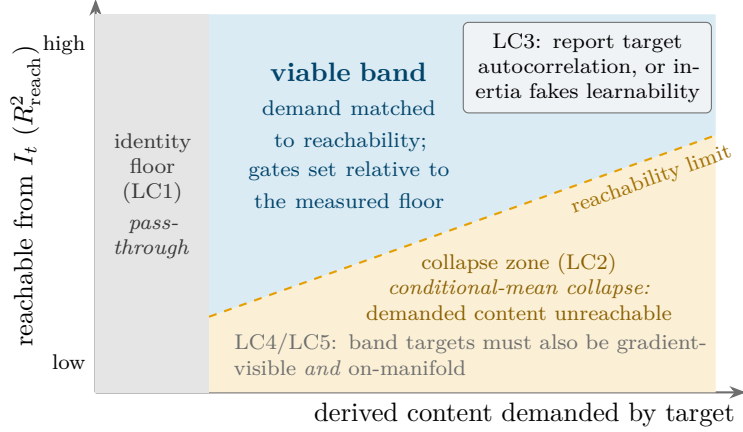


Figure 7: The viable band for latent targets. Horizontal: how much derived content the target demands beyond the input. Vertical: how much of the target is reachable from the information state (R^2_{reach}). The identity floor (LC1) and the collapse zone—where demanded content exceeds the reachable fraction (LC2)—bound the band; persistence mirages (LC3) fake learnability along the time axis; LC4/LC5 govern whether a target inside the band is *seen* by the optimizer at all.

Condition violated	Failure mode	Signature
LC1 non-triviality	identity solution	near-unity probe fit from input <i>or</i> carried state
LC1 non-triviality (closure)	generator closure [†]	near-unity fit only at the generator’s own capacity class
LC2 reachability	conditional-mean collapse	content falls as target “richness” rises
LC3 persistence	pseudo-learnability	fit $\leq$ persistence baseline
LC4 gradient visibility	silent neglect	negligible or cancelled per-term gradient
LC5 on-manifold	unproducibility / displacement	target statistics off the manifold; other segments degrade
LC2 reachability (aux. head)	starved interpretation	head fit at chance; conditioning output near-constant

Table 2: Failure taxonomy for latent supervision targets and for the auxiliary heads that condition them (last row, Section 7.4). [†]Generator closure is the one entry that is not a defect of the learned representation: it limits what a result on that target may be taken as evidence for, and is reported alongside the result rather than disqualifying it (Section 7.3).

7.4 The supervised-interpretation principle

Beyond what targets must avoid, the framework makes one positive commitment about where broad competence comes from at small scale.

Derived situation interpretations—appraisal-style summaries of what the current situation is—should be supervised explicitly and fed back as conditioning, rather than left to emerge.

The *supervised interpretation* principle has a psychological and an optimization face. Psychologically, appraisal theory holds that emotion episodes are organized around an evaluation of the situation against the agent’s concerns [95]; an architecture that wants situated affect needs situated interpretation as an explicit quantity. From the optimization side, an auxiliary supervised task whose labels are derivable from the data (situation type, appraisal features) provides dense, well-posed gradient toward exactly the distinctions the core needs.

The two prescriptions govern different objects. LC1 governs *latent targets*, which supervise coordinates of the core state itself: a target derivable from the input makes those coordinates a pass-through, and the state is the product. An auxiliary head is a different object—a disposable output whose labels are *meant* to be derivable, because its purpose is not to add content but to route gradient toward distinctions the main objective underweights. The head is discarded after training; its coordinates are not part of the state under evaluation. The boundary is therefore:

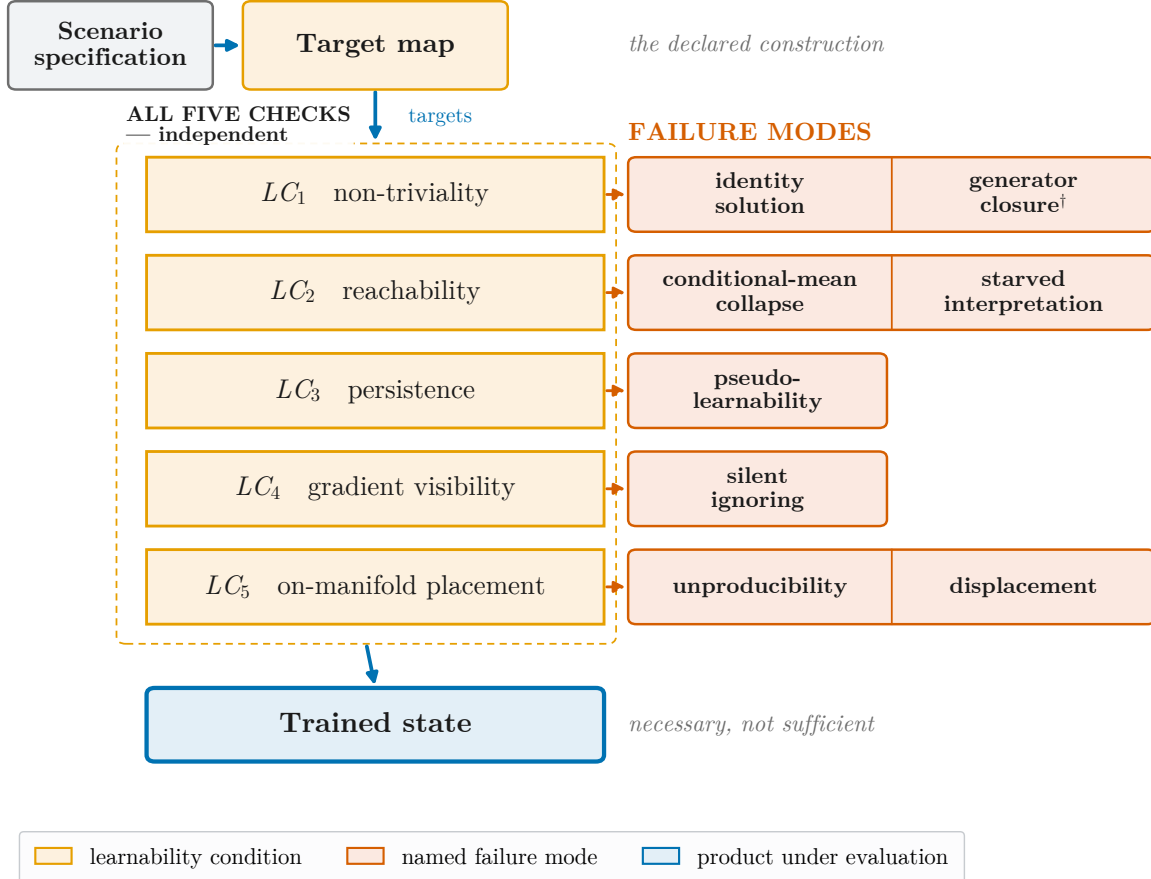


Figure 8: The latent-target pipeline. Each condition is a question asked *ahead of* the failure it prevents rather than a diagnosis afterwards, and each has an exit—the vermilion connectors on the right—to the named failure that results from passing it unchecked. The schedule differs by condition and is fixed in D1: LC1, LC3 and LC5 before training, LC4 from initialization onward, LC2 both before training against an oracle information state and again on the trained model. Top to bottom is a reading order, not a dependency: all five obligations must be discharged, and none presupposes another. LC2 carries two exits because the auxiliary heads that condition a target are reachability-bound in their own right (last row of Table 2). Generator closure carries a dagger because it limits what a result may be taken as evidence for; unlike the other boxes, it is not a defect of the learned representation. The conditions are not of one epistemic kind—LC1 and LC2 rest on optimality arguments about the Bayes-optimal predictor, LC3 is a reporting requirement that changes no target and no gradient, LC4 and LC5 are engineering conditions on the objective and on the target’s statistics. Reaching the bottom box is necessary, not sufficient: the checks exclude the degeneracies named here and say nothing about whether the representation that survives them is worth having.

derivability disqualifies a *latent target* and qualifies an *auxiliary label*. The conditioning channel that carries the appraisal signal back into the tick is a third case and follows the same logic: it is an *input* to the update, not a supervised coordinate of the state, so LC1 does not apply to it—what applies is LC4, since an unsupervised conditioning channel carries no gradient of its own and is the silently ignored case the next paragraph describes. This is the classic mechanism by which related supervised tasks act as inductive bias for a shared representation [13]—whereas hoping the same distinctions emerge from the main latent-target objective runs through LC1/LC2 territory with no guarantee at all. Unsupervised conditioning—feeding back

a derived context *without* supervising it—provides the mechanism without the pressure and is expected to underperform its supervised twin; H5 makes the comparison explicit.

What the interpretation head reads. The principle prescribes supervision and a feedback path; it does not by itself say which part of the input the interpretation head should read, and that silence is where the construction is easiest to get wrong. An auxiliary head is exempt from LC1—derivability is its purpose—but it is not exempt from LC2. Reachability applies to the head’s own conditioning set: a head asked to predict situation classes from a representation in which those classes are not cheaply separable cannot produce them, and it does not fail loudly. It converges toward the conditional mean of its label distribution, and the conditioning channel then carries a near-constant vector back into the tick. Two costs compound. The channel occupies input width the operator must process regardless of what it carries, and the auxiliary loss keeps claiming gradient for a head with nothing to contribute. We name this mode *starved interpretation*: correct supervision, correct feedback path, insufficient input.

The consequence is uncomfortable enough to state as a prediction rather than as advice. Where the input translator makes typed signal channels available alongside the continuous embedding (Section 3.5), an interpretation head that reads those channels should outperform one reading the embedding alone—and the starved head should fall *below* the no-interpretation control, not merely below its well-fed twin, because it pays the two costs above and collects none of the benefit. This predicts that the deciding input can be a handful of typed scalars—event class and intensity—against an embedding two or three orders of magnitude wider. If a small explicit block outranks a large semantic one for this purpose, the reason is not information content but *separability at the head’s capacity*, which is what LC2 measures and what dimension counts do not. The diagnostic is correspondingly cheap: report the head’s own fit and the variance of the conditioning output it produces, not just the downstream score. A conditioning channel whose output barely varies across situations is starved, whatever the training loss shows (Section 9.2, M9).

The principle has a price: the *supervision paradox*. Whatever is supervised is, by construction, not emergent—a supervised appraisal channel carries the data pipeline’s theory of situations, not one the system found. Push the principle to its limit and the state becomes a well-trained rendering of its label schema; refuse it entirely and, at accessible scale, the distinctions may simply never form. The framework’s position is a deliberate middle: supervise the *interpretive scaffold* (situation types, appraisal features), leave the *content* that flows through it to learning, and treat the right extent of that scaffold as an open problem—catalogued in Section 12.3—rather than a solved one.

7.5 Goodhart separation

The framework’s evaluation suite (Section 9) scores the qualities the program actually cares about, which creates a standing temptation: use the score as a training signal. The reason not to is architectural rather than moral—the evaluation suite is the program’s instrument for *noticing* that training has gone somewhere unintended, and spending it as a training signal destroys the instrument precisely where it is needed most; a measure ceases to be a good measure once it becomes a target, in Strathern’s formulation of Goodhart’s law [105].

A blanket prohibition, however, is unworkable and would be dishonest to state. Some qualities are legitimately both training objectives and evaluation dimensions: confidence calibration is the clearest case—the framework *requires* it to be trained (Section 5.4) and *also* evaluates it. Forbidding the overlap would forbid the architecture. *Goodhart separation* is therefore a stratification of metrics rather than a wall:

Development metrics. May inform training, hyperparameter choice, and architecture iteration. They are expected to be optimized against, and results reported on them carry no

evidential weight beyond “the recipe converged.”

Sealed confirmatory metrics. Fixed before the runs they will judge and touched only at final evaluation. No training signal, no surrogate tuned against them, no architecture selection informed by them. These carry the evidential weight, and their seal is what gives it force.

External metrics. Defined and ideally executed by parties outside the program (M8), and therefore beyond the reach of any local optimization pressure, intentional or not.

The stratification must be declared before the runs, and a metric may move only in one direction—a sealed metric that informs a training decision becomes a development metric permanently, and the results conditioned on it are relabeled accordingly. What the framework rules out is not overlap between training and evaluation vocabulary, but the silent migration of a confirmatory metric into the development loop.

7.6 Scale-transfer skepticism

Small-scale pilot runs are indispensable for iteration speed and treacherous for inference. Regularization pressure, data diversity, and optimization dynamics all change with scale, and an intervention’s sign can change with them. The framework therefore assigns small-scale runs an asymmetric epistemic role: a *screen* may *falsify* a candidate, but may not *qualify* one. Qualification happens at the target scale, under the full evaluation suite.

The asymmetry has a condition attached, and stating it prevents the rule from becoming its own error. A screen kills legitimately when the failure mode is *scale-invariant*—numerical instability, representational collapse, dead gradients, a mechanism that cannot be trained at all—because these are properties of the construction rather than of the regime. A screen does *not* kill legitimately when the mechanism plausibly requires capacity or data diversity the pilot lacks: competitive selection among specialists, memory recruitment under task pressure, and anything whose benefit is defined over a distribution the pilot does not span can all fail at small scale for reasons that say nothing about the target scale. Discarding such a candidate on a screen is itself a scale-transfer error, in the opposite direction from the one the rule guards against. The operational form: a screen’s verdict is binding when the observed failure has a scale-invariant signature, and advisory otherwise—and which one applies is declared with the screen, not after seeing its result.

7.7 Curriculum and schedule requirements

Two requirements govern how training unfolds over time.

Anchor-based curriculum. Training progresses from short, fully supervised horizons to long autoregressive rollouts punctuated by *anchor events*—externally specified state targets at intervals—so that the system learns to run on its own outputs (Section 3.3) without drifting unmoored. Memory-demanding structure enters the curriculum explicitly, per the task-pressure condition of Section 6.3: recall distances are a curriculum variable, not an accident of the data.

Step-normalized schedules. Optimization schedules—learning-rate decay above all—must be *step-normalized*: parameterized in optimizer steps, not in epochs. An epoch-parameterized schedule silently rescales with dataset size: doubling the data doubles the steps spent in every schedule phase, which changes the optimization trajectory even when nothing about the task changed. Under epoch parameterization, a data-scaling experiment is never the single-variable comparison it appears to be. The requirement extends to a discipline: *the schedule is part of the*

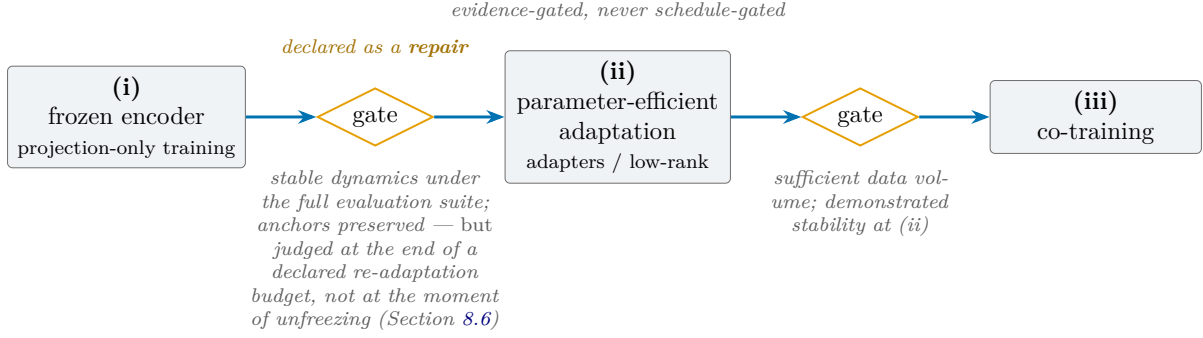


Figure 9: The staged unfreezing path: staged relaxation of the frozen-first policy, each transition gated by stability evidence under the full evaluation suite—never by schedule.

recipe’s identity. Because a schedule’s phase lengths set the optimization trajectory, comparing runs whose schedules differ in normalization compares schedules, not data or architecture.

7.8 The staged unfreezing path

Finally, the frozen-first policy of Section 3.5 must end somewhere: a fixed encoder bounds what the system can ever distinguish, and LC2 makes the bound quantitative—content that the encoder does not separate is unreachable target variance for every downstream stage. The framework prescribes a *staged unfreezing path* (Figure 9): (i) *frozen encoder*, projection-only training, until the core’s dynamics are stable under the full evaluation suite; (ii) *parameter-efficient adaptation* of the encoder (adapters or low-rank updates), gated on preserved anchors and non-degraded evaluation; (iii) *co-training*, gated on data volume and demonstrated stability at stage (ii). Each transition is gated by evidence, not by schedule—unfreezing is earned in the same sense that growth is earned in Section 8. Note that stage (ii) is exactly the situation the repair paradox of Section 8.6 describes: unfreezing the encoder perturbs an equilibrium the core was fitted to, so a gate read at the moment of unfreezing will reject the transition on principle. The transition is therefore declared with a re-adaptation budget in advance, and judged at its end.

The same bound motivates grounding. Text-derived targets inherit the text encoder’s ceiling; observation streams from interactive environments carry target variance that no text encoding provides—state transitions with causal structure, consequences of action. H14 states the resulting claim: grounded streams change what is learnable, at the price of lower reachable fractions of their own, and text-only paradigms meet LC2 limits earlier than mixed ones.

8 Growth Governance

A system built to accumulate experience must eventually learn from it—and continual learning in an autonomous, affectively coupled system is exactly where quiet failure modes concentrate: forgetting, self-reinforcement, drift that no single step makes visible. TASA therefore treats growth not as a training mode but as a *governed process*: a set of standing invariants, gates, and monitoring duties that any continual-learning instantiation must satisfy. The section’s organizing idea is that in continual operation, *the burden of proof is on the change*.

8.1 Seed before growth

The framework separates two regimes. The *seed program* produces a stable, fully evaluated base system—trained offline, measured under the complete evaluation suite, its invariants verified. The *growth program* is the continual regime that operates on top of that seed system: ingesting

experience, consolidating memory, and, rarely, changing weights. The separation is asymmetric on purpose: growth presupposes a seed system whose behavior is characterized well enough that *deviation* is detectable. A system that grows before it is measurable grows blind. Growth, in short, is earned.

8.2 Memory-first consolidation

The first standing invariant governs *where* new content lands.

New knowledge enters memory first. Parametric consolidation—writing into the core’s weights—is rare, individually verified, and reversible.

The *memory-first* rule maps the complementary-learning-systems insight onto governance: fast episodic acquisition and slow structural integration are different processes with different risk profiles [78]. A memory write is local, inspectable, attributable, and revocable by deletion at the store level; a weight update is none of these by default—its effects disperse across parameters, its interference with existing competence is the classic catastrophic-forgetting risk [60], and the reversibility the ladder’s top rung demands must be engineered explicitly (checkpointed rollback) rather than inherited. The governance ladder follows: memory writes are cheap and gated lightly; consolidation between shelves is explicit and criterion-gated (Section 6.2); consolidation into weights sits at the top, requiring verification against held-out competence, a replay guarantee, and a rollback path. What regularization methods attempt inside the objective, the framework additionally enforces *around* it—because at this layer, failure is operational, not statistical.

These concerns have direct recent prior art: Text2Mem frames agent memory as a self-evolving system, MemOS treats memory management as an operating-system layer, and verifiable-memory work foregrounds evidence, conflict, and uncertainty [69, 108, 113]. The contribution of this section is therefore their integration and operationalization within the paper’s conformance, provenance, rollback, and gate contract—not the invention of governed or verifiable agent memory.

Revocability has to survive the ladder. The claim that a memory write is revocable holds for the write itself and stops there, which is not enough for what the rest of this section rests on. By the time a source episode is deleted, it may already have contributed to a shelf promotion, to a derived concept, and through those to a parametric update; deleting the leaf leaves the derivatives standing, and the deleted content continues to act through them. Global checkpoint rollback does not repair this either—it discards every valid change made since, which makes it unusable as a routine remedy and therefore, in practice, unused. The framework’s requirement is consequently a data structure rather than a policy: consolidation maintains a *provenance graph* whose edges run from source memories to derived shelf entries and from those to any parametric update that consumed them. Three obligations follow. Disqualifying a source—by deletion, by a provenance-quota violation, or by a failed canary—marks every dependent artifact as invalid rather than silently keeping it. Reversal means *selective* rollback or deterministic reconstruction of the affected artifacts without the disqualified source, not restoration of an earlier global state. And an update whose provenance cannot be represented in the graph is not admissible for consolidation, because its effects could not be undone if its source were later disqualified. Without this, the no-self-sealing rule (Section 8.3) and the provenance quotas it depends on are auditable only at the moment of writing, and not afterwards—which is the only moment that matters.

8.3 The no-self-sealing rule

The second invariant governs *what may teach the system*.

The system’s own outputs, and any single feedback partner, must not become a dominant learning source.

No-self-sealing is the operational counterpart of the circularity risk of Section 7.2. A continual learner that ingests its own productions recycles its conventions with compounding bias—the degenerative dynamics of training on recursively generated data [101]—and a learner dominated by one interlocutor converges on that interlocutor. The rule is enforceable only because memory carries provenance (Section 6.2): source quotas can be computed, self-sourced content can be blocked from consolidation, and the block itself can be tested. The framework requires exactly that test: blocked pathways must be *proven* blocked by *negative probes*—operations that attempt the forbidden write and must fail—run as standing regression checks, not assumed from code inspection. The output translator’s no-backflow rule (Section 3.6) is this invariant’s boundary case at the language interface.

8.4 The multi-source governor

Growth decisions are made by a *governor*: the decision layer that admits or blocks each growth step. Its defining constraint is epistemic humility enforced by design:

Governors may advise and may block; they may never autonomously green-light. A green light requires the confluence of independent evidence sources—and for irreversible steps, a human.

The governor reads several independent instruments: *canary* evaluations (a fixed, deterministic probe set that must not regress), held-out competence checks, the gold-standard set of Section 7.2, and *drift criteria*—fixed in advance watchdog conditions over the growth process itself (shelf growth rates, concept similarity, identity drift, parity status). No single instrument is trusted alone: each covers the others’ blind spots, and the confluence requirement means a growth step proceeds only when nothing objects. The asymmetry—any instrument can stop a step, none can approve one alone—ensures that no single instrument’s blind spot, and not the governor’s own approval logic either, can silently green-light growth (Figure 10).

The asymmetry has a cost the framework does not hide: a governor that can only block is biased toward stasis. Every instrument added multiplies the ways a legitimate growth step can be vetoed, and an over-instrumented governor converges on a system that is perfectly safe and never learns. Where the right operating point lies between recklessness and stasis is not derivable from first principles—it depends on the deployment’s risk tolerance and the instruments’ false-alarm rates—and the framework accordingly treats governance strictness as a tunable with a named failure mode on *both* ends (Section 12.3), with intake coverage accounting (Section 8.7) as the instrument that makes the stasis end visible.

8.5 Criterion reachability

Watchdog criteria have their own failure mode, subtle enough to deserve a named requirement. In the common pattern, a criterion is validated twice: a negative control (on clean operation it stays silent) and a positive control (on a synthetic fault it fires). Both can pass while the criterion is still worthless, because a synthetic-fault control typically *injects* the fault into the guarded quantity directly, bypassing the causal path through which real operation would have to produce it. If that path never moves the quantity in the deployed regime, the criterion is structurally unable to fire in operation—and its permanent silence is indistinguishable from health. The framework therefore requires a *reachability proof* for every watchdog criterion. Operational variance is not enough for that proof, and stating it as though it were is a mistake: a quantity can vary reliably between 0.00 and 0.10 while its alarm sits at 0.90, and it will pass a variance test every time. What the proof must show is that fault-realistic operation moves the guarded

statistic through a pre-specified fraction of its *threshold margin*—the same three quantities the variation evidence of Section 7.3 already asks of any derived instrument: range, spread, and the visited fraction of the decision range. A criterion that cannot fire is not protection; it is only the appearance of protection, and a criterion certified by variance alone can still be one.

A second, related duty: criteria whose statistics vary with population size must be validated against *N-matched random baselines*. Extreme-value statistics—maximum or nearest-neighbor similarity, for instance—drift upward with the number of items even for independent random vectors, and even statistics whose expectation is *N*-invariant change their sampling spread as the population grows. A threshold on such a statistic, validated only at one population size, converts growth itself into a false alarm—or masks a real one. The general rule: *a criterion validated only in the regime where its failure cannot occur has not been validated*. H11 states the empirical form of both duties.

8.6 The repair paradox

Gates that admit a change only if no measured quality declines are the natural expression of the governance stance, and they carry a defect that follows directly from their form. Any repair to a load-bearing part of the system—correcting a channel that was silently inactive, restoring a schedule that never completed, changing what a segment means—perturbs an equilibrium the current parameters were fitted to. The trained system should be expected to be worse immediately after such a change and better only after it has re-adapted. A gate evaluated at the moment of the change therefore rejects precisely the class of changes that fix the system’s real defects, while admitting additive changes that disturb nothing—and the resulting trajectory looks disciplined in every report. The paradox is not an argument against non-degradation gates; it is an argument that a gate needs a declared *evaluation horizon* as much as it needs a threshold.

The framework’s requirement is therefore procedural rather than numerical. A change whose author expects a transient regression is declared as a *repair* before it is run, and the declaration fixes four things in advance: the mechanism claimed to be defective, the re-adaptation budget after which the comparison is made, the acceptance criterion at that point, and the revert condition. Outside a declared repair, the ordinary gate applies unchanged. Two safeguards keep the exemption from becoming a loophole, and both are consequences of the no-self-sealing rule (Section 8.3): the declaration is written before the outcome is known, and the acceptance criterion is not the metric the repair was aimed at. A repair that can only be judged by the quantity it was designed to move is indistinguishable from a change that was tuned to that quantity.

8.7 Input-novelty governance

Growth loops have a failure mode that no per-step gate detects: *standing still while appearing to move*. A loop that selects experience deterministically from a static pool re-consumes the same episodes forever; shelves grow, counters rise, every individual step is valid—and nothing new is ever learned, because the loop’s *input* is exhausted. The framework therefore requires *intake coverage* accounting as a first-class growth metric: what fraction of available experience has actually been consumed, how much of each growth cycle’s intake is first-time content, and whether selection excludes what is already consolidated. Deduplication at ingestion and consolidation-aware selection are the mechanisms; the degenerate-key guard of Section 6.5 is their precondition, since degenerate entries corrupt exactly these statistics. Shelf-size growth without intake coverage growth is the signature to alarm on: it means the loop is re-digesting, not learning. Coverage is accounted per growth cycle—one full pass of ingestion, re-experience, and consolidation—so that first-time intake is distinguishable from repetition at the granularity governance operates on.

8.8 Train-serve parity

All governance above assumes one thing silently: that the system being operated is the system that was evaluated. That assumption fails easily and quietly—a deployed path that reconstructs the model differently, feeds a channel with placeholder values, or uses a different encoder produces behavior that no evaluation ever certified, while every dashboard still shows the certified numbers. The framework elevates *train-serve parity* to a standing invariant: the deployed processing path must be verified equivalent to the evaluated path—ideally bit-exact under a flags-off configuration—by automated probes that run as regression checks, not as a one-time audit. Parity is also a drift criterion in the governor’s instrument set: a parity break during growth blocks everything else, because once parity is gone, no other measurement means what it claims to mean.

What a parity check must compare. Parity means *the same tick computed from the same inputs yields the same state*—not that the same components were found and loaded. A check that verifies the presence of the expected parameter groups is a loading test; it will pass while the two paths compute different functions, and it will pass loudly enough to be mistaken for the invariant. Two common failure paths leave no trace. First, anything the training loop applies to the forward pass *outside* the model modules—a routing mask, a mixture stage, a clamp on intermediate magnitudes—is part of the computed function but not part of the saved parameters, so it does not travel with the checkpoint and its absence at serving time produces no missing key and no warning. The rule that follows is architectural: whatever shapes the state belongs *inside* the operator, and whatever is genuinely training-only (loss terms, gradient surgery, regularizers) must be demonstrably outside the forward pass, not merely believed to be. Second, a module can load completely and still be inert because the caller never supplies one of its inputs; the parameters are present, the check finds them, and the branch never runs. The only check immune to both is a differential one: run one training-side tick and one serving-side tick on identical inputs and compare the resulting states coordinate-wise, under a configuration in which all optional paths are off. Where that comparison is not bit-exact, the difference is the thing to explain before any number from either path is quoted.

8.9 Simulation-first exposure

Finally, the framework constrains growth’s *environment*. Early growth operates on simulation, replay of curated experience, and anchored scenarios—not on open interaction with users. The reasons compound: open interaction delivers an uncontrolled learning distribution (against Section 7.1), concentrates exposure on few partners (against the no-self-sealing rule), and couples the system’s dispositional development to feedback loops nobody audits. *Simulation-first* exposure is not caution for its own sake; it preserves the experimental property that makes governance possible at all—a known input distribution against which drift is definable. H10 summarizes the section’s joint claim: under these invariants, continual growth preserves the seed system’s stability envelope over many growth cycles *while the system acquires content it did not previously have*—both endpoints, since either alone is satisfiable trivially. The complementary question—under which gates interaction surfaces may later *open*—is a policy question; the framework’s stance on it is stated in Section 12.9.

9 Evaluation and Evidence Theory

A framework whose object is an internal state cannot borrow its evaluation from task benchmarks: there is no external answer key for a persistent internal state. TASA therefore ships with its own evaluation theory—what must be measured, what any measurement of a stateful, stochastic system must satisfy before its numbers mean anything, and how claims about “understanding”

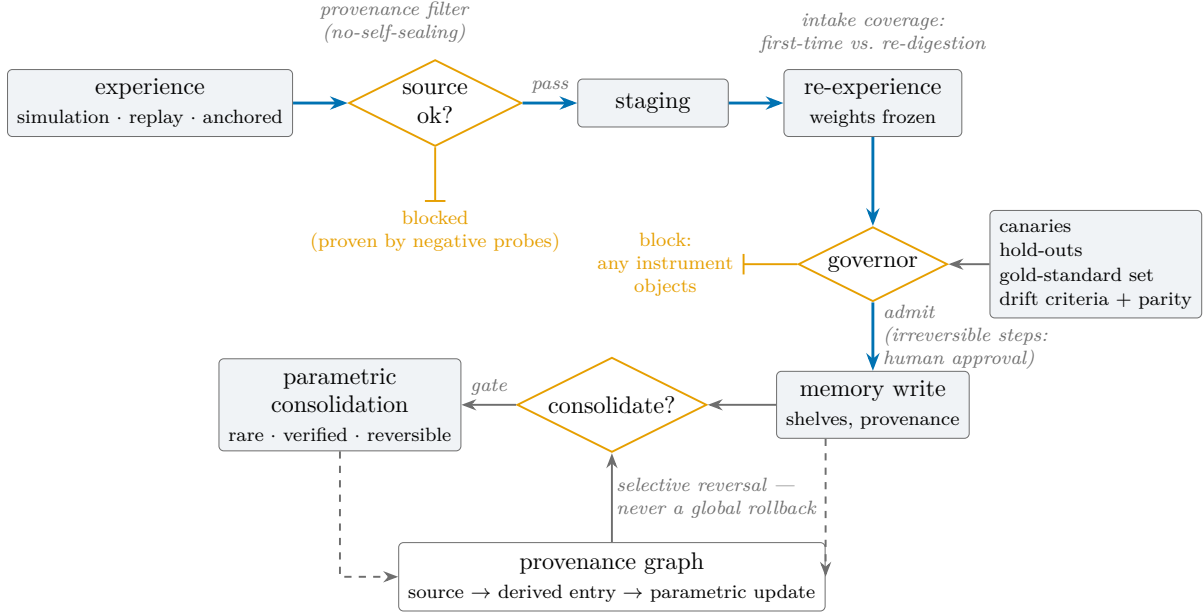


Figure 10: The governed growth loop. Experience enters through provenance filtering and staging; re-experiencing runs with frozen weights; the governor admits or blocks each step under the confluence of canaries, hold-outs, the gold-standard set, and drift criteria (including parity); memory writes precede any rare, reversible parametric consolidation. Blocked pathways are proven blocked by standing negative probes. Both write paths record their edges into the provenance graph, and that is what “reversible” rests on: disqualifying a source marks its dependants invalid and the affected artifacts are rolled back or reconstructed *selectively*. A global checkpoint restore is deliberately not drawn—it would discard every valid change made since, which is why Section 8.2 rules it out as a routine remedy.

and consciousness-adjacent structure can be made honest. Several of its requirements exist precisely because scenario-based evaluation of one’s own architecture is an invitation to self-deception.

9.1 What must be measured

Evaluation probes the running system with synthetic scenario batteries—parameterized event sequences (threat, ambiguity, novelty, and recovery after a negative event) driven through the real tick path—and scores the resulting *state trajectories*, not text outputs. The dimensions follow from the architecture: long-horizon *state coherence* (no collapse, no chaos under autoregressive operation—the behavioral face of H1); *affect-dynamics plausibility* (contextually appropriate reactions, recovery, homeostasis); *meta calibration* (Section 5.4); *memory continuity* (write–retrieve–use round trips); *reactivity* (event sensitivity without hair-trigger instability); *differentiation* (distinct situations produce distinct trajectories); *decision convergence* (intention distributions resolve under decision pressure); and *deliberation depth* (consumed processing depth against input demand, Section 5.5).

Two aggregation duties apply before any score is reported. First, scenario scores are combined into aggregates whose composition is itself a design decision with failure modes (Section 9.2). Second, *robust aggregation*: verdicts are taken from windowed or median-based aggregates of repeated measurements, never from single peaks. A maximum over noisy epochs measures the noise’s upper tail—the winner’s-curse bias familiar from empirical algorithm comparisons in deep reinforcement learning [50]—and rewards configurations that fluctuate over configurations that work.

9.2 Measurement requirements M1–M9

The following requirements apply to *any* quantitative claim about a system built on this framework—and nothing in them depends on that, which is why Section 3.8 carries them as the measurement discipline D2 rather than as a condition of membership: they bind any claim about a stateful, stochastic system, and a program outside this framework’s class inherits them unchanged. They are stated as obligations because each guards against a documented class of self-deception in ML evaluation [10, 50, 71]. Those sources motivate variance accounting and replication discipline; they do not by themselves determine the correct unit or test for every claim below.

M1 — Noise floor first, decomposed by source. Before any verdict, measure the metric’s resolution. The naive version of this requirement—repeat the identical configuration and record the spread—is necessary but not sufficient, and reporting only that number understates the floor that most comparisons actually face. Same-seed repeats measure numerical nondeterminism alone: modern GPU stacks are typically not bit-deterministic under default settings, so identical configurations produce genuinely different models, but the resulting spread is the *smallest* of several.

The requirement is therefore a hierarchy. Measure and report the spread at each level that a claim crosses:

- (i) identical configuration, identical seed and data—numerical nondeterminism;
- (ii) different training seeds—initialization and ordering;
- (iii) different data-generation seeds—which sequences the pipeline produced;
- (iv) different data shards or splits—sampling of the corpus;
- (v) different scenario variants within the evaluation suite—instrument variance;
- (vi) where relevant, different hardware or kernel paths.

The levels are expected to be ordered, each contributing at least as much variance as the one before—an ordering that is itself an empirical claim, verified per metric rather than assumed. The floor applicable to a claim is the one at the level the claim generalizes over: a comparison between two recipes must clear the seed-level floor, not the same-seed floor, and a claim about a class of systems must clear the data-level floor. Quoting the lowest level as *the* noise floor is the error this requirement exists to prevent. Variance from seeds and other run-level perturbations routinely rivals the effect sizes papers report [10]. Floors are metric-specific and must be re-measured when the metric changes.

Corollary: fine-grained components (per-scenario scores) can have far larger floors than their aggregate—identical runs can solve the same aggregate score by different component mixtures—so component-level claims need multi-run agreement, not one run. H12 states both parts empirically.

M2 — Aggregate uniqueness. Every aggregate has exactly one name and one formula. Aggregation choices—mean versus median, window length, which components enter, arithmetic versus geometric combination—change the number and its failure modes; two aggregates that share a name will eventually be compared to each other, and the comparison will be meaningless while looking precise. The aggregate’s definition is part of the metric’s identity, and renaming or redefining it is a breaking change, versioned like one.

M3 — Anchor decomposability. Composite scores and guard thresholds must be reported with their components. A sum or weighted mean is not injective: a composite can hold its value while a critical component collapses or inverts—a calibration score can remain above threshold while its correlation component sits at zero or negative (the anti-calibration failure of Section 5.4). A guard that reads only the composite passes precisely the failures it was built to catch.

M4 — Structurally equivalent positive controls. Every null result requires a *structurally equivalent positive control*: a control of the *same structure* as the test condition—same probe format, same statistical pipeline, differing only in that the effect is known to be present. A null with a failing positive control is not evidence of absence; it is evidence that the instrument cannot see. The framework therefore mandates a three-way verdict vocabulary: *positive*, *null* (control passed), and *not measurable* (control failed)—and treats conflating the last two as a reporting error (Figure 12).

The phrase “known to be present” is where this requirement is most often satisfied in name only, so it needs a sharper reading. Two different things get called a positive control. An *instrument-positive control* supplies a trace in which the measured effect is present *independently of the system under test*—a synthetic trajectory with the effect written into it, a recorded reference case, an injected perturbation of known size. A *sensitivity condition* merely makes the effect easier for the system to produce: an easier item set, a stronger stimulus, a cleaner ordering. Only the first discharges M4, and the reason is the whole point of the requirement. If the control depends on the system exhibiting the capability, then a failing control is compatible with both an insensitive instrument and an incapable system, which is exactly the distinction the three-way verdict exists to draw. Sensitivity conditions remain useful—they diagnose the system and bound its operating range—but they must be labeled as such and may not license a *null*. Where no instrument-positive control can be constructed for a marker, the framework requires that this be stated in the marker itself, and any negative result from it is reported as *not measurable*.

M5 — Unit of analysis. Statistics run at the level of independent items, not repeated measurements. Scoring one stimulus many times and treating the repetitions as samples inflates the effective n and shrinks confidence intervals around what is largely measurement noise—pseudo-replication. The unit of analysis is the independent *item*—a stimulus, scenario, or concept—not the unit of observation; repetitions aggregate *within* items before any test runs *across* them. This guard follows the classical diagnosis of pseudoreplication [54].

Where several configurations, metrics, or data sets are compared, the analysis plan must also declare the comparison family, correction, and test choice. Multi-data-set classifier comparisons and NLP significance testing provide concrete decision guidance [25, 28]; neither a single omnibus score nor uncorrected pairwise testing is a default.

M6 — Correlated-similarity hygiene. Claims built on differences of similarity measures (cosine similarities to related references, before/after distances) must account for the measures’ shared structure: raw differences of correlated similarities produce artifactual effects with convincing magnitudes, driven by baseline offsets rather than the phenomenon. Residualize against the shared component, and treat one diagnostic as decisive: when two independent questions produce the same effect signature through the same pipeline, suspect the pipeline before the phenomenon.

M7 — Pre-registration. Expected outcomes are written down before measurement—direction, rough magnitude, and the interpretation matrix (what each possible outcome will be taken to mean). Wrong predictions are reported as wrong, and thresholds are never adjusted

after seeing data that would fail them. Pre-registration is cheap at this scale and buys the one thing post-hoc analysis cannot: the ability to distinguish prediction from postdiction [86]—and thereby a finding from a story [71].

M8 — External anchoring. The requirements above discipline a system’s self-assessment; they do not make it externally valid. An architecture that defines its own state segments, its own plausible trajectories, its own probes, and its own aggregation, and then scores how well the system reproduces that structure, is measuring internal consistency—a necessary result, and a circular one. M1–M7 bound self-deception; they cannot manufacture external reference. The framework therefore requires that a claimed capability be anchored outside the program that defined it, by at least one of:

- (i) *Frozen public instrument.* The evaluation suite is published and version- frozen before the systems it will judge are trained, so that instrument and subject cannot co-evolve.
- (ii) *Independent authorship.* Probes, or a designated subset, are written by people who did not design the architecture—the strongest available guard against testing what one built rather than what one claimed.
- (iii) *Human reference with reliability.* Where targets encode human judgments (affect trajectories, appraisal labels), multiple raters supply them, inter-rater reliability is reported, and the system’s agreement is compared against rater-to-rater agreement rather than against a single annotator’s ground truth.
- (iv) *Consequential external tasks.* At least one evaluation in which success is defined outside the framework’s vocabulary—a task with an objective outcome that the state must serve rather than describe.
- (v) *Independent replication.* Another group reproduces a claimed result from the published specification, which is the only anchor that tests the specification itself.

Two ancillary obligations follow. The mapping from raw measurements to any bounded score scale must be published with the instrument, since an unstated mapping makes a number uninterpretable and silently comparable to nothing. And adversarial probes—scenarios written to make the system look good while failing the underlying capability—belong in the suite by construction, not as an afterthought. This requirement is the one the present framework is furthest from satisfying—and anchor (i) is foreclosed retrospectively for any system already trained, so for those, anchors (ii)–(v) are the available routes. Section 12.2 treats the gap as a limitation rather than a formality.

M9 — Effective-configuration verification. The requirements so far discipline what a number means once a mechanism has run. This one concerns whether it ran. A declared mechanism can be inert for reasons that leave no trace in any loss curve, any evaluation score, or any error log, because inertness is not an error: the system simply computes something slightly different from what its specification says, and reports it cleanly. Three such failures recur often enough to be named, and each is checkable in minutes.

Capacity that receives gradient. Report the parameter count that actually takes optimizer steps, not the count the architecture defines—and note that the two obvious tests do not establish this. A module can be marked trainable, be registered in the optimizer’s parameter set, and still never move, because it is unreachable in the forward pass: it then receives no gradient, and a modern optimizer skips it entirely rather than complaining. Such a module passes every configuration inspection, appears in the model summary, and inflates the reported size. The tests that do decide are behavioral: whether the optimizer holds accumulated state for the parameter,

or—cheapest of all—whether its values differ between two checkpoints of the same run. Any argument that reasons from model size, including scaling comparisons, capacity-matched controls, and the conclusion that a given scale was insufficient, is void if it counts parameters that never moved.

Schedules that complete. Report the *realized* endpoint of every schedule, not its configured target. A schedule parameterized in units the run does not reach—a decay that would finish after more steps than the run contains, an annealing that begins later than the run ends—leaves the quantity it controls near its initial value for the entire run, while the configuration file states an intent that reads as fact. Step normalization (Section 7.7) prevents the data-volume version of this failure; the general form needs the realized value.

Channel occupancy per regime. Report, for each input pathway, how often it carries content in each regime the system runs in—training, evaluation, and deployment separately. Train-serve parity (Section 8.8) compares two computed paths on one fixed set of inputs; it does not compare how often a pathway carries content across regimes. A pathway that is densely populated while training and structurally empty under the evaluation harness turns every number that harness produces into a measurement of a regime the system was not trained for, and the parity check passes throughout, because both sides of it are equally empty. The complementary duty falls on the parity check itself, which must compare computed states rather than loaded components (Section 8.8).

The general form behind all three: *a declaration is not a measurement*. Where a mechanism’s activity can be observed directly, observe it, and report the observation next to the claim that depends on it.

9.3 The staged understanding model

Claims about what an inner-state system “understands” need a scale. The framework defines four *understanding stages*, each with an operationalization hook (Figure 11). One scoping note first: the stages are an *engineering progression for this class of systems*—an ordering of measurable capabilities by their architectural prerequisites—not a claim about human or animal development, whose empirical ordering is a separate matter this paper does not enter. The stages are also not neutral bookkeeping: ascent through them, S4 included, is the program’s declared aim (Section 1), and the gating thesis of Section 12.6 states what that ascent is predicted to require.

S1 Affective reaction. Contextually appropriate affect responses to event types—measurable as reaction-profile separation across scenario classes.

S2 Pattern recognition. Sensitivity to regularities across situations—measurable as differential trajectories for structurally distinct inputs (e.g., pragmatic categories), beyond surface features.

S3 Prospective consequence sensitivity. Anticipatory state shifts before an announced consequence, measurable as pre-event trajectory divergence conditioned on the announced outcome (B.11). The name is deliberately weaker than the phenomenon it is sometimes taken for. When the outcome is stated in the announcement, a system that merely reads affective valence out of the announcement text passes; nothing in the marker requires a causal model or expectation formed from causes. Establishing *causal* expectation would need transfer to unfamiliar causal structures and prediction from premises that do not name the outcome, and the framework does not currently specify those tests.

S4 Content understanding. Transfer, contradiction sensitivity, and compositional use of content. Contradiction sensitivity currently has a marker (B.4); transfer and compositional use do not. S4 therefore remains *not claimable*, and the highest claimable stage is S3, until

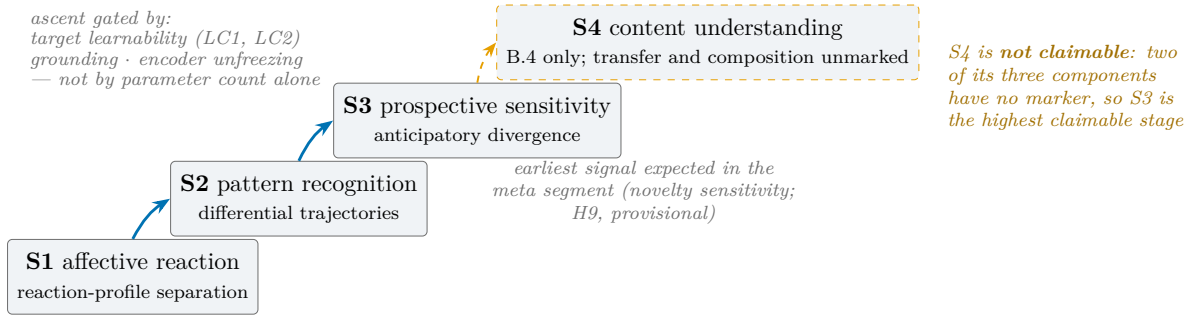


Figure 11: The staged understanding model. Each stage names a capability and an operationalization; ascent is gated by target learnability (LC1, LC2), grounding, and encoder unfreezing—not by parameter count alone. S4 is drawn dashed because it is specified but *not claimable*: of its three components only contradiction sensitivity has a marker (B.4), and until transfer and composition markers exist the highest claimable stage is S3. The dash marks a hole in the instrument set, not a stage the framework is unsure about.

dedicated markers with instrument-positive controls are specified for all three components. Those markers must be fixed before the corresponding results are observed.

The progression concerns functional claims, not subjective depth. Improvement within a stage and ascent to another are different verdicts: the former changes performance on the current stage’s declared operationalization, whereas the latter requires satisfying the next stage’s declared operationalization under the measurement requirements of this section. A higher aggregate, a longer trajectory, or more parameters is therefore never sufficient on its own to establish ascent. Because S4 currently lacks dedicated transfer and compositional-use markers, no amount of lower-stage improvement can license that claim.

The model’s substantive claim is about *gating*, not just ordering: ascent is enabled by the learnability conditions of Section 7.3 and by grounding and encoder unfreezing—not by parameter count alone. The framework predicts that a system whose latent targets violate LC1 or LC2 can saturate S1 while none of the levers named here is available to lift it to S4, and that raw scale is not among those levers. H9 adds an ordering claim with a metacognitive twist: novelty sensitivity—visible in the meta segment—is expected to precede contradiction competence, making the meta channel the right place to look for the *earliest* signs of content sensitivity. Ordering relative to transfer is left open, and Section 11 says why.

9.4 Evidence markers for structural claims

The framework evaluates structural properties through a battery of *evidence markers*, in the spirit of the indicator-property methodology that has recently organized this debate for AI systems [12]. The markers are of two kinds: *theory-derived* markers (workspace ignition dynamics, meta-confidence calibration, contradiction-sensitive revision) descend from the cited theories and would be evidence about any system; *design-verification* markers (reportability, temporal continuity) confirm that the built system actually has the structure this paper specifies—necessary, but not evidence beyond the design. Together they can support claims that a system instantiates specific structural properties considered relevant in consciousness science; they do not compose a test of consciousness. The framework’s marker set spans: reportability (state reports track the state), temporal continuity, self/world distinction, contradiction-sensitive revision, meta-confidence calibration, perturbation robustness, introspection–behavior consistency, workspace ignition dynamics, narrative continuity, causal efficacy of the meta-state (meta variables influence downstream processing, not merely report it), anticipatory state shifts (the S3 signature: pre-event divergence under announced outcomes), deliberation depth (processing depth varies with

demand—and pays), exposed-state sufficiency (the inspectable state, not an internal carrier, drives downstream behavior), and non-broadcast influence (whether losing workspace proposals still shape later processing—a question only realizations that give them a path into the state can answer, Section 5.1), and grounding fidelity (the emitted reply is licensed by the state report rather than by the language model’s own priors; retrieved content contributes only after the operator has incorporated it into C_t).

Marker discipline is where the section’s machinery converges (Figure 12): each marker carries an operationalization, a *failure criterion fixed in advance* (M7), a structurally equivalent positive control (M4), item-level statistics (M5), and named confounds—above all *retrieval shortcut* (correct behavior via lookup rather than the claimed structure) and *prompt echo* (reports that paraphrase the probe rather than read the state). The recent adversarial-collaboration test of leading consciousness theories illustrates the value of this discipline: under a pre-registered protocol, the study reported challenges to predictions of both theories [20]. A subsequent GNW response contests parts of the operationalization and inference [83]; the episode therefore illustrates both the value of severe testing and the need to keep theory-to-marker mappings open to documented dispute. Appendix B specifies each marker; H13 covers the aggregate-visibility face of the same discipline—whether an evaluation’s aggregates can see representational content at all is itself a design property, to be verified rather than assumed.

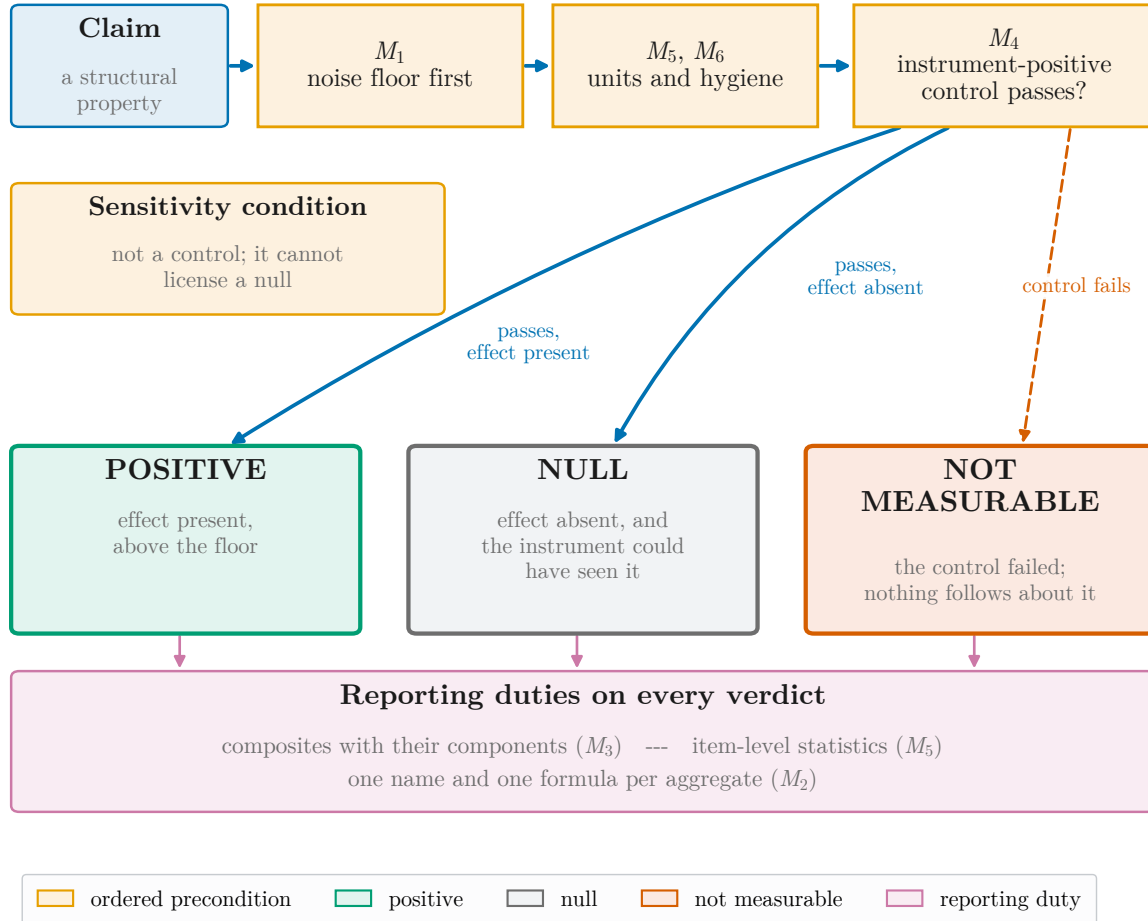


Figure 12: From a claim to a verdict. The preconditions are ordered and each can stop the process, but not in the same way: a noise floor is measured before any comparison and units are fixed before any number is compared, and failing either leaves the comparison uninterpretable rather than producing a verdict—which is why only M_4 has a drawn exit. Its failure *is* a verdict. The three-way vocabulary exists because a genuine null and a blind instrument are indistinguishable from the outside, and a failing control forces the third verdict, never the second. The condition on the left is the distinction the requirement turns on: an easier item set or a stronger stimulus diagnoses the system but cannot license a null, because it fails for reasons that involve the system. The bar along the bottom quotes three requirements that attach to every verdict regardless of which one it is.

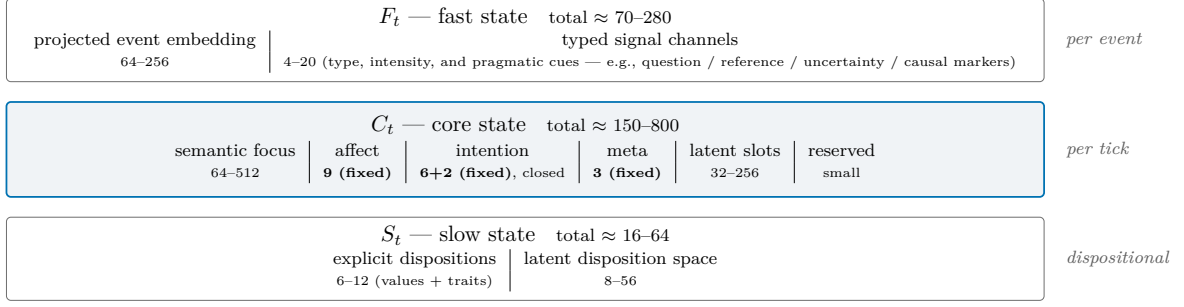
10 A Reference Instantiation

A framework earns part of its credibility by showing that it can be built. This section sketches *one* concrete instantiation that the preceding theory admits—state layout, operator choice, training recipe, data pipeline, and evaluation suite—in enough detail that an experienced practitioner could implement it. Three reading rules apply throughout. First, the sketch is *abstracted, not invented*: it distills the reference implementation mentioned in Section 1 into ranges and alternatives rather than reporting it, and where alternatives are competitive we say so. It is a reference *design* rather than a record: some elements follow from the requirements of the preceding sections and are included because the framework demands them, whether or not any existing system already realizes them—so the sketch should be read as what the theory asks for, not as an inventory of what has been built. Second, quantities are given as *ranges* with their selection logic, not as fixed values—every choice below is one of several the operator contract (Section 3.3) admits, and the framework constrains orders of magnitude and ratios, not decimals; Section 3.8 says which of them cannot wait, and why. Third, nothing in this section is an empirical claim: the sketch argues implementability and cost plausibility, while performance questions remain exactly where Section 11 puts them. Comparative judgements below—about operator families, what stabilizes at small scale, and runtime cost—are design judgements and reported practice; none is a measurement taken for this paper.

10.1 State layout

Figure 13 instantiates the state space of Section 3.2 with concrete segment budgets. The guiding ratio: the semantic focus and latent capacity dominate the core state’s volume, while the role-bearing segments are small. Under the reference ontology of Section 3.2 they come out at 9 affect dimensions (valence, arousal, dominance; five basic-emotion intensities; one epistemic drive), 6 explicit intention categories (act, plan, explore, reflect, retreat, wait) plus two scalars (confidence, action readiness), and 3 meta read-outs (confidence, conflict, complexity). These counts are the reference ontology’s resolution, not a theoretical necessity: an instantiation that covers the same roles with a different inventory—six categorical emotions rather than five, an uncertainty read-out separate from confidence—remains within the framework, provided the difference is declared. What follows fixes the reference counts so that the rest of the sketch has something definite to build on. Around them:

- **Fast state:** a projected event embedding of 64–256 dimensions plus 4–20 typed signal channels. Workable channels include an event-type one-hot (e.g., neutral, positive, negative, or conflicting) and an intensity scalar, together with a handful of pragmatic cues—for instance markers for questions, for uncertainty or causal framing, or for whether the event refers to the self, another agent, or the world. The precise cue inventory is instantiation freedom and should follow what the data pipeline can reliably annotate: channels the data cannot yet populate should be omitted rather than zero-filled, since a permanently silent input channel should be expected to teach the operator to ignore it—the input-side analogue of LC4.
- **Core state:** 150–800 dimensions total. Semantic focus 64–512 (larger preserves more of the encoder’s relational structure; the reduction factor from the raw embedding should stay moderate—anywhere from identity width down to roughly 4× compression—and is chosen jointly with the encoder width); intention comprising the declared closed distribution over the six action tendencies plus its readiness and confidence scalars, and nothing else—uncommitted capacity belongs to the latent workspace slots below, where it is declared as untyped rather than sitting inside a typed segment as unnamed action tendencies; latent workspace slots 32–256 (the segment most worth over-provisioning, per the segmentation bet of Section 3.2); a small reserved block for later typed extensions.



bold = framework-default ontology (Sections 3.2, 4 and 5.4); *ranges* = instantiation freedom with selection logic in Section 10.1

Figure 13: Instantiated state layout with dimension ranges. The role-bearing segments (affect 9, intention 6+2, meta 3) are fixed by the reference ontology (Section 3.2); all other budgets are ranges the framework admits, with the selection logic of Section 10.1. The intention segment is *closed*: it carries the six action tendencies and the two scalars and nothing further, because uncommitted capacity placed inside a typed segment would be unnamed action tendencies in all but name. Spare capacity belongs to the latent workspace slots, where it is declared untyped.

- **Slow state:** 16–64 dimensions — a handful of explicit dispositional dimensions (e.g., 6–12), split between value orientations (such as truth-orientation, harm-avoidance, or social orientation) and processing traits (such as risk tolerance or deliberative tendency), plus a latent dispositional remainder. The split and the exact dimension inventory are instantiation choices; the theory fixes only that explicit dispositions exist and stay few.

The layout embodies the segmentation bet of Section 3.2 deliberately: enough typed structure to evaluate and govern, enough latent capacity that the typing does not strangle representation.

10.2 The tick operator: candidate architectures and sizing

The operator contract requires a differentiable map with a typed readout whose recurrence is carried explicitly by C_{t-1} , and optionally by an internal carrier as well. Four architecture families are the natural candidates. All four can satisfy the contract—including the feed-forward transition, since Equation (3) takes C_{t-1} as an argument and a map over it is state-recurrent at the system level even with no carrier of its own. It is a weak baseline for its *horizon*, not for its conformance: its memory is exactly the declared state and its explicit summaries, which is also what makes it the cleanest staging and ablation control. Table 3 summarizes their suitability. The comparison is genuinely open—the framework does not need the strongest sequence model, it needs the most *stabilizable* one at the data scales such training can realistically reach.

The table varies the operator’s *family*; it does not settle its *cardinality*. A core realized as several sub-networks—one per segment group, or one per timescale partition as R2 already admits—satisfies the same contract, provided the composition is one computation graph and each part’s readout is typed. Whether such a composition is worth its coupling is the fork of Section 12.5, not a question this table answers.

Default and upgrade path. The sketch’s default is deliberately conservative: begin with an MLP transition for debugging, move to a *gated recurrent operator* as the working baseline, and treat selective state-space or extended-LSTM variants as an upgrade once rollouts lengthen past a few hundred ticks and data volume supports them. At small data scales we expect gated RNNs to be the most straightforward of these families to stabilize under a many-term objective, and the sketch accordingly treats stability rather than expressiveness as the binding constraint early on. A hidden width of a few hundred to a few thousand units over one or two layers, with the typed readout as a linear head plus segment activations, places the operator at $10^6\text{--}5\times 10^7$

Family	Typical params	Strengths for this role	Weaknesses / risks
MLP (no recurrence)	10^5 – 10^7	Trivial to train and debug; useful as a first stage and as an ablation control	Conforming in the no-carrier regime, since recurrence runs through C_{t-1} ; horizon limited to the declared state and its explicit summaries
Gated RNN (GRU/LSTM) [18, 51]	10^6 – 5×10^7	Few parameters per unit of sequential capacity and a strong recurrence bias—analytically, fewer directions along which competing objectives can interfere; per-tick semantics map onto recurrence without added machinery	Long-horizon gradient issues (contained by truncated BPTT at the price of credit horizon); sequential compute
Small Trans-former [111]	5×10^6 – 10^8	Expressive; attention gives free working-memory-like access	Needs more data; no native persistent state (must be added); overkill for short tick contexts
Selective SSM / mod. recurrent (Mamba [40], xLSTM [7])	10^6 – 5×10^7	Efficient long-sequence scaling; a natural upgrade path for long autoregressive roll-outs	Less accumulated tuning experience than the gated families; the efficiency advantage is defined over sequence lengths that early stages do not reach

Table 3: Candidate operator families against the tick-operator contract. The sketch’s default is a gated RNN with a staged upgrade path; the choice is an engineering trade-off the framework leaves open, and H1/H2 are family-agnostic.

parameters—a band spanned by the complete tick machine (workspace modules, projection, and heads included), not by the recurrent cell alone. Which point in the band an instantiation actually occupies should be read as the effective count of M9 rather than as the architecture’s nominal size. The two can differ by a factor of several, and the discrepancy is one-sided: it always makes the system look larger than it is, which is the direction that turns a training-recipe problem into an apparent capacity ceiling.

Sizing follows data, not hardware. At these scales, parameter and optimizer-state memory is not an obvious prototype barrier: under the stated accounting assumptions, a 5×10^7 -parameter operator with full optimizer state occupies roughly a gigabyte. The real constraint is the *transition count*: a corpus of N sequences averaging T ticks supplies $N \cdot T$ supervised transitions, and the parameter budget should track it. A workable ladder: 10^6 – 5×10^6 parameters at $\sim 10^4$ transitions, 5×10^6 – 2×10^7 at $\sim 5 \times 10^4$, and 2×10^7 – 5×10^7 from $\sim 10^5$ transitions upward, with validation loss divergence as the overfit indicator that halts the ladder. This supports only a resource-plausibility claim for a prototype core on one GPU. It does not establish that the full training loop meets a wall-clock, latency, energy, or repeat-run budget; hardware-specific throughput and the affordability of noise-floor and multi-seed studies remain to be measured.

10.3 The input translator

The frozen-first policy (Section 3.5) instantiates naturally as: a benchmark-selected sentence-embedding model [82], preferring open, reproducible weights [87]; where the encoder supports matryoshka-style nested representations [64], truncate first to an intermediate width (e.g., 256–512) and project the rest of the way with a small learned MLP, keeping the per-stage

compression factor moderate. The projection is pre-trained briefly with a supervised contrastive objective over scenario clusters [59]—sequences from the same scenario family pull together, others apart—then co-trained with the core so the projection adapts to the state space while the encoder itself stays frozen. A linear/PCA projection serves as the debuggable baseline the learned projection must beat.

10.4 Training recipe

Figure 14 assembles the recipe; Appendix E gives the full term-by-term table. The shape is a *many-term supervised objective with scheduled emphasis*, built from four groups:

1. **State supervision.** The core term supervises the next core state against the latent targets, as a hybrid of squared error and cosine distance (weighted toward squared error; the exact ratio is instantiation freedom, and the trade-off that follows says which way to move it)—squared error anchors magnitudes, the cosine share resists the mean-collapse pull toward “safe average vectors” that pure MSE invites on continuous outputs—at the price of norm-growth pressure, since cosine terms are scale-invariant and leave norm growth unopposed—countered by an explicit state-norm penalty with radius scaling as $c\sqrt{d}$ in the state dimension.
2. **Affective structure.** A separately weighted affect term; the soft consistency rules of Section 4.2 as hinge penalties; an *adaptive smoothness* penalty whose tolerated per-tick affect change grows with event intensity (calm ticks demand calm affect, intense events license jumps); and the affect–intention prior of Section 4.3 as an initialization-plus-decaying-regularizer, active over an initial fraction of training and then released to learning.
3. **Distributional heads.** The intention distribution trained with KL divergence under softmax (with an ϵ -floor against $\log 0$); an event/scenario prediction head (predicting the *class* of what comes next, which is predictable, rather than the raw next embedding, which largely is not); and the supervised appraisal-context head of Section 7.4.
4. **Loop-breaking and stability.** Scheduled output noise (a small, slowly decaying σ that never fully vanishes) counteracts the trivial fixed-trajectory solution of self-prediction; truncated backpropagation with a 50–100-tick window bounds memory and gradient pathologies on long rollouts; layer normalization inside, segment activations outside.

Weights are scheduled, not static: event-prediction emphasis decays as state- supervision emphasis rises, so the system first learns world contact and then inner coherence. With this many terms, per-term gradient-norm logging is mandatory instrumentation, and automatic balancing methods—gradient normalization [17] or gradient surgery [121]—are the designated upgrade once manual schedules strain. Two of the standing watchdogs carry explicit machinery: a *mode-collapse detector* (batch-wise mean pairwise output similarity with a warning and an alarm threshold, plus an input–output diversity correlation check), answered by a three-stage mitigation ladder—raise the output-noise floor, rebalance the cosine share of the state loss, and, only on alarm, a contrastive dispersion term; and the schedule discipline of Section 7.7—step-normalized, treated as part of the recipe’s identity.

Curriculum. Training proceeds in the anchor-based stages of Section 7.7: short fully supervised sequences (tens of ticks); concatenated sequences with rest-tick transitions; then autoregressive rollouts of hundreds of ticks punctuated by anchor events with *soft* anchoring (anchor loss weighted down as the free trajectory drifts, so anchors correct rather than yank) and adaptive anchor density; finally long rollouts with sparse anchors. Each transition is gated by explicit

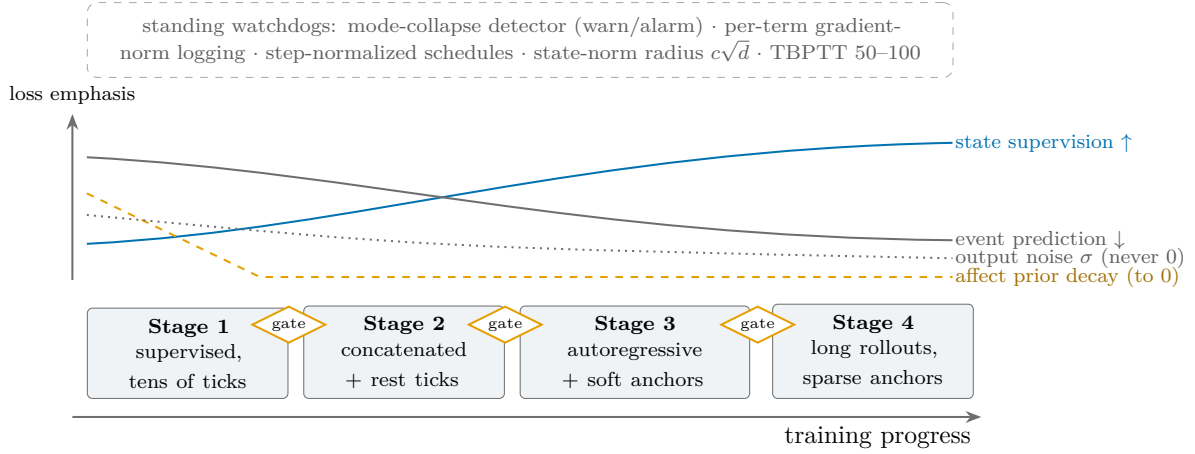


Figure 14: Training recipe at a glance: curriculum stages over training time, the scheduled shift from event-prediction to state-supervision emphasis, the decaying affect-prior and output-noise schedules, and the standing watchdogs (mode-collapse detector, per-term gradient-norm logging, step-normalized schedule discipline, state-norm radius, truncated BPTT).

criteria (convergence of state supervision, coherence scores above floor, no collapse signatures)—and each sequence ends with programmatic *recovery ticks* (quiet events with a decaying state target) so that stillness and decay are learned distributions, not out-of-distribution surprises. Appendix E tabulates the stages and gate criteria.

10.5 Data pipeline and disposition profiles

Figure 15 instantiates Sections 7.1 and 7.2. A training item is a *scenario sequence*: a JSON document holding, per tick, an event (text, type, intensity) and the expected state (affect values, intention distribution, meta values, a free-text semantic focus), plus a sequence-level disposition profile; the numeric layers convert deterministically to the latent targets, the event text enters through the input translator, and the semantic-focus text is encoded by the same encoder to form the semantic segment’s target—two distinct strings, per Section 7.2 (Appendix E shows an abridged schema). Around this format:

- **Scenario taxonomy.** A closed set of roughly ten scenario families (e.g., curiosity, dilemma, loss, monotony, recovery, social interaction, success, surprise, threat, and everyday-neutral episodes—the exact family set is instantiation freedom), optionally refined into a few tens of classes at the taxonomy’s finest granularity, supplies cluster labels—the supervision signal for the appraisal head and the contrastive structure for the projection, and the stratification unit for splits and hold-outs.
- **Generation, split by kind.** The scalable route lets a language model write the *texts*—events and semantic-focus lines—while a deterministic local target generator computes the *numbers* (affect trajectories, distributions, smoothness-checked), with schema validity enforced by constrained decoding rather than assumed. The split addresses generator circularity at the point where it matters most: in this instantiation no numeric target is chosen by the generator. It does not remove generator-induced psychology: scenario content, class assignments, intensity descriptions and the semantic-focus texts still carry the generator’s conventions, and Section 12 states what remains. The third construction of Section 7.2 trades part of that guarantee for interpretive coverage, and an instantiation choosing it says so.

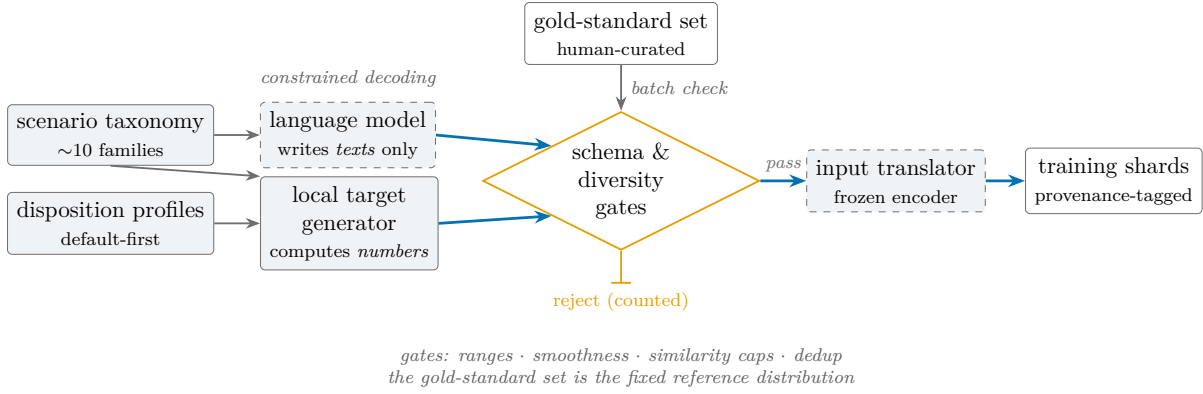


Figure 15: Data pipeline: scenario taxonomy and disposition profiles parameterize text-first generation (the language model writes texts; a deterministic local target generator computes the numeric targets); constrained decoding enforces the schema; diversity and smoothness gates filter; the gold-standard set provides the fixed human-curated reference; encoded shards feed training with provenance tags.

- **Quality gates.** Range checks; per-tick smoothness limits; diversity thresholds on embedding similarity (within-sequence and between-sequence caps, and cluster separability floors) with rejection of monotone sequences; deduplication; and a small human-curated gold-standard set (Section 7.2) as the fixed reference distribution every generated batch is checked against.
- **Disposition profiles.** The slow state’s explicit dimensions admit a small library (three to eight) of named profiles—archetypes such as *venturesome* (high risk tolerance, high exploration), *deliberate-systematic*, *socially warm*, or *steady-pragmatic*—the lever by which the *same* event trains *different* reactions (Section 7.1: the training distribution *is* the disposition specification). One recommendation the framework’s own logic supports: **begin with a single default profile.** Profile diversity multiplies the learning problem and confounds early evaluation (every effect must be checked per profile); a single disposition removes that confound while the basic dynamics stabilize—at the stated cost that dispositional variation is postponed, so dispositional probes measure a channel that barely varies and are expected to be uninformative until the library activates (the reachability caution of Table 10). The library becomes valuable exactly when dispositional differentiation itself is the phenomenon under study.
- **Stream widening.** Beyond the emotional core: knowledge-experience sequences (learning, insight, dilemma, causal-chain episodes—the system learns what acquiring knowledge *does to the state*, not facts as strings), mid- and long-horizon episodes, adversarial probes, grounded trajectories from simulators where available, each stream provenance-tagged per Section 8.3.

10.6 An experience-evaluation suite

The evaluation theory of Section 9 instantiates as a fixed *experience-evaluation suite* (Figure 16): a battery of 12–20 scenario probes, each driving the live tick path with a parameterized synthetic scenario and scoring the state trajectory on a bounded scale (here 0–100; the scale is a presentation choice, the published raw-to-score mapping of M8 is what makes it interpretable). The framework fixes the *duties* such a battery must cover, not a canonical probe list; one workable set, grouped by duty: *affective adequacy* (a narrative affect arc through a story-like scenario; mixed and sequential affect; homeostasis paired with reactivity); *differentiation and responsiveness* (situation

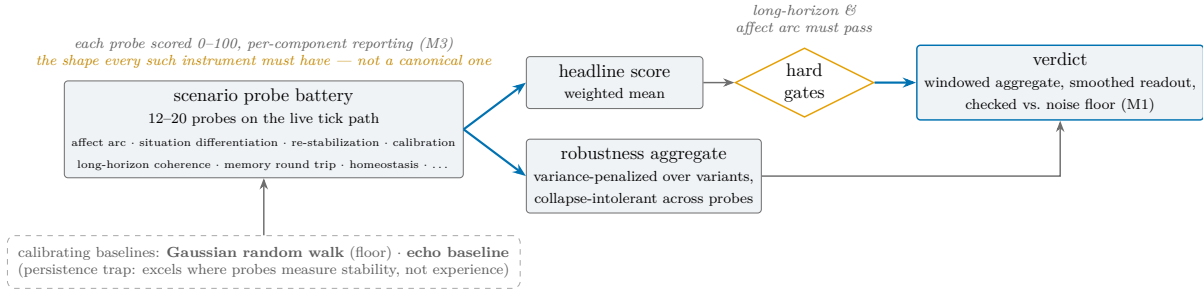


Figure 16: The experience-evaluation suite: scenario probes drive the live tick path; scores aggregate through a weighted mean with hard gates and, separately, a variance-penalized, collapse-intolerant robustness aggregate over scenario variants; Gaussian random-walk and echo baselines calibrate the instrument; verdicts are windowed aggregates over a smoothed readout, checked against the measured noise floor. What the figure fixes is the *shape*: probe list, weights, robustness construction, aggregation window, and raw-to-score mapping are all open, so this is the structure any conforming instrument must have rather than a canonical instrument (Section 10.9).

differentiation—distinct situations, distinct trajectories; event proportionality); *stability* (long-horizon coherence—no collapse or chaos over hundreds of ticks; topic coherence across an arc; re-stabilization after adversity; dispositional stability); *metacognition* (a calibration probe scoring, e.g., confidence–error correlation, calibration error, and reactivity or resolution components—reported per component, per M3, never only as a sum); *memory* (write–retrieve–use round trips; retention under interference); *decision* (intention convergence under decision pressure); and *deliberation and anticipation* (consumed processing depth versus input demand, against a fixed-depth control; pre-event trajectory divergence under announced outcomes—the probe forms of markers B.12 and B.11, Section B).

Aggregation follows the theory’s duties. The headline score is a weighted mean whose largest weights belong to the probes without which an inner state is not worth scoring—stability and the affective core—enforced also as *hard gates*: a run failing long-horizon coherence or the narrative affect arc cannot pass overall regardless of its mean. Beside the headline score stands a separately named *robustness aggregate* (its own instrument in the sense of M2), per the robust-aggregation duty of Section 9.1. The framework admits a family of constructions here: probe scores penalized for variance across scenario *variants* (a mean-minus-fraction-of-spread score, floored at zero, or a lower quantile), combined across probes by a collapse-intolerant aggregate (a geometric mean, a soft minimum, or worst- k averaging) so that no single strong axis can carry a collapsed one; probes saturated at their ceiling contribute no resolution and are better tracked separately than averaged in. Verdicts come from windowed aggregates over the smoothed score trajectory across training epochs—trailing means or best- k within a window, never single-epoch peaks—and every claimed difference is checked against the measured noise floor at the level the comparison generalizes over (M1).

Two baselines calibrate the whole instrument, and the second guards against a specific trap: a *Gaussian random-walk* baseline (per-dimension random walk with data-matched variance) marks the floor any functioning system must clear—and an *echo baseline* (previous state plus small noise) marks the trap: by construction, a maximally persistent baseline excels on any stability-flavored probe, so any probe on which the echo baseline scores well is measuring persistence, not experience, and must be gated by reactivity requirements. Score bands (not functional, functional, good, excellent) are fixed with the suite, and the suite itself is versioned per M2: any change to probes, weights, or aggregation is a new instrument, not a new number.

10.7 Runtime sketch

Four runtime elements complete the instantiation. **Cold start:** the initial core state is neutral and alert—neutral valence, low arousal, mid dominance, minimal emotion activations, a reflect-leaning intention distribution, mid confidence—with small per-sequence noise so the operator never overfits to one exact origin; a data-driven alternative (the training distribution’s state mean) is the natural comparison. **Tick economics:** the argument the framework needs here is a *ratio*, not a duration. A tick is a single forward pass through an operator of the size sketched above; input encoding is a forward pass through a sentence encoder, which is the larger of the two by roughly an order of magnitude. Multi-tick deliberation of 5–10 ticks therefore costs a small multiple of one operator pass, not of one encoder pass, and that ratio is what makes the deliberation policy (Section 5.5) practical rather than decorative. Absolute latency remains unmeasured and depends on hardware, precision, batching, and implementation. A tick is a logical step—triggered by input, by a timer, or by the system’s own continuation decision—not a fixed wall-clock interval. **Within-tick reasoning:** the deliberation policy admits a concrete refinement mechanism: a small residual state-to-state module applied 2–5 times within a tick when meta-confidence is low, halting when confidence recovers—inspired by latent-space reasoning [48], with learnable halting in the style of PonderNet [5] as a candidate upgrade over thresholds. Refinement iterations are internal to a single state update and leave tick semantics untouched: an *intra-tick* realization of processing depth, complementing—not replacing—the multi-tick deliberation policy in which the framework counts depth. **Curiosity-driven acquisition:** the epistemic drive closes its loop operationally: low meta-confidence plus a memory miss triggers an external knowledge query; the answer enters as an ordinary event, so the acquisition passes through the full state machinery—affect responses and memory writes included—rather than being injected as a cold fact; the episode lands in the knowledge shelf under provenance and rate limits—instantiating the knowledge-shelf lifecycle of Section 6.1 and, per the no-self-sealing rule, never becoming a dominant learning source. The staged unfreezing path’s middle stage likewise gains a concrete form: a small residual adapter branch beside the frozen *encoder*, blended in with a factor that starts at zero and grows only while evaluation holds (Section 7.8).

10.8 The interaction loop

The runtime elements assemble into a complete interaction loop (Figure 17). One exchange, end to end:

1. **Ingest.** The user’s message is encoded by the input translator and annotated with its signal channels (question cue, reference cues, intensity); it enters as one event tick, exactly like any other event.
2. **Deliberate—zero to a handful of ticks** in the common case, with the tick budget bounding the tail (counted here in addition to the ingest tick). The deliberation policy (Section 5.5) decides whether the state is ready to act: readiness or meta-confidence crosses its threshold, or the budget ends deliberation. The minimal viable loop responds directly from the post-ingest state; deeper configurations interpose deliberation ticks whose count is content-dependent—and observable.
3. **Report.** The exposed state renders into a state report (Section 3.6): current affect, dominant intention, semantic focus, meta read-outs, dispositional tone, and what holds the broadcast. Memory-derived content appears only where the operator has already written it into the core state, so the line below reports what the system now holds, not what the store returned. Each field is tagged [direct] or [decoded] as Section 3.6 requires, so a reader can see which values were read and which were inferred by an instrument. An illustrative report, structurally (values invented):

```

[direct] affect:  valence +0.1, arousal 0.35;
                  curiosity (epistemic drive) elevated;
                  emotion intensities low
[direct] intention: reflect; not ready to act
[direct] meta:   confidence 0.55, conflict low, complexity elevated
[decoded] semantic focus: "the stated plan conflicts
                          with an earlier commitment"
[direct] broadcast: conflict-relevant content won this tick
[direct] disposition: steady default profile

```

The example excludes two natural additions because neither is derivable from X_t . There are no *trend* labels—“arousal rising”, “explore gaining”—since a direction of change is a statement about two ticks and the report reads one; an instantiation that wants them must carry the trend as an explicit state field, at which point it is an ordinary direct read-out. And there is no retrieval *provenance* line: that a focus entered from the episodic shelf is a fact about M_t , which Figure 4 puts outside the renderer’s reach. Both omissions are the report-only rule of Section 3.6 taking effect at the one place a developer would otherwise copy the weaker version into code.

4. **Translate.** The output translator verbalizes the report under the grounding contract of Section 3.6. Conversational surface coherence is obtained from *non-propositional* conditioning only—register, grammatical person, formality, target length—which the report itself can carry as fields; raw prior exchanges are not supplied, since a translator that can read them can assert them. Where an instantiation needs an antecedent for coreference, the antecedent belongs in the report as a declared field. The report thus remains the sole assertable content, and the grounding check of Section 3.6 scores the reply against the report. Its declared coverage is what makes the score readable, and the cheapest workable instrument is narrow rather than shallow: an entailment check restricted to the report’s typed fields—numeric values, the named intention, the stated confidence—which decides those classes reliably and *declares* that it does not decide claims about the semantic field. Surface string matching is not an option here, since it accepts a paraphrase that adds a claim and rejects a faithful rewording. Role: a per-reply diagnostic at minimum, a release gate at strictest; B.15 measures the checker before it measures the system.

Narrow coverage buys reliability at a price that has to be paid rather than declared away. The semantic focus is the report field from which most propositional content is drawn, so a checker that skips it leaves the framework’s central claim—speech is a function of state—unverified on the channel that carries the most meaning, while every scalar passes. The rule is therefore a floor rather than a preference, and Section 3.6 states it: a *grounding claim* requires checker coverage of every content-bearing field the reply drew propositions from, decoded semantic content included. Fields outside the declared coverage may be displayed as diagnostics and may inform a human reader, but they may not support an emitted assertion until an instrument covers them. An instantiation whose checker is typed-fields-only is conforming and honest, and it may not claim verified grounding—only verified grounding *of its typed fields*.

5. **Close the loop.** The reply’s emission registers as an event on the output side: the canonical *self-action event* of Section 3.6—act type, addressee, and a pointer to the report it was grounded in—enters at step T1 of the following tick, exactly like an external event, with the no-backflow rule keeping the generated *text* out of the core. This tick is not optional and not merged with the next user turn: it is what keeps the system’s own history inside its state. The user’s next message then enters as the next input event. Between exchanges the system is not suspended: *settling ticks* let the state relax after an exchange, timer-driven ticks keep the slow dynamics moving, and the curiosity loop of Section 10.7 may issue governed knowledge queries—idle is a state, not a pause.

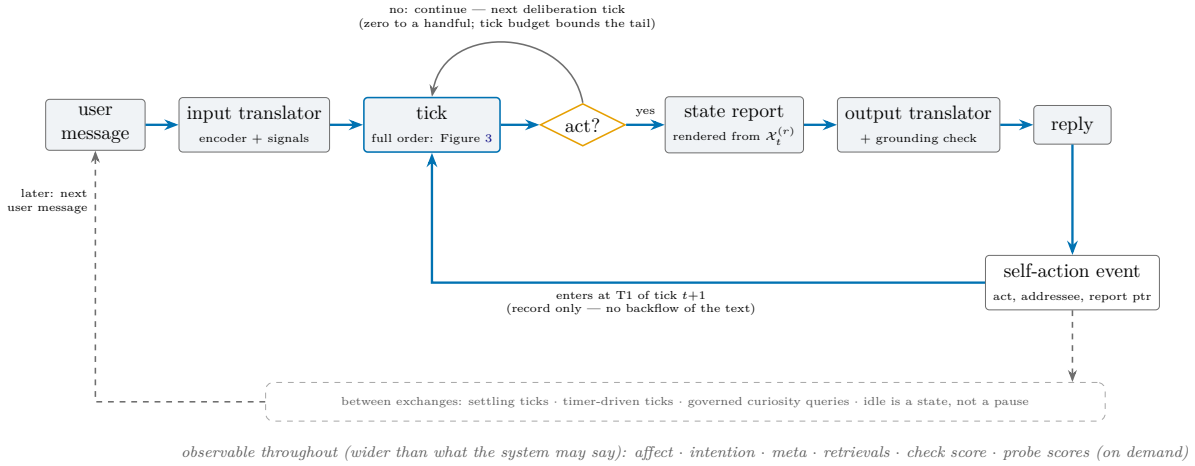


Figure 17: The interaction loop, at the grain of one exchange; Figure 3 carries the per-tick detail this figure does not repeat. A user message enters as an event tick; on *continue*, the deliberation policy repeats the same tick mechanism with a null event rather than invoking a second one, zero to a handful of times before the budget bounds the tail; on *act*, the typed state renders into a state report that the output translator verbalizes under the grounding check of Section 3.6. Two return paths close the loop, and they are not the same path. The short, solid one is mandatory: the canonical self-action record—act type, addressee, a pointer to the report, never the generated text itself (no backflow)—enters as an ordinary event at T1 of the very next tick. The long, dashed one is indeterminately timed: the user’s next message, which may be preceded by settling ticks, timer-driven ticks, and governed curiosity queries—idle is a state, not a pause. The tick stream is live-readable throughout.

Live observability. The architectural principle that the typed state is readable at any tick pays out at interaction time: an operator console can render the inner trajectory *live*—the affect segment as a continuously updating profile, the intention distribution with its per-tick shifts, the meta gauges, memory-retrieval events, and the per-reply grounding-check score. The same probe battery of Section 10.6 can in principle run against a live session’s tick stream, turning a sustained exchange into an evaluation opportunity—subject to the measurement cautions of Section 9: read out during input phases rather than neutral settling, respect item-level units (M5), and treat live readings as approximations until verified against the certified evaluation path (train–serve parity, Section 8.8). What the loop excludes is self-initiated user address. When a system may speak unprompted is a governance question, not an engineering one, and it is deferred to the governance discussion (Section 12.9).

10.9 What the sketch argues—and what it cannot establish

The sketch argues three things: that a prototype core has plausible parameter-memory and computational-complexity bounds for a single GPU, without a throughput or repeat-run affordability claim; that its theoretical requirements translate into concrete, checkable engineering choices (gradient-visibility arithmetic, schedule normalization, baseline calibration, watchdog thresholds); and that its evaluation theory defines the requirements and the construction space for a workable instrument.

The third claim is limited by M2. Section 10.6 leaves the probe list, the weights, the robustness formula, the aggregation window, and the raw-to-score mapping open, and an aggregate whose composition is open is not one instrument but a family of them—two groups could build suites that share nothing but the section they cite and compare their numbers in good faith. What would close the gap is a *versioned manifest* fixing probe identifiers, raw-to-score maps, weights,

hard gates, the robustness construction, the aggregation window, and the verdict thresholds; until one exists, this paper specifies what such an instrument must satisfy and not which instrument it is. It establishes nothing about performance: whether an instantiation in these ranges sustains coherent trajectories, develops content-selective memory, shows a measurable benefit from additional deliberation ticks over matched-compute controls, or climbs the understanding stages is precisely the subject of Section 11. An executable interaction loop likewise does not by itself establish coherent or practically useful interaction.

11 The Research Program: Seventeen Hypotheses

The framework of the preceding sections is falsifiable, and this section says how. Each hypothesis below names its rationale (the section that motivates it) and an operationalization sketch; all are formulated against the measurement machinery of Section 9—noise floors established first, positive controls in place, item-level statistics, criteria fixed in advance. All seventeen are slated for staged empirical testing, though not all are ready for it today. For the fourteen non-provisional entries, “bidirectional” is a study-specific design obligation, not a claim that this paper supplies one universal executable suite: before a run, the study versions the estimand, instrument control, sensitivity condition, decision rule, and negative criterion that instantiate the sketch. Numbering is stable across drafts and therefore append-only; the ordering below is thematic, not numeric.

H8, H9, and H13 are *provisional* because a quantity they require is not yet represented by a defined estimand: H8 and H13 rest on *content share* and *effective dimensionality*, H9 on *novelty sensitivity*, and none of the three has a versioned definition or a marker with an instrument-positive control. Such a hypothesis can be argued for but is not yet eligible for that study-specific bidirectional test, which is the property the rest of this catalog is designed to support. Confirmatory testing of a provisional hypothesis begins only once its estimand is defined, its negative criterion fixed in advance, and—where the quantity is measured rather than computed—an instrument-positive control specified.

A scoping note that Section 3.8 states in full, and it separates three groups rather than two. Some hypotheses range over conforming systems generally (H1, H2, H9, and H17a); the paired H17b arm ranges over a predeclared non-conforming comparison variant. Two are not architecture-specific at all: H12 and H13 are claims about measuring stateful, stochastic systems, they answer to the measurement discipline D2 rather than to the conformance register, and a result on either bears on any program that uses composite metrics on a non-deterministic stack. The rest depend on declarable choices and therefore test a reported configuration rather than the framework—H3 on the numeric target map, H4 on the write path, H15 on the halting inputs, H10 on the governance package as a whole. Results for the third group carry their declarations or they carry nothing.

An empirical program already exists. **This catalog is therefore not a preregistration:** some hypotheses were formulated after measurements that bear on them, whereas preregistration fixes criteria before the relevant results are known. The catalog is instead a *prospective public specification before result disclosure*. It fixes statements and operationalizations before corresponding results are disclosed and requires every hypothesis—negative halves included—to be reported against those formulations. Each companion result must state whether its hypothesis, thresholds, and analysis plan were fixed before data collection, before analysis, or only before publication.

Core dynamics.

H1 (State coherence). A tick-based recurrent core with a segmented, multi-timescale state can sustain non-trivial, non-collapsing state trajectories over horizons far exceeding its

training-sequence length, under bounded input noise. *Rationale:* Section 3.3. *Test:* long-horizon autoregressive rollouts; collapse and chaos detectors (pairwise-similarity and variance criteria); trajectory statistics against degenerate baselines.

H2 (Timescale separation). Two claims of different standing, tested separately. (*2a, mechanism check.*) Enforced fast/core/slow separation yields measurably distinct dynamics—variance and drift-rate orderings across the three carriers. This is largely constructive: the update rules impose the ordering, so 2a verifies that the construction holds in a trained system rather than predicting a discovery, and a failure indicates the separation has collapsed. (*2b, the empirical claim.*) That separation improves long-horizon stability over a single-timescale control of matched capacity. *Rationale:* Section 3.7. *Test:* per-carrier trajectory statistics for 2a; ablation against a collapsed-timescale variant, with stability measured over rollouts far exceeding training length, for 2b.

Affect and interpretation.

H3 (Affect dynamics). Structured affect with consistency constraints and an affect–intention prior produces contextually plausible reaction patterns across scenario classes (threat, success, loss, ambiguity) without per-scenario rules—including trajectory-level robustness: recovery and re-stabilization after negative states, as installed by contrast-bearing training distributions. *Rationale:* Sections 4 and 7.1. *Test:* reaction-profile separation across scenario classes; recovery-trajectory statistics after adverse events. *Admissibility condition—this hypothesis is void without it:* the confirmatory evaluation must exclude target generators that map a scenario class directly onto an affect trajectory, the template construction of Section 7.2 among them. Such a generator makes the hypothesis unfalsifiable in the least visible way: the model reproduces the target generator, and the result is reported as architecture-produced plausibility. Generating the numbers locally rather than with a language model does not help—it relocates the circularity into the generator instead of removing it. Confirmatory scenario families are therefore held out *at the level of generator rules*, not merely of items, and plausibility is anchored to independent human ratings or to a consequential external outcome rather than to the target scheme that produced the training data.

H4 (Competitive selection and disjointness). Under matched capacity, workspace integration with competitive selection and disjoint output responsibility retains specialist diversity, while soft averaging over shared outputs collapses it. The two interventions are separable and must be separated: the hypothesis is tested as a 2×2 design over $\{\text{competition on/off}\} \times \{\text{disjoint outputs on/off}\}$, which distinguishes a main effect of either from their interaction. The framework’s own expectation is that disjointness carries most of the effect and that competition without it degrades toward averaging—but that ordering is the open question, not an assumption. The write path is part of the design declaration: a study reports whether it ran the broadcast-only or the two-phase form (Section 5.1), since the factorial cells mean different things in each. *Rationale:* Sections 5.2 and 5.3. *Test:* specialist output similarity over training in all four cells; downstream differentiation measures; alternative diversity-preserving mechanisms (load balancing, orthogonality penalties, expert-specific losses) as additional comparison arms where capacity permits.

H5 (Supervised interpretation). Supervised, derived situation- interpretation signals (appraisal-style auxiliary supervision feeding a conditioning context) improve broad competence reliably at small scale, whereas identical conditioning without supervision does not, and may even hurt. The sharper half concerns what the head reads: a supervised head denied the typed signal channels is predicted to fall below the no-interpretation control, not merely below its well-fed twin (starved interpretation). *Rationale:* Section 7.4.

Test: single-variable comparison of supervised, unsupervised, and absent interpretation channels on the full evaluation suite, plus a fourth arm that varies only the head’s input source; report the head’s own fit and the variance of its conditioning output in every arm.

Memory.

H6 (Task pressure). *In the absence of competing incentives to retrieve*—no explicit retrieval supervision, no severe recurrent-capacity limit, no interference pressure, no precision requirements the recurrent state cannot hold, no regularization that penalizes carrying content in the recurrent state—content-selective memory use emerges only when task-relevant information lies beyond the effective recurrent horizon; with short-horizon tasks under those controls, the memory pathway stays functionally inert regardless of mechanism health. *Rationale:* Section 6.3, which lists the competing incentives this hypothesis controls for—an unqualified “only” would be refuted by the framework’s own text. *Test:* recall accuracy as a function of information–use distance, with mechanism health verified in isolation and each competing incentive held at a declared, reported level.

H7 (Representation learnability). Episodic content channels require gradient visibility (LC4) and on-manifold placement (LC5); violating either leaves recall at chance even under large upstream signal amplification. *Rationale:* Section 6.4. *Test:* recall under systematic variation of channel loss share and target scale.

Learning targets.

H8 (Target degeneracy)—*provisional*. Latent supervision targets violating LC1 or LC2 produce degenerate solutions—identity solutions under LC1, conditional-mean collapse under LC2; in particular, enriching a target without adding input-side information *reduces* representational content. Generator closure is deliberately *not* part of this prediction: it limits what a result may be taken as evidence for, and does not by itself make the learned representation degenerate (Section 7.3). *Rationale:* Section 7.3. *Test:* content share and effective dimensionality of the learned state under controlled target substitutions; input→target probes and reachability estimates, each a probe-relative lower bound and each stated as design-time or realized (Section 7.3). *Scope note:* LC3 is deliberately absent from this hypothesis. It is a reporting requirement, and omitting a persistence baseline changes no target, no gradient, and no learned system—it can only make inertia look like predictive skill. Its violation therefore invalidates a *claim*, and there is nothing in it to manipulate experimentally; treating it as a cause would be a category error.

H14 (Grounding). Simulator-grounded observation streams contribute target variance unreachable from text-only streams and thereby change what is learnable, at the cost of lower reachable fractions of their own; text-only paradigms meet LC2 limits earlier than mixed ones. *Rationale:* Section 7.8. *Test:* reachability estimates ($\hat{R}^2$, probe-relative, reported separately design-time and realized per Section 7.3) and persistence baselines per stream type; learnability comparisons across stream mixtures.

Governance.

H10 (Governed growth). Memory-first, gated consolidation with the no-self-sealing rule and provenance quotas maintain the seed system’s stability invariants (parity, anchors, canaries) over many growth cycles *while the system acquires content it did not previously have*. Both endpoints are required, and reporting one without the other is uninformative: a governor that blocks every change satisfies the stability endpoint trivially, so *stability preservation* must be paired with *verified competence gain*—new content demonstrably

present after a cycle and absent before it. The negative half worth naming is exactly this degenerate solution: governance that preserves the envelope by preventing growth. Because the intervention is a package (memory-first, gates, provenance, no-self-sealing), a failure localizes to the package, not to a component; component attribution requires the ablation arms listed below. *Rationale:* Section 8. *Test:* long multi-cycle growth operation with per-cycle invariant checks, intake coverage accounting, and a per-cycle novelty ledger (what entered that was not already held); single-component ablations where the cycle budget allows.

H11 (Criterion reachability). Watchdog criteria validated only by negative and injection-based positive controls can be structurally unreachable in operation; requiring reachability proofs changes governance outcomes (criteria get redesigned or retired). Additionally, criteria whose statistics vary with population size fire falsely or mask real faults unless validated against N-matched random baselines. *Rationale:* Section 8.5. *Test:* audit of deployed criteria for threshold-margin traversal of the guarded quantities under fault-realistic operation—not variation alone, which a structurally unreachable criterion also shows; N-sweep validation of population-sensitive statistics.

Measurement.

H12 (Noise floor). Composite experience metrics on non-deterministic training stacks have same-configuration noise floors large enough to bound claimable effect sizes, and fine-grained components are not reproducible from single runs: identical configurations can realize the same aggregate through different component mixtures. *Rationale:* Section 9.2, M1. *Test:* repeated identical-configuration runs; component-level agreement analysis.

H13 (Aggregate visibility)—*provisional*. Aggregate metric design determines whether representational content is visible at all: scenario-scored experience aggregates can be insensitive to large changes in the state’s content share, so content-sensitive axes must be measured separately. *Rationale:* Section 9.2, M2/M3. *Test:* joint measurement of experience aggregates and content-share statistics across models that differ widely in the latter.

Development.

H9 (Partial stage ordering)—*provisional*. Novelty sensitivity—a metacognitive-channel effect—emerges before contradiction competence; ascent to later stages is gated by target richness (LC1, LC2) and encoder unfreezing, not by parameter count alone. The hypothesis is deliberately partial: ordering relative to *transfer* is left open, because no transfer marker with an instrument-positive control exists (Section 9.3), and a predicted ordering over a quantity the instrument set cannot measure would be the kind of unredeemed testability promise this catalog exists to avoid. Stating the transfer half becomes admissible when the marker does. *Rationale:* Section 9.3. *Test:* the marker battery of Section 9.4 with structurally equivalent positive controls per marker; stage placement against target-richness manipulations.

State exposure.

H17 (Exposed-state sufficiency; two predeclared arms). *Conforming arm (H17a).* Under principle A9, every non-operator downstream route is mediated by the regime-specific declared exposed state $\mathcal{X}_t^{(r)}$. Regimes without a persistent hidden carrier satisfy route closure by construction. In regime (iii), B.13 reports attribution between the typed core X_t and the declared carrier h_t within $\mathcal{X}_t^{(r)}$; in regime (iv), the prediction is that the

typed core dominates the constrained, undeclared carrier as the determinant of next state, retrieval, halting, and report. *Rationale:* Section 3.4. *Test:* marker B.13 (Section B); symmetric clamp-and-perturb interventions with per-carrier normalization, interpreted under the declared regime. *Unconstrained-carrier comparison arm (H17b).* In the explicitly non-conforming comparison variant that keeps a persistent carrier outside $\mathcal{X}_t^{(r)}$, permits downstream routing through it, and applies no pressure against it, the carrier is predicted to accumulate task-relevant content and dominate at least one downstream channel even while supervised readouts remain accurate. This is a negative self-test of the framework boundary, not a hypothesis quantified over conforming instantiations. *Test:* the same B.13 pipeline and dominance margins, with the variant labeled outside R4 before the run.

Deliberation and anticipation.

H15 (Deliberation dividend). Under a trained deliberation policy, consumed processing depth varies with input demand and *pays*: variable-depth processing outperforms fixed-depth controls of matched average compute on outcome quality. Depth here means inter-tick depth, counted in ticks; intra-tick refinement steps are a separate quantity and are reported separately, since the matched-compute comparison changes its answer if a refinement step is priced as a tick (B.12). The sharp negative half: absent demand-graded training pressure, the halting inputs saturate and the policy degenerates to constant depth—the reachability failure of Section 8.5 in miniature—deliberation machinery without deliberation. *Rationale:* Section 5.5. *Test:* marker B.12 (Section B)—reachability proof of the halting inputs first, then depth–demand correlation and matched-compute outcome comparisons.

H16 (Anticipatory shifts). A system trained on consequence-bearing sequences shows the S3 signature: announced consequences shift the state before the event, measurably and direction-appropriately, relative to matched unannounced controls. Text-only training bounds the effect (per H14’s reachability logic); consequence-bearing grounded streams should strengthen it. *Rationale:* Sections 7.8 and 9.3. *Test:* marker B.11 (Section B); announcement batteries with valence-matched phrasing, unannounced controls, and the echo-baseline comparison of that marker.

A closing remark on posture. Several of these hypotheses have a sharp negative half: H6 predicts inertness, H8 predicts impoverishment, H11 predicts unreachable safeguards, H12 bounds what can be claimed at all, H15 names the degenerate outcome in its own statement, and H17b predicts that the framework’s typed-core claim fails under an unconstrained-carrier comparison. The program treats those halves as first-class outcomes—the evaluation theory of Section 9 exists so that null and negative results are reportable with the same rigor as positive ones—and the program’s staged reporting is designed accordingly.

12 Discussion and Limitations

12.1 Claim boundary and aggregation policy

The non-claims of Section 1 remain in force. The evidence markers of Section 9.4 form a profile rather than a consciousness aggregate: no arithmetic over structural indicators produces a phenomenal claim, because summing a profile changes its units, not its subject. This standing prohibition is distinct from M2; aggregate uniqueness would require a consciousness score to be named and versioned, whereas this framework defines no such score. The same boundary applies to biological analogy: where the framework departs from established accounts of biological cognition for engineering reasons, the neighboring science supplies no support for that departure.

What the framework does claim is narrower and harder than what it disclaims: that persistent, inspectable inner state admits precise specification, and that the conditions under which it can

be built from differentiable parts, grown under governance, and *measured honestly* are statable in advance.

12.2 Principal risks of the framework itself

Three risks are structural to the approach, not incidental to any implementation.

Generator circularity. Training data for inner-state trajectories must largely be generated, and the generators are language models with their own conventions about affect and inner speech. The mitigations of Section 7.2—the gold-standard set, grounded streams, schema constraints—bound but do not eliminate the risk that a system built on this framework learns a *portrait of interiority* painted by its data generator. The residual risk is real, and the framework’s position is that only grounded, non-linguistic experience streams (Section 7.8) shrink it fundamentally.

Evaluation self-reference. The framework evaluates inner-state systems with instruments designed by the same program that builds them—scenario batteries, markers, aggregates. M1–M7 exist precisely because this loop invites self-deception, and the marker discipline’s three-way verdict makes instrument failure a reportable outcome rather than a silent one. But internal requirements reduce the risk; they cannot supply external reference, which is why M8 states the anchoring obligations—frozen public instruments, independent probe authorship, human reference with reported reliability, consequential external tasks, independent replication—as requirements rather than aspirations. It is the requirement this paper is furthest from satisfying: nothing here has been externally anchored, and until it is, results obtained under M1–M7 should be read as internally disciplined rather than externally valid.

The open ceiling of the paradigm. It is an open question whether S4 content understanding (Section 9.3) is reachable inside this paradigm at any scale: whether latent targets satisfying LC1–LC5, grounding, and encoder unfreezing jointly suffice, or whether mechanisms outside the framework are required. The hypothesis catalog is designed so that this question is answered by measurement rather than by argument. The learnability conditions, governance model, and evaluation theory bear on latent-state systems generally, whatever the ceiling turns out to be.

12.3 Open problems and anticipated difficulties

Beyond the structural risks above, the framework’s own theory predicts a set of difficulties that any instantiation should expect to meet. We collect them here with each derived from the framework’s structure and paired with an available handle or marked open. Each entry states what failure would look like, and the hypothesis catalog (Section 11) binds the program to report those outcomes if they occur.

O1 — Multi-objective antagonism. *Difficulty:* convergence-flavored and differentiation-flavored desiderata read the same core state (Section 5.3); improving one axis in isolation is locally zero-sum against its antagonists, and multi-objective training on a shared carrier should be expected to settle into compromises. *Handle:* of the three devices available here, only output-disjointness acts on the mechanism, since it is the one that removes the shared write path the tension runs through; segment typing and capacity limits localize it without removing it—typing fixes what a coordinate *means* rather than who writes it, and a capacity bound changes how much contention there is rather than its geometry. Robust aggregation (Section 9.1) keeps single-axis peaks from masquerading as progress. *Open:* whether any architectural device dissolves—rather than manages—the tension.

Measurement caveat. Establishing that a particular antagonism exists is harder than it looks, and two traps are worth naming because they push in opposite directions. The first is the *floor*

effect: bounded scores can sit at their floor, and two quantities that are both pinned there cannot be shown to trade off—their correlation is a statement about the measurement range, not about the objectives. The guard is to report, before any correlation, the fraction of observations in which *both* quantities are off their floor, and to restrict the estimate to those; the surviving sample can be a small minority of the runs, which is itself the finding. The second is aggregation across axes: an antagonism estimated between two *groups* of scores averages over pairs that may behave quite differently, and a group-level correlation near zero is compatible with one strongly antagonistic pair inside it. Antagonism claims should therefore be made per pair, with the group-level number reported as a summary rather than as the evidence. Both traps also affect the negative form of the claim, which is the direction an architecture program is most tempted to trust: “we found no trade-off here” is not a result until the floors and the pairs have been checked.

O2 — Mode multiplicity. *Difficulty*: a small stochastic system trained under many interacting objectives has multiple solution modes; because aggregates are not injective (M3), runs of identical configuration can realize similar aggregate quality through different component trade-offs—different modes behind the same number (Section 9.2, M1). Fine-grained claims from single runs are therefore structurally unreliable. *Handle*: noise-floor measurement, multi-run agreement for component claims, H12. *Open*: whether mode selection can be controlled rather than merely measured.

O3 — The supervision paradox. *Difficulty*: supervised interpretation is the framework’s argued lever at small scale (Section 7.4, H5), but whatever is supervised is by construction not emergent; the interpretive scaffold carries the pipeline’s theory of situations. *Handle*: supervise the scaffold, not the content; keep the scaffold small and auditable. *Open*: the right extent of the scaffold—and whether it can be faded as scale grows without the distinctions collapsing.

O4 — The segmentation-ontology bet. *Difficulty*: typed segments presuppose a factorization of inner-state content that cognition may not have (Section 3.2); latent slots restore freedom but escape inspection. The bet has a second face, treated as its own problem in Section 3.4: the tick operator’s internal carrier h_t (Section 3.3) is untyped, and least-effort optimization can route content through it rather than through the typed readout. Principle A9 and marker B.13 narrow and measure that risk; what they do not do is remove it in regimes that keep an undeclared persistent carrier. *Handle*: the bet is explicit; latent capacity is declared, not hidden; evaluation reads the typed portion as specified; supervision and evaluation pressure on the typed state are the counterweight to the h_t loophole, their sufficiency an empirical question rather than a guarantee. *Open*: the readability–richness balance, and whether learned content respects the declared boundaries at all.

O5 — Slow-channel vacuity. *Difficulty*: low-frequency carriers are learning-starved by construction (Section 3.7); dispositional and self-descriptive channels risk becoming typed, present, and empty—the state-level analogue of vacuous meta variables. *Handle*: curricular pressure that forces dispositional variation—deferred, by the framework’s own staging recommendation, until the profile library activates (Sections 3.3 and 10.5), so early instantiations should expect this channel to be quiet and report that as staged rather than as measured; and the reachability discipline (Section 8.5) applied to every monitor of a slow channel, so an empty channel cannot hide behind a green watchdog. *Open*: how much genuine dispositional dynamics small-scale training can induce at all.

O6 — Tick-semantics under-determination. *Difficulty*: the tick is defined functionally (one state update), but its relation to interaction time and world time, and the scaling of tick

Position	Standing	Basis
Conditional-mean collapse under unreachable targets (LC2)	committed	MSE optimality argument
Identity solution under trivial targets (LC1)	committed	optimal-solution argument
Degenerate-key pathology in cosine stores	committed	mathematical property
Persistence baselines required for predictive claims (LC3)	committed	time-series argument
Noise floors bound claimable effects (M1)	committed	measurement logic
Competitive selection over soft averaging	argued	Goyal et al. [38]; H4
Output-disjointness preserves specialization	argued	gradient argument; H4
Supervised interpretation as small-scale lever	argued	Caruana [13]; H5
Task pressure required for memory use	argued	shortcut-learning argument; H6
Exposed-state sufficiency under A9	argued	construction constraint; H17
Contrast-installed stability of affect dynamics	open	conjecture; H3
Multi-timescale benefit over collapsed timescales	argued	H2b
Governed growth preserves stability envelope	argued	construction-level invariants; H10
Stage ascent gated by levers, not raw scale	argued	H9/H16; Section 12.6
Stage-S4 reachability within the paradigm	open	Section 12.2
Right extent of the interpretive scaffold (O3)	open	Section 12.3
Phenomenal experience	outside scope	non-claims (Section 1)

Table 4: Epistemic standing of the framework’s core positions.

budgets with task demands, are not theoretically fixed (Section 3.7). *Handle*: the deliberation policy makes depth adaptive without requiring the semantics to be settled. *Open*: a principled account of tick granularity relative to world and interaction time—arguably the framework’s least developed corner.

O7 — Governance conservatism. *Difficulty*: a confluence-gated, block-only governor is biased toward stasis (Section 8.4); safety instrumentation compounds into a system that never learns. *Handle*: intake coverage accounting makes the stasis end visible; strictness is treated as a tunable with failure modes named on both ends. *Open*: principled placement of the operating point.

O8 — Theoretical contingency of the affect ontology. *Difficulty*: the dimensional–categorical hybrid takes a working position in an unsettled theoretical debate (Section 4.1); constructionist accounts [6] would reject the categorical layer’s naturalness. *Handle*: the hybrid is modular—a constructionist replacement (appraisal-derived patterns over the dimensional core) would leave the architecture standing. *Open*: which affect ontology best serves artificial inner worlds is an empirical question this framework can host but not settle.

12.4 What is claimed, argued, and left open

Finally, the paper’s own epistemic bookkeeping. Table 4 classifies the framework’s core positions by their standing—the summary a reader in a hurry should read first, and the one Section 1 points at. *Committed* positions rest on mathematical or construction-level derivations, or on measurement logic that holds independently of any particular system—they are refutable by finding an error in the derivation or its premises, not by any single system’s behavior; *argued* positions rest on structural reasoning and cited literature but await the empirical program; *open* questions are exactly that. The hypothesis catalog (Section 11) operationalizes the argued tier.

12.5 The composition as a choice

The table above records the epistemic standing of the framework’s positions. A separate design question is where the framework could have taken another form. Five commitments carry most of the structure, and each had a live alternative.

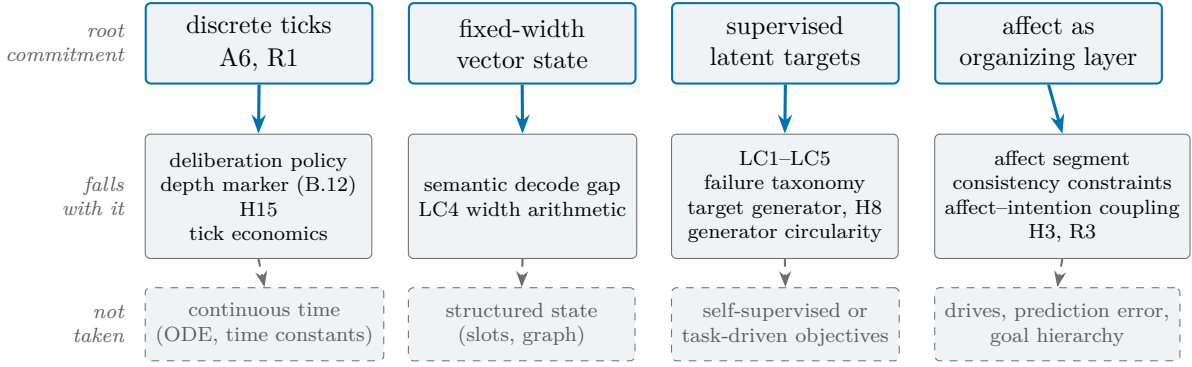
Discrete ticks (A6). The alternative is continuous-time dynamics, where the state evolves under a differential equation and distinct time constants replace the update-frequency partition of Section 3.7. That alternative would obtain the multi-timescale separation as a parameter family rather than as a construction, and it would dissolve the tick-semantics question of O6 instead of leaving it open. We chose discrete ticks because processing depth then becomes a *countable* quantity—the basis of the deliberation policy, of B.12, and of H15—and a measurable depth seems to us worth more than a cleaner temporal formalism. The trade remains open.

A fixed-width vector state (Equation (1)). The alternative is a structured state: typed slots, a graph, a set of propositions. It would make the semantic decode gap of Section 3.6 disappear, since structured content is directly readable, and it would remove the width arithmetic LC4 has to contend with. It would trade fixed-vector simplicity for markedly more complex differentiable interfaces—soft slots, differentiable graphs and relational tensors are all differentiable in principle, so the choice is a trade in interface complexity and inductive bias rather than the end of differentiability. Nor does structure buy readability automatically: a learned slot is as opaque as a learned coordinate unless something anchors it. We took the vector because the state is meant to be *learned* and the interface stays simple, and paid for it with the decode gap.

Supervised latent targets (Section 7). This is the deepest of the four, because the learnability chapter exists only under it. Section 2.3 already notes that predictive and task-driven objectives obtain target well-posedness for free; LC1–LC5 are the price of not using them. We pay it because those objectives shape a state without giving it roles, and a state without declared roles cannot be the object of study this framework is about—it would hand the inspectability question back to Section 3.4 unanswered. One consequence should be stated for readers who choose otherwise: pursued with self-supervised objectives, LC1–LC5 are *inapplicable* rather than refuted. They are conditional on coordinate-wise supervision, and that condition is part of their statement—which is why D1 (Section 3.8) carries the alternative explicitly rather than excluding it. Of the five commitments on this page, this is the one that is not a departure: a system taking the other route conforms, and declares the regime.

Affect as the organizing layer (Section 4). Homeostatic drives, prediction error, or a goal hierarchy could each occupy that position, and each has an active literature. What is notable is how little else would change: the learnability conditions, the governance model, and the evaluation theory are indifferent to which quantity organizes the core. O8 records that the affect *ontology* is theoretically contingent; this entry records that the affect *axis* is a choice as well. It is the commitment we would expect a successor framework to revisit first, and Section 3.8 provides the form such a successor should use to say so: name the requirement dropped, say what plays its role, and inherit the disciplines rather than the hypotheses.

A single trainable core (A1, R7). The alternative is a federation: several recurrent cores, each responsible for a group of segments, coupled through a declared interface. It would give each objective family its own gradient budget and would attack the shared-carrier tension of O1 at its root rather than localizing it. We took the single core for two reasons, and neither is a derivation. The first is that the exposed-state contract is checkable at one place only as long as there is one place: under a federation, a core that reads another core’s state is either a consumer in the sense of R4—and then the coupling must enter Figure 4 as a channel with permissions harder to state than any existing row—or it is not, and the notion of an internal carrier needs redrawing. The second is that a coupling channel is also a mechanism for resolving tension by transfer rather than by capacity: a core can relieve pressure on its own objectives by drawing on what another core maintains, and an aggregate that sums over both will not show the difference. Neither reason forecloses the option, and both are engineering judgements rather than results.



Indifferent to all four: the governance model and the evaluation theory. Everything not shown is either derived from one of these commitments or declared free under conformance.

Figure 18: Four of the five commitments of Section 12.5 and what depends on them. Solid: framework content that would have to be rebuilt if the commitment were dropped. Dashed grey: the alternative not taken. The fifth—a single trainable core against a federation—is absent because its dependent column would be nearly empty, which is the point made there rather than an omission. The figure deliberately shows only *root* commitments; everything else is either derived from one of these or declared free in Section 3.8.

Of the five commitments on this page, this is the one that would change the least elsewhere in the paper—R7 and the single-carrier form of the exposure check are close to the whole of it—which is why it does not appear in Figure 18 beside the four. That a commitment carries little is not an argument that it is right.

The remaining commitments are either argued where they arise—language at the periphery against language as the state (Section 2.7), state trajectories against task benchmarks (M8)—or declared free in Section 3.8. Figure 18 shows which parts of the framework depend on which commitment, which is also the picture of what this paper is: a composition, not a derivation.

12.6 Relation to scaling-first AI, and a conditional view of emergence

The framework is complementary to, not competitive with, the scaling paradigm. Scaling asks what emerges when capability is pushed; this program asks what organization is required for a persistent, inspectable inner state. A plausible reading of the division of labor: scaling-first systems supply the peripheral competence (encoding, verbalization, data generation) that this framework deliberately outsources, while the framework supplies what scaling does not automatically produce—typed persistent state, governed growth, and an evaluation theory for persistent inner state.

On emergence itself the framework takes a *conditional* position rather than an expectation. In that restricted sense, emergence remains an admissible outcome, but the label alone does not explain the transition; the relevant conditions must be varied and tested. The ascent through the understanding stages of Section 9.3 is predicted to be gated by specific, nameable levers—the learnability of the latent targets (LC1, LC2), the richness the data pipeline actually supplies, encoder unfreezing (Section 7.8), and consequence-bearing grounded streams (H14)—and *not* by raw scale: H9 predicts that parameter count alone does not lift a system to later stages, and Section 9.3 states the same commitment in its sharpest form: a system whose targets violate the learnability conditions is predicted to saturate early stages with none of the named levers available to lift it further, raw scale included. This yields falsifiable predictions in both directions. Scaling data or parameters *without* moving the gates should produce flat stage profiles—at most more polish on the aggregates, no ascent. Moving the gates at modest scale—richer reachable targets, a co-adapting encoder, grounded streams—should move the stage markers (H16; and,

where the pipeline’s richness includes demand-graded pressure, the deliberation marker of H15) even where raw scale does not. And if stage ascent were observed under pure scaling with fixed gates, the gating thesis itself would be falsified—a result the evaluation theory is built to detect rather than obscure. Both predictions are decided by the markers at the scales actually run (M7), and the levers named above are the thesis as stated: “not scaled enough” and “a gate we did not name” are not admissible post-hoc verdicts. A gate-moving experiment that fails to move the stage markers counts against the gating thesis, not for a longer list of gates.

12.7 Long-horizon questions and conceptual boundaries

The questions in this subsection are implications and open research directions, not additional conformance claims. A persistent state supplies candidate carriers for history, interventions, and between-instance comparisons, but the framework does not answer the questions by naming those carriers. A negative result is interpretable only after the relevant reachability checks, noise floors, and instrument-positive controls pass; otherwise the verdict is not measurable rather than negative.

History-dependent differentiation. The framework contains no theory of personal identity in the philosophical sense, and none is required for conformance. It does, however, make a weaker question precise. A conforming system carries forward a slow dispositional state, typed memory records, provenance-linked consolidation, and canonical records of its own actions (Sections 3.6, 6.1 and 8). Two initially matched instances exposed to different controlled histories may therefore cease to be functionally interchangeable even when their architecture and initial parameters are the same. The target is persistent divergence under a later common battery in declared carriers such as disposition, retrieval, state integration, or action tendency. A valid test must distinguish load-bearing history from mere record difference and establish that its analysis can recover a planted divergence before interpreting a null. If those instrument conditions pass, a positive result would establish history dependence at the tested carriers; a null would bound their causal role at that scale.

History-conditioned relevance. A related question is whether a held-fixed event has system-specific consequences when carried histories differ. Memory, slow state, current affect, intention, and prior processing are candidate mediators; their presence alone does not establish causal relevance. The stronger test asks whether controlled interventions on carried history change how the same event is appraised, integrated, remembered, or acted upon, with each intervention restricted to the valid support of the affected carrier and each endpoint to a declared and reachable path. Here “understanding” is used operationally: successful transfer, intervention, and compositional use under declared tests. Causal-abstraction methods offer one route for testing whether internal variables implement a hypothesized higher-level mechanism [35], while the form–meaning distinction cautions that behavioral success on linguistic form alone does not establish meaning [8]. Generator closure (Section 7.2) remains a central confound: history-conditioned differences may reproduce distinctions installed by the data generator rather than organization developed beyond its schema.

Self, world, and causal contact. The present meta segment is intentionally minimal: calibrated monitoring variables are not a rich self-model (Section 5.4). The architecture likewise specifies no comparably explicit persistent world model; environmental information may be carried through semantic state, memory, and grounded streams, and a world-model backbone with a typed inner-state overlay remains a possible hybrid (Section 2.3). This scope leaves open whether richer self-representation can develop in that asymmetric setting, or whether representations of self, other agents, and world dynamics must deepen together. Linguistic

histories, simulator trajectories, sensorimotor interaction, and physical embodiment offer different routes to externally constrained consequences; the framework does not rank them by necessity or sufficiency. H14 assigns grounded streams a specific learnability role; it does not establish that embodiment is necessary or sufficient. The broader empirical question is which forms of development require persistent causal contact with consequences not generated by the same source that supplies the system’s descriptions.

The ceiling and the phenomenal boundary. The open ceiling of Section 12.2 is therefore substantive. Persistent state, typed memory, metacognition, grounding, and governed development may produce increasingly differentiated organization while still failing to support robust transfer or compositional content use. Such a failure counts against the sufficiency of the present composition only under a predeclared, criterion-reachable gate-moving test whose relevant instruments and controls pass; otherwise the conclusion remains underdetermined. A positive result is likewise bounded by the tested carriers, tasks, and scale. Even if every structural marker passed, the evidence would establish the measured functional properties and their causal organization; it would not establish phenomenal experience. What further evidence, if any, could justify a claim about subjective experience is a separate philosophical and scientific problem for which this framework supplies no criterion. The framework leaves that problem outside its scope; its measurements concern the functional organization beneath that boundary.

12.8 Ethics note

Systems that present affective behavior invite affective attribution. A deployed inner-state system—precisely because its reactions are persistent, contextual, and consistent—should be expected to be read by users as feeling, regardless of the non-claims in its documentation. Deployments of systems built on this framework should disclose their functional nature explicitly, avoid designs that exploit attribution (attachment loops, dependency dynamics), and treat the state reports of Section 3.6 as a transparency instrument for users, not only for developers. Inspectability helps here: a system whose typed state is readable can display the variables its reports are drawn from—subject to the exposure condition of Section 3.4. Most objects of anthropomorphic attribution offer no such display. One limit belongs in the same paragraph, because presenting the report as full transparency would overstate it: the report is deliberately narrower than the system’s processing. Influences that did not reach the core state—a retrieval that changed nothing, a losing workspace proposal—do not appear in it (Section 3.6). That gap is defensible, since the alternative is a system narrating machinery it did not use, but a transparency instrument should be described as what it is: a faithful account of the state the system acted from, not a log of everything that happened inside it.

This first risk category concerns *users*: attachment, dependency, inappropriate reliance, and clinically unsafe responses. Empirical work on AI companions reports associations between interaction patterns and psychological well-being [122], and clinically oriented auditing shows why mental-health interactions require behavior-specific safety tests [118]. These studies do not show that every persistent-state system causes harm; they justify treating companion deployment and mental-health use as separate, evidence-bearing risk surfaces rather than generic interface choices.

Deployments that retain state associated with a user should also specify retention and deletion policies, mechanisms for user control, and the treatment of derived memory entries or parameter updates when a source record is removed (Section 8.2).

A distinct question concerns the *system’s possible moral status*. Functional affect variables, self-reports, persistence, or passed structural markers do not by themselves establish sentience, welfare, interests, or moral patienthood. Conversely, the paper’s phenomenal non-claim is not evidence that moral status is impossible. Responsible AI-consciousness research therefore recommends explicit uncertainty, precaution proportionate to evidence, and avoidance of unsupported

consciousness claims [11]. This framework supplies no moral-status criterion; any future status assessment must be conducted separately from user-safety evaluation and must not treat the architecture’s labels as ontological findings.

12.9 Outlook

Three lines of development follow from the framework, in ascending order of governance weight. First, systems of this kind are *research instruments* before they are anything else: a persistent, inspectable state that can be probed live turns questions about affect dynamics, memory continuity, and deliberation from positions into measurements (Section 10.8). Second, the interaction loop defines a possible instantiation for *sustained interaction*, in which state persists across exchanges and sessions, memory accumulates under provenance, and the state report serves as the user-facing transparency instrument of the ethics note above. Companion-style interfaces are one natural embodiment of that setting—and would inherit every governance gate of Section 8 unchanged, simulation-first discipline included. Third, and deliberately last: whether and when such a system may act *unprompted*—initiate contact, pursue a curiosity query beyond its standing limits, speak when its own state crosses a report-worthiness threshold—is a governance question, not an engineering one. The framework’s position is that it must be answered by the same gate machinery that governs growth (Section 8.4; exposure doctrine: Section 8.9), with silence as the default and every opened surface carrying the full gate set—canaries, hold-outs, the gold-standard set, train-serve parity—plus a reachability-proofed criterion of its own that can close the surface again.

A concrete long-horizon study would implement the matched-instance design of Section 12.7: hold architecture, parameters, initial state, and initial memory fixed; vary only controlled histories; then apply a shared battery with complete lineage for interventions and consolidation. Predeclared swap, ablation, restoration, and delayed-retest arms, together with checks that each intervention remains within the valid support of the affected carrier, reachability checks, and instrument-positive controls, would test whether any later divergence is persistent and attributable to carried history rather than transient context or store contents alone. The framework supplies candidate carriers and intervention points for this test, not a guarantee that the divergence is measurable or present. A valid positive result would establish history dependence at the tested carriers, not personhood.

Behind all three strands stands the long horizon of Section 1: a system whose functional inner state deepens across growth cycles—affect that stays coherent over longer arcs, deliberation that pays at greater depths, memory that carries history across sessions, metacognition that tracks more of what the system is doing. Nothing about that horizon is asserted here; the point of the framework is that each step toward it must be earned through the same gates and instruments as the first. The companion papers will report the reference implementation and the empirical program against the hypothesis catalog of Section 11, negative halves included.

13 Conclusion

We have presented TASA, a theoretical framework for persistent, inspectable inner state. Its tick-based, affect-centric architecture (C1) is paired with learnability conditions for supervised latent targets (C2), governance for continual growth (C3), and an evaluation theory for stateful stochastic systems (C4). A worked instantiation connects these commitments to an end-to-end interaction loop (C5), while seventeen hypotheses with operationalization sketches—fourteen non-provisional and three provisional—define the staged empirical program (C6).

The framework’s wager is the structural stance: that “inner world” can be an engineering term with testable content before it is a philosophical one, and that its claimed organization should be exposed to measurement and perturbation. The long-horizon question is whether persistent state

can become more differentiated, coherent, and history-dependent through governed operation, with each claimed change tied to a predeclared instrument and failure criterion.

Under valid instruments, positive results would support the tested functional properties and their causal organization; negative results would bound the present composition under the tested conditions. Neither verdict addresses subjective experience. Empirical measurement under this program is the next step.

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Code and Data Availability

The material accompanying this paper is available at

<https://github.com/SebastianFoxxx/tasa-framework-paper>.

An archival snapshot with a persistent identifier is provided at Zenodo:

doi:10.5281/zenodo.22283059.

References

- [1] M. Amanlou, P. Abed Azad, F. Davoodi, M. Masumi, B. Bahrak, and A.-H. Vahabie. *PsychoAgent: An Affect-Sensitive Cognitive Architecture for Conflict-Aware Memory in LLM Agents*. Preprint. 2026. DOI: [10.48550/arXiv.2608.07438](https://doi.org/10.48550/arXiv.2608.07438). arXiv: [2608.07438](https://arxiv.org/abs/2608.07438) [cs.AI].
- [2] J. R. Anderson, D. Bothell, M. D. Byrne, S. Douglass, C. Lebiere, and Y. Qin. “An Integrated Theory of the Mind”. In: *Psychological Review* 111.4 (2004), pp. 1036–1060. DOI: [10.1037/0033-295X.111.4.1036](https://doi.org/10.1037/0033-295X.111.4.1036).
- [3] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas. “Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. arXiv:2301.08243. 2023, pp. 15619–15629. DOI: [10.1109/CVPR52729.2023.01500](https://doi.org/10.1109/CVPR52729.2023.01500).
- [4] B. J. Baars. *A Cognitive Theory of Consciousness*. New York, NY: Cambridge University Press, 1988. ISBN: 9780521301336.
- [5] A. Banino, J. Balaguer, and C. Blundell. *PonderNet: Learning to Ponder*. Presented at the 8th ICML Workshop on Automated Machine Learning. 2021. arXiv: [2107.05407](https://arxiv.org/abs/2107.05407) [cs.LG].
- [6] L. F. Barrett. “The theory of constructed emotion: an active inference account of interoception and categorization”. In: *Social Cognitive and Affective Neuroscience* 12.1 (2017). Corrigendum: doi:10.1093/scan/nsx060; cited main claim unchanged, pp. 1–23. DOI: [10.1093/scan/nsw154](https://doi.org/10.1093/scan/nsw154).
- [7] M. Beck, K. Pöppel, M. Spanring, A. Auer, O. Prudnikova, M. Kopp, G. Klambauer, J. Brandstetter, and S. Hochreiter. “xLSTM: Extended Long Short-Term Memory”. In: *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. arXiv:2405.04517. 2024, pp. 107547–107603. DOI: [10.52202/079017-3417](https://doi.org/10.52202/079017-3417).

- [8] E. M. Bender and A. Koller. “Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data”. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 2020, pp. 5185–5198. DOI: [10.18653/v1/2020.acl-main.463](https://doi.org/10.18653/v1/2020.acl-main.463).
- [9] C. M. Bishop. *Mixture Density Networks*. Tech. rep. NCRG/94/004. Aston University, 1994.
- [10] X. Bouthillier et al. “Accounting for Variance in Machine Learning Benchmarks”. In: *Proceedings of Machine Learning and Systems (MLSys)*. Vol. 3. Official MLSys metadata author list; arXiv:2103.03098. 2021. URL: <https://proceedings.mlsys.org/paper/2021/hash/0184b0cd3cfb185989f858a1d9f5c1eb-Abstract.html>.
- [11] P. Butlin and T. Lappas. *Principles for Responsible AI Consciousness Research*. 2025. DOI: [10.48550/arXiv.2501.07290](https://doi.org/10.48550/arXiv.2501.07290). arXiv: [2501.07290](https://arxiv.org/abs/2501.07290).
- [12] P. Butlin et al. *Consciousness in Artificial Intelligence: Insights from the Science of Consciousness*. 2023. arXiv: [2308.08708](https://arxiv.org/abs/2308.08708) [cs.AI]. URL: <https://arxiv.org/abs/2308.08708>.
- [13] R. Caruana. “Multitask Learning”. In: *Machine Learning* 28 (1997), pp. 41–75. DOI: [10.1023/A:1007379606734](https://doi.org/10.1023/A:1007379606734).
- [14] H. A. Chapman and A. K. Anderson. “Things rank and gross in nature: A review and synthesis of moral disgust”. In: *Psychological Bulletin* 139.2 (2013), pp. 300–327. DOI: [10.1037/a0030964](https://doi.org/10.1037/a0030964).
- [15] H. Chateau-Laurent and R. VanRullen. *Learning to Chain Operations by Routing Information Through a Global Workspace*. 2025. arXiv: [2503.01906](https://arxiv.org/abs/2503.01906) [cs.LG]. URL: <https://arxiv.org/abs/2503.01906>.
- [16] X. Chen, C. T. Leong, W. Guo, J. Wang, W. Li, and X. Shen. *What Makes Effective Supervision in Latent Chain-of-Thought: An Information-Theoretic Analysis*. Preprint; accepted at ICML 2026. 2026. DOI: [10.48550/arXiv.2606.20075](https://doi.org/10.48550/arXiv.2606.20075). arXiv: [2606.20075](https://arxiv.org/abs/2606.20075) [cs.LG].
- [17] Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Rabinovich. “GradNorm: Gradient Normalization for Adaptive Loss Balancing in Deep Multitask Networks”. In: *Proceedings of the 35th International Conference on Machine Learning (ICML)*. Vol. 80. Proceedings of Machine Learning Research. arXiv:1711.02257. 2018, pp. 794–803.
- [18] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio. “Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation”. In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. arXiv:1406.1078. Doha, Qatar: Association for Computational Linguistics, 2014, pp. 1724–1734. DOI: [10.3115/v1/D14-1179](https://doi.org/10.3115/v1/D14-1179). URL: <https://aclanthology.org/D14-1179/>.
- [19] J. Chung, S. Ahn, and Y. Bengio. “Hierarchical Multiscale Recurrent Neural Networks”. In: *International Conference on Learning Representations (ICLR)*. arXiv:1609.01704. 2017.
- [20] Cogitate Consortium, O. Ferrante, U. Gorska-Klimowska, S. Henin, R. Hirschhorn, A. Khalaf, et al. “Adversarial testing of global neuronal workspace and integrated information theories of consciousness”. In: *Nature* 642.8066 (2025), pp. 133–142. DOI: [10.1038/s41586-025-08888-1](https://doi.org/10.1038/s41586-025-08888-1).
- [21] A. S. Cowen and D. Keltner. “Self-report captures 27 distinct categories of emotion bridged by continuous gradients”. In: *Proceedings of the National Academy of Sciences* 114.38 (2017), E7900–E7909. DOI: [10.1073/pnas.1702247114](https://doi.org/10.1073/pnas.1702247114).

- [22] S. Dehaene, M. Kerszberg, and J.-P. Changeux. “A neuronal model of a global workspace in effortful cognitive tasks”. In: *Proceedings of the National Academy of Sciences* 95.24 (1998), pp. 14529–14534. DOI: [10.1073/pnas.95.24.14529](https://doi.org/10.1073/pnas.95.24.14529).
- [23] S. Dehaene and L. Naccache. “Towards a cognitive neuroscience of consciousness: basic evidence and a workspace framework”. In: *Cognition* 79.1–2 (2001), pp. 1–37. DOI: [10.1016/S0010-0277\(00\)00123-2](https://doi.org/10.1016/S0010-0277(00)00123-2).
- [24] S. Dehaene, C. Sergent, and J.-P. Changeux. “A neuronal network model linking subjective reports and objective physiological data during conscious perception”. In: *Proceedings of the National Academy of Sciences* 100.14 (2003), pp. 8520–8525. DOI: [10.1073/pnas.1332574100](https://doi.org/10.1073/pnas.1332574100).
- [25] J. Demšar. “Statistical Comparisons of Classifiers over Multiple Data Sets”. In: *Journal of Machine Learning Research* 7 (2006), pp. 1–30. URL: <https://www.jmlr.org/papers/v7/demsar06a.html>.
- [26] T. Ding, A. Nannapaneni, B. Liu, and L. Zhang. *Always-On Agents: A Survey of Persistent Memory, State, and Governance in LLM Agents*. Preprint; survey of 435 coded works. 2026. DOI: [10.48550/arXiv.2606.30306](https://doi.org/10.48550/arXiv.2606.30306). arXiv: [2606.30306](https://arxiv.org/abs/2606.30306) [cs.MA].
- [27] R. F. J. Dossa, K. Arulkumaran, A. Juliani, S. Sasai, and R. Kanai. “Design and evaluation of a global workspace agent embodied in a realistic multimodal environment”. In: *Frontiers in Computational Neuroscience* 18 (2024), p. 1352685. DOI: [10.3389/fncom.2024.1352685](https://doi.org/10.3389/fncom.2024.1352685).
- [28] R. Dror, G. Baumer, S. Shlomov, and R. Reichart. “The Hitchhiker’s Guide to Testing Statistical Significance in Natural Language Processing”. In: *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 2018, pp. 1383–1392. DOI: [10.18653/v1/P18-1128](https://doi.org/10.18653/v1/P18-1128).
- [29] P. Ekman. “An argument for basic emotions”. In: *Cognition and Emotion* 6.3–4 (1992), pp. 169–200. DOI: [10.1080/02699939208411068](https://doi.org/10.1080/02699939208411068).
- [30] S. El Hihi and Y. Bengio. “Hierarchical Recurrent Neural Networks for Long-Term Dependencies”. In: *Advances in Neural Information Processing Systems 8 (NIPS)*. 1995, pp. 493–499.
- [31] S. M. Fleming. “Awareness as inference in a higher-order state space”. In: *Neuroscience of Consciousness* 2020.1 (2020), niz020. DOI: [10.1093/nc/niz020](https://doi.org/10.1093/nc/niz020).
- [32] S. M. Fleming and H. C. Lau. “How to measure metacognition”. In: *Frontiers in Human Neuroscience* 8 (2014), p. 443. DOI: [10.3389/fnhum.2014.00443](https://doi.org/10.3389/fnhum.2014.00443).
- [33] S. Franklin, T. Madl, S. D’Mello, and J. Snider. “LIDA: A Systems-level Architecture for Cognition, Emotion, and Learning”. In: *IEEE Transactions on Autonomous Mental Development* 6.1 (2014), pp. 19–41. DOI: [10.1109/TAMD.2013.2277589](https://doi.org/10.1109/TAMD.2013.2277589).
- [34] K. Friston. “The free-energy principle: a unified brain theory?” In: *Nature Reviews Neuroscience* 11.2 (2010), pp. 127–138. DOI: [10.1038/nrn2787](https://doi.org/10.1038/nrn2787).
- [35] A. Geiger, H. Lu, T. Icard, and C. Potts. “Causal Abstractions of Neural Networks”. In: *Advances in Neural Information Processing Systems*. Vol. 34. 2021, pp. 9574–9586.
- [36] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann. “Shortcut learning in deep neural networks”. In: *Nature Machine Intelligence* 2 (2020), pp. 665–673. DOI: [10.1038/s42256-020-00257-z](https://doi.org/10.1038/s42256-020-00257-z).
- [37] B. Goertzel, C. Pennachin, and N. Geisweiller. *Engineering General Intelligence, Part 2: The CogPrime Architecture for Integrative, Embodied AGI*. Vol. 6. Atlantis Thinking Machines. Atlantis Press, 2014. DOI: [10.2991/978-94-6239-030-0](https://doi.org/10.2991/978-94-6239-030-0).

- [38] A. Goyal, A. Didolkar, A. Lamb, K. Badola, N. R. Ke, N. Rahaman, J. Binas, C. Blundell, M. Mozer, and Y. Bengio. “Coordination Among Neural Modules Through a Shared Global Workspace”. In: *International Conference on Learning Representations (ICLR)*. Oral. arXiv:2103.01197. 2022. URL: <https://arxiv.org/abs/2103.01197>.
- [39] A. Graves et al. “Hybrid Computing Using a Neural Network with Dynamic External Memory”. In: *Nature* 538.7626 (2016), pp. 471–476. DOI: [10.1038/nature20101](https://doi.org/10.1038/nature20101).
- [40] A. Gu and T. Dao. “Mamba: Linear-Time Sequence Modeling with Selective State Spaces”. In: *Proceedings of the Conference on Language Modeling (COLM)*. Preprint arXiv:2312.00752. 2024. URL: <https://openreview.net/forum?id=tEYskw1VY2>.
- [41] A. Gu, K. Goel, and C. Ré. “Efficiently Modeling Long Sequences with Structured State Spaces”. In: *International Conference on Learning Representations*. 2022. URL: <https://openreview.net/forum?id=uYLFoz1v1AC>.
- [42] W. Gurnee et al. *Verbalizable Representations Form a Global Workspace in Language Models*. Preprint. 2026. DOI: [10.48550/arXiv.2607.15495](https://doi.org/10.48550/arXiv.2607.15495). arXiv: [2607.15495](https://arxiv.org/abs/2607.15495) [cs.CL].
- [43] D. Ha and J. Schmidhuber. “Recurrent World Models Facilitate Policy Evolution”. In: *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*. arXiv:1809.01999. Montréal, Canada, 2018, pp. 2455–2467. URL: <https://proceedings.neurips.cc/paper/2018/hash/2de5d16682c3c35007e4e92982f1a2ba-Abstract.html>.
- [44] D. Ha and J. Schmidhuber. “World Models”. In: *arXiv preprint arXiv:1803.10122* (2018). Interactive version: <https://worldmodels.github.io>. URL: <https://arxiv.org/abs/1803.10122>.
- [45] D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, H. Lee, and J. Davidson. “Learning Latent Dynamics for Planning from Pixels”. In: *Proceedings of the 36th International Conference on Machine Learning*. Vol. 97. Proceedings of Machine Learning Research. 2019, pp. 2555–2565. URL: <https://proceedings.mlr.press/v97/hafner19a.html>.
- [46] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. “Mastering Diverse Domains through World Models”. In: *arXiv preprint arXiv:2301.04104* (2023). URL: <https://arxiv.org/abs/2301.04104>.
- [47] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. “Mastering diverse control tasks through world models”. In: *Nature* 640 (2025), pp. 647–653. DOI: [10.1038/s41586-025-08744-2](https://doi.org/10.1038/s41586-025-08744-2).
- [48] S. Hao, S. Sukhbaatar, D. Su, X. Li, Z. Hu, J. Weston, and Y. Tian. “Training Large Language Models to Reason in a Continuous Latent Space”. In: *Proceedings of the Conference on Language Modeling (COLM)*. Preprint arXiv:2412.06769, version 4 (2026). 2025. URL: <https://openreview.net/forum?id=Itxz7S4Ip3>.
- [49] D. Hart and B. Goertzel. “OpenCog: A Software Framework for Integrative Artificial General Intelligence”. In: *Artificial General Intelligence 2008: Proceedings of the First AGI Conference*. Vol. 171. Frontiers in Artificial Intelligence and Applications. IOS Press, 2008, pp. 468–472. DOI: [10.1007/978-3-540-68677-4_46](https://doi.org/10.1007/978-3-540-68677-4_46).
- [50] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, and D. Meger. “Deep Reinforcement Learning That Matters”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 32. 1. arXiv:1709.06560. 2018. DOI: [10.1609/aaai.v32i1.11694](https://doi.org/10.1609/aaai.v32i1.11694).
- [51] S. Hochreiter and J. Schmidhuber. “Long Short-Term Memory”. In: *Neural Computation* 9.8 (1997), pp. 1735–1780. DOI: [10.1162/neco.1997.9.8.1735](https://doi.org/10.1162/neco.1997.9.8.1735).
- [52] N. Hsing. *MIRROR: Converging Cognitive Principles as Computational Mechanisms for AI Reasoning*. Version v3 (April 2026); earlier versions carry different titles. Workshop version: ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems. 2026. DOI: [10.48550/arXiv.2506.00430](https://doi.org/10.48550/arXiv.2506.00430). arXiv: [2506.00430](https://arxiv.org/abs/2506.00430) [cs.AI].

- [53] Y. Hu, Y. Wang, and J. McAuley. “Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions”. In: *International Conference on Learning Representations*. 2026. URL: <https://openreview.net/forum?id=DT7JyQC3MR>.
- [54] S. H. Hurlbert. “Pseudoreplication and the Design of Ecological Field Experiments”. In: *Ecological Monographs* 54.2 (1984), pp. 187–211. DOI: [10.2307/1942661](https://doi.org/10.2307/1942661).
- [55] E. Jang, S. Gu, and B. Poole. “Categorical Reparameterization with Gumbel-Softmax”. In: *International Conference on Learning Representations (ICLR)*. arXiv:1611.01144. 2017.
- [56] S. Ji. *Ortho-Hydra: Orthogonalized Experts for DiT LoRA*. Preprint. 2026. DOI: [10.48550/arXiv.2605.03252](https://doi.org/10.48550/arXiv.2605.03252). arXiv: [2605.03252](https://arxiv.org/abs/2605.03252) [cs.LG].
- [57] Z. Jin, B. Wang, J. Li, R. Xu, and M. Zhang. *Mitigating Provenance-Role Collapse in Long-Term Agents via Typed Memory Representation*. Preprint. 2026. DOI: [10.48550/arXiv.2605.25869](https://doi.org/10.48550/arXiv.2605.25869). arXiv: [2605.25869](https://arxiv.org/abs/2605.25869) [cs.CL].
- [58] D. Kerestecioglu, A. Robsky, C. Vasters, A. Sharma, and Y. Kesselman. *Human-Inspired Memory Architecture for LLM Agents*. 2026. DOI: [10.48550/arXiv.2605.08538](https://doi.org/10.48550/arXiv.2605.08538). arXiv: [2605.08538](https://arxiv.org/abs/2605.08538).
- [59] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan. “Supervised Contrastive Learning”. In: *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. arXiv:2004.11362. 2020.
- [60] J. Kirkpatrick et al. “Overcoming Catastrophic Forgetting in Neural Networks”. In: *Proceedings of the National Academy of Sciences* 114.13 (2017). arXiv:1612.00796, pp. 3521–3526. DOI: [10.1073/pnas.1611835114](https://doi.org/10.1073/pnas.1611835114).
- [61] I. Kotseruba and J. K. Tsotsos. “40 Years of Cognitive Architectures: Core Cognitive Abilities and Practical Applications”. In: *Artificial Intelligence Review* 53.1 (2020). arXiv:1610.08602, pp. 17–94. DOI: [10.1007/s10462-018-9646-y](https://doi.org/10.1007/s10462-018-9646-y).
- [62] J. Koutník, K. Greff, F. Gomez, and J. Schmidhuber. “A Clockwork RNN”. In: *Proceedings of the 31st International Conference on Machine Learning*. Vol. 32. Proceedings of Machine Learning Research. arXiv:1402.3511. PMLR, 2014, pp. 1863–1871.
- [63] D. Kumaran, D. Hassabis, and J. L. McClelland. “What Learning Systems do Intelligent Agents Need? Complementary Learning Systems Theory Updated”. In: *Trends in Cognitive Sciences* 20.7 (2016), pp. 512–534. DOI: [10.1016/j.tics.2016.05.004](https://doi.org/10.1016/j.tics.2016.05.004).
- [64] A. Kusupati et al. “Matryoshka Representation Learning”. In: *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*. arXiv:2205.13147. 2022. DOI: [10.52202/068431-2192](https://doi.org/10.52202/068431-2192).
- [65] J. E. Laird. *The Soar Cognitive Architecture*. Cambridge, MA: MIT Press, 2012. ISBN: 978-0-262-12296-2.
- [66] H. Lau and D. Rosenthal. “Empirical support for higher-order theories of conscious awareness”. In: *Trends in Cognitive Sciences* 15.8 (2011), pp. 365–373. DOI: [10.1016/j.tics.2011.05.009](https://doi.org/10.1016/j.tics.2011.05.009).
- [67] Y. LeCun. *A Path Towards Autonomous Machine Intelligence*. OpenReview position paper, Version 0.9.2, 2022-06-27. 2022. URL: <https://openreview.net/pdf?id=BZ5a1r-kVsf>.
- [68] J.-A. Li, H. Xiong, R. C. Wilson, M. G. Mattar, and M. K. Benna. “Language Models Are Capable of Metacognitive Monitoring and Control of Their Internal Activations”. In: *Advances in Neural Information Processing Systems*. 2025. DOI: [10.52202/085713-2009](https://doi.org/10.52202/085713-2009).
- [69] Z. Li, C. Xi, C. Li, D. Chen, B. Chen, et al. *MemOS: A Memory Operating System for AI System*. Preprint, version 4. 2025. DOI: [10.48550/arXiv.2507.03724](https://doi.org/10.48550/arXiv.2507.03724). arXiv: [2507.03724](https://arxiv.org/abs/2507.03724).

- [70] J. Liang and L. Pan. *Do Latent-CoT Models Think Step-by-Step? A Mechanistic Study on Sequential Reasoning Tasks*. 2026. DOI: [10.48550/arXiv.2602.00449](https://doi.org/10.48550/arXiv.2602.00449). arXiv: [2602.00449](https://arxiv.org/abs/2602.00449).
- [71] Z. C. Lipton and J. Steinhardt. “Troubling Trends in Machine Learning Scholarship”. In: *ACM Queue* 17.1 (2019). Also arXiv:1807.03341 (2018), pp. 45–77. DOI: [10.1145/3317287.3328534](https://doi.org/10.1145/3317287.3328534).
- [72] G. Loewenstein. “The psychology of curiosity: A review and reinterpretation”. In: *Psychological Bulletin* 116.1 (1994), pp. 75–98. DOI: [10.1037/0033-2909.116.1.75](https://doi.org/10.1037/0033-2909.116.1.75).
- [73] J. Ma, Z. Zhao, X. Yi, J. Chen, L. Hong, and E. H. Chi. “Modeling Task Relationships in Multi-task Learning with Multi-gate Mixture-of-Experts”. In: *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*. London, UK, 2018, pp. 1930–1939. DOI: [10.1145/3219819.3220007](https://doi.org/10.1145/3219819.3220007).
- [74] C. J. Maddison, A. Mnih, and Y. W. Teh. “The Concrete Distribution: A Continuous Relaxation of Discrete Random Variables”. In: *International Conference on Learning Representations (ICLR)*. arXiv:1611.00712. 2017.
- [75] B. Maniscalco and H. Lau. “A signal detection theoretic approach for estimating metacognitive sensitivity from confidence ratings”. In: *Consciousness and Cognition* 21.1 (2012), pp. 422–430. DOI: [10.1016/j.concog.2011.09.021](https://doi.org/10.1016/j.concog.2011.09.021).
- [76] G. A. Mashour, P. Roelfsema, J.-P. Changeux, and S. Dehaene. “Conscious Processing and the Global Neuronal Workspace Hypothesis”. In: *Neuron* 105.5 (2020), pp. 776–798. DOI: [10.1016/j.neuron.2020.01.026](https://doi.org/10.1016/j.neuron.2020.01.026).
- [77] A. Masumori, I. Doi, N. Maruyama, R. Takata, and T. Ikegami. “OpenLife: Toward Open-World Artificial Life with Autonomous LLM Agents”. In: *Proceedings of the Artificial Life Conference (ALIFE 2026)*. arXiv:2606.31046. 2026.
- [78] J. L. McClelland, B. L. McNaughton, and R. C. O’Reilly. “Why There Are Complementary Learning Systems in the Hippocampus and Neocortex: Insights from the Successes and Failures of Connectionist Models of Learning and Memory”. In: *Psychological Review* 102.3 (1995), pp. 419–457. DOI: [10.1037/0033-295X.102.3.419](https://doi.org/10.1037/0033-295X.102.3.419).
- [79] J. L. McGaugh. “The amygdala modulates the consolidation of memories of emotionally arousing experiences”. In: *Annual Review of Neuroscience* 27 (2004), pp. 1–28. DOI: [10.1146/annurev.neuro.27.070203.144157](https://doi.org/10.1146/annurev.neuro.27.070203.144157).
- [80] A. Mikedo and B. Goertzel. “A Motivational Architecture for Conversational AGI”. In: *Proceedings of the 19th Conference on Artificial General Intelligence (AGI-26)*. arXiv:2606.05411. 2026.
- [81] T. M. Moerland, J. Broekens, and C. M. Jonker. “Emotion in reinforcement learning agents and robots: a survey”. In: *Machine Learning* 107.2 (2018). arXiv:1705.05172, pp. 443–480. DOI: [10.1007/s10994-017-5666-0](https://doi.org/10.1007/s10994-017-5666-0).
- [82] N. Muennighoff, N. Tazi, L. Magne, and N. Reimers. “MTEB: Massive Text Embedding Benchmark”. In: *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*. arXiv:2210.07316. Dubrovnik, Croatia: Association for Computational Linguistics, 2023, pp. 2014–2037. DOI: [10.18653/v1/2023.eacl-main.148](https://doi.org/10.18653/v1/2023.eacl-main.148). URL: <https://aclanthology.org/2023.eacl-main.148/>.
- [83] L. Naccache, C. Sergent, S. Dehaene, X.-J. Wang, M. Farisco, and J.-P. Changeux. “GNW theoretical framework and the “adversarial testing of global neuronal workspace and integrated information theories of consciousness””. In: *Neuroscience of Consciousness* 2025.1 (2025). Author correction: doi:10.1093/nc/niag020; fourth author corrected to Xiao-Jing Wang; substantive content unchanged, niaf037. DOI: [10.1093/nc/niag037](https://doi.org/10.1093/nc/niag037).

- [84] J. Nakanishi, J. Baba, Y. Yoshikawa, H. Kamide, and H. Ishiguro. “Hypothesis on the functional advantages of the selection-broadcast cycle structure: global workspace theory and dealing with a real-time world”. In: *Frontiers in Robotics and AI* 12 (2025). arXiv:2505.13969, p. 1607190. DOI: [10.3389/frobt.2025.1607190](https://doi.org/10.3389/frobt.2025.1607190).
- [85] T. Ni, B. Eysenbach, E. Seyedsalehi, M. Ma, C. Gehring, A. Mahajan, and P.-L. Bacon. “Bridging State and History Representations: Understanding Self-Predictive RL”. In: *International Conference on Learning Representations*. Preprint arXiv:2401.08898. 2024. URL: <https://openreview.net/forum?id=agPpmEgf8C>.
- [86] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor. “The preregistration revolution”. In: *Proceedings of the National Academy of Sciences* 115.11 (2018), pp. 2600–2606. DOI: [10.1073/pnas.1708274114](https://doi.org/10.1073/pnas.1708274114).
- [87] Z. Nussbaum, J. X. Morris, B. Duderstadt, and A. Mulyar. *Nomic Embed: Training a Reproducible Long Context Text Embedder*. 2024. arXiv: [2402.01613](https://arxiv.org/abs/2402.01613). URL: <https://arxiv.org/abs/2402.01613>.
- [88] R. W. Picard. *Affective Computing*. Cambridge, MA: MIT Press, 1997. ISBN: 978-0-262-16170-1.
- [89] A. Pritzel, B. Uria, S. Srinivasan, A. Puigdomènech Badia, O. Vinyals, D. Hassabis, D. Wierstra, and C. Blundell. “Neural Episodic Control”. In: *Proceedings of the 34th International Conference on Machine Learning*. Vol. 70. Proceedings of Machine Learning Research. arXiv:1703.01988. PMLR, 2017, pp. 2827–2836. URL: <https://proceedings.mlr.press/v70/pritzel17a.html>.
- [90] M. Ran et al. *CaveAgent: Transforming LLMs into Stateful Runtime Operators*. Preprint, version v4. 2026. DOI: [10.48550/arXiv.2601.01569](https://doi.org/10.48550/arXiv.2601.01569). arXiv: [2601.01569](https://arxiv.org/abs/2601.01569) [cs.AI].
- [91] N. Reimers and I. Gurevych. “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks”. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. arXiv:1908.10084. Hong Kong, China: Association for Computational Linguistics, 2019, pp. 3982–3992. DOI: [10.18653/v1/D19-1410](https://doi.org/10.18653/v1/D19-1410). URL: <https://aclanthology.org/D19-1410/>.
- [92] D. M. Rosenthal. *Consciousness and Mind*. Oxford: Clarendon Press / Oxford University Press, 2005. ISBN: 9780198236962.
- [93] J. A. Russell. “A circumplex model of affect”. In: *Journal of Personality and Social Psychology* 39.6 (1980), pp. 1161–1178. DOI: [10.1037/h0077714](https://doi.org/10.1037/h0077714).
- [94] J. A. Russell and A. Mehrabian. “Evidence for a three-factor theory of emotions”. In: *Journal of Research in Personality* 11.3 (1977), pp. 273–294. DOI: [10.1016/0092-6566\(77\)90037-X](https://doi.org/10.1016/0092-6566(77)90037-X).
- [95] K. R. Scherer. “The dynamic architecture of emotion: Evidence for the component process model”. In: *Cognition and Emotion* 23.7 (2009), pp. 1307–1351. DOI: [10.1080/02699930902928969](https://doi.org/10.1080/02699930902928969).
- [96] M. Shanahan. “A cognitive architecture that combines internal simulation with a global workspace”. In: *Consciousness and Cognition* 15.2 (2006), pp. 433–449. DOI: [10.1016/j.concog.2005.11.005](https://doi.org/10.1016/j.concog.2005.11.005).
- [97] W. Shang. “Theater of Mind” for LLMs: A Cognitive Architecture Based on Global Workspace Theory. Preprint. 2026. DOI: [10.48550/arXiv.2604.08206](https://doi.org/10.48550/arXiv.2604.08206). arXiv: [2604.08206](https://arxiv.org/abs/2604.08206) [cs.MA].
- [98] C. Shao, D. Li, F. Meng, and J. Zhou. “Continuous Autoregressive Language Models”. In: *arXiv preprint arXiv:2510.27688* (2025). URL: <https://arxiv.org/abs/2510.27688>.

- [99] D. Sharma. *Gubernaut: A Deterministic Homeostatic Controller for Affect-Regulated LLM Agents, Validated Across Independent Model Families*. Preprint. 2026. DOI: [10.48550/arXiv.2607.24339](https://doi.org/10.48550/arXiv.2607.24339). arXiv: [2607.24339](https://arxiv.org/abs/2607.24339) [cs.AI].
- [100] Z. Shen, H. Yan, L. Zhang, Z. Hu, Y. Du, and Y. He. “CODI: Compressing Chain-of-Thought into Continuous Space via Self-Distillation”. In: *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2025, pp. 677–693. DOI: [10.18653/v1/2025.emnlp-main.36](https://doi.org/10.18653/v1/2025.emnlp-main.36).
- [101] I. Shumailov, Z. Shumaylov, Y. Zhao, N. Papernot, R. Anderson, and Y. Gal. “AI models collapse when trained on recursively generated data”. In: *Nature* 631 (2024). Author Correction: *Nature* 2025, doi:10.1038/s41586-025-08905-3, pp. 755–759. DOI: [10.1038/s41586-024-07566-y](https://doi.org/10.1038/s41586-024-07566-y).
- [102] C. A. Smith and P. C. Ellsworth. “Patterns of cognitive appraisal in emotion”. In: *Journal of Personality and Social Psychology* 48.4 (1985), pp. 813–838. DOI: [10.1037/0022-3514.48.4.813](https://doi.org/10.1037/0022-3514.48.4.813).
- [103] J. Snider and S. Franklin. “Vector LIDA”. In: *Procedia Computer Science* 41 (2014). 5th Annual International Conference on Biologically Inspired Cognitive Architectures (BICA 2014), pp. 188–203. DOI: [10.1016/j.procs.2014.11.103](https://doi.org/10.1016/j.procs.2014.11.103).
- [104] Z. Song et al. *Branch-JEPA: Finite-Support Predictive Distributions for JEPA World Models*. Preprint; v1 appeared as “MoP-JEPA”. 2026. DOI: [10.48550/arXiv.2607.05238](https://doi.org/10.48550/arXiv.2607.05238). arXiv: [2607.05238](https://arxiv.org/abs/2607.05238) [cs.AI].
- [105] M. Strathern. “‘Improving ratings’: audit in the British University system”. In: *European Review* 5.3 (1997), pp. 305–321. DOI: [10.1017/S1062798700002660](https://doi.org/10.1017/S1062798700002660).
- [106] H. Su. *Simulating Human Cognition: Heartbeat-Driven Autonomous Thinking Activity Scheduling for LLM-based AI Systems*. Preprint. 2026. DOI: [10.48550/arXiv.2604.14178](https://doi.org/10.48550/arXiv.2604.14178). arXiv: [2604.14178](https://arxiv.org/abs/2604.14178) [cs.AI].
- [107] T. R. Sumers, S. Yao, K. Narasimhan, and T. L. Griffiths. “Cognitive Architectures for Language Agents”. In: *Transactions on Machine Learning Research* (2024). URL: <https://openreview.net/forum?id=1i6ZCyf1QJ>.
- [108] X. Sun, Q. Wang, H. Li, and L. Chen. *Verifiable Memory for Long-Lived Agents: Evidence, Conflict, and Uncertainty*. 2026. DOI: [10.48550/arXiv.2608.03137](https://doi.org/10.48550/arXiv.2608.03137). arXiv: [2608.03137](https://arxiv.org/abs/2608.03137).
- [109] N. Taguchi and Y. Tsuruoka. *Partially Non-Recurrent Controllers for Memory-Augmented Neural Networks*. 2018. DOI: [10.48550/arXiv.1812.11485](https://doi.org/10.48550/arXiv.1812.11485). arXiv: [1812.11485](https://arxiv.org/abs/1812.11485).
- [110] R. VanRullen and R. Kanai. “Deep learning and the Global Workspace Theory”. In: *Trends in Neurosciences* 44.9 (2021), pp. 692–704. DOI: [10.1016/j.tins.2021.04.005](https://doi.org/10.1016/j.tins.2021.04.005).
- [111] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. “Attention Is All You Need”. In: *Advances in Neural Information Processing Systems 30 (NIPS 2017)*. arXiv:1706.03762. 2017.
- [112] C. Walsh and E. Barkett. *Representation Without Control: Testing the Realization Effect in Language Models*. Preprint. 2026. DOI: [10.48550/arXiv.2605.25151](https://doi.org/10.48550/arXiv.2605.25151). arXiv: [2605.25151](https://arxiv.org/abs/2605.25151) [cs.AI].
- [113] L. Wang, L. Yang, B. Chen, K. Xu, G. Zou, B. Tang, F. Xiong, S. Chen, and Z. Li. “Text2Mem: A General Framework for Self-Evolving Agent Memory”. In: *Findings of the Association for Computational Linguistics: ACL 2026*. Association for Computational Linguistics, 2026, pp. 2105–2119. DOI: [10.18653/v1/2026.findings-acl.100](https://doi.org/10.18653/v1/2026.findings-acl.100).
- [114] P. Wang. *Rigid Flexibility: The Logic of Intelligence*. Vol. 34. Applied Logic Series. Dordrecht: Springer, 2006. ISBN: 978-1-4020-5044-2. DOI: [10.1007/1-4020-5045-3](https://doi.org/10.1007/1-4020-5045-3).

- [115] P. Wang. *Non-Axiomatic Logic: A Model of Intelligent Reasoning*. Singapore: World Scientific, 2013. ISBN: 978-981-4440-27-1. DOI: [10.1142/8665](https://doi.org/10.1142/8665).
- [116] X. Wang, S. Hayou, and E. Nalisnick. *The Myth of Expert Specialization in MoEs: Why Routing Reflects Geometry, Not Necessarily Domain Expertise*. 2026. DOI: [10.48550/arXiv.2604.09780](https://doi.org/10.48550/arXiv.2604.09780). arXiv: [2604.09780](https://arxiv.org/abs/2604.09780).
- [117] A. B. Warriner, V. Kuperman, and M. Brysbaert. “Norms of valence, arousal, and dominance for 13,915 English lemmas”. In: *Behavior Research Methods* 45 (2013), pp. 1191–1207. DOI: [10.3758/s13428-012-0314-x](https://doi.org/10.3758/s13428-012-0314-x).
- [118] V. Weinhhammer, K. Y. C. Hou, L. Luettgau, C. Summerfield, R. Dolan, and M. M. Nour. “A clinically validated framework for auditing AI chatbot behavior in mental health interactions”. In: *Nature Medicine* (2026). DOI: [10.1038/s41591-026-04577-2](https://doi.org/10.1038/s41591-026-04577-2).
- [119] D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, and D. Yu. “LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory”. In: *International Conference on Learning Representations*. 2025. URL: <https://openreview.net/forum?id=pZiyCaVuti>.
- [120] K. Yang, T.-Q. Peng, S. Lee, and H. Liu. *Think-Before-Speak: From Internal Evaluation to Public Expression in Multi-Agent Social Simulation*. Preprint; accepted at the KDD 2026 SciSoc Workshop. 2026. DOI: [10.48550/arXiv.2606.03137](https://doi.org/10.48550/arXiv.2606.03137). arXiv: [2606.03137](https://arxiv.org/abs/2606.03137) [cs.AI].
- [121] T. Yu, S. Kumar, A. Gupta, S. Levine, K. Hausman, and C. Finn. “Gradient Surgery for Multi-Task Learning”. In: *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. arXiv:2001.06782. 2020.
- [122] Y. Zhang, D. Zhao, J. T. Hancock, R. Kraut, and D. Yang. “Interaction with AI companions and psychological well-being”. In: *Nature Human Behaviour* (2026). DOI: [10.1038/s41562-026-02516-2](https://doi.org/10.1038/s41562-026-02516-2).
- [123] C. Zhou et al. *Externalization in LLM Agents: A Unified Review of Memory, Skills, Protocols and Harness Engineering*. Preprint. 2026. DOI: [10.48550/arXiv.2604.08224](https://doi.org/10.48550/arXiv.2604.08224). arXiv: [2604.08224](https://arxiv.org/abs/2604.08224) [cs.SE].

A Notation and Identifier Register

This appendix answers two neighboring questions that the paper otherwise leaves to search. Section A.1 says what a symbol denotes; Section A.2 says where a family of identifiers is fixed and what, if anything, tests it.

A.1 Symbols

Table 5 lists the symbols used throughout, with the section that introduces each.

A.2 Identifier register

The paper carries ten identifier families and refers to their members across every section and appendix, so where each is fixed and what, if anything, tests it are questions worth answering in one table rather than by search. Table 6 does that, and Figure 19 draws the same families as a dependence graph—which family constrains which, which one decides membership, and which is the only one that can come out false. The table’s last column records the framework’s instrumentation *gaps* in the same place as its instruments, because a register that listed only what is covered would read as completeness. Four gaps are visible there—the tick order has no instrument of its own, A7 and A8 deliberately bind no requirement, stage S4 is not claimable for want of transfer and composition markers, and the affect paths R3 requires into integration and

Symbol	Meaning	Introduced
X_t	fixed-width typed exposed core at tick t , (F_t, C_t, S_t, g_t)	Section 3.2
$\mathcal{X}_t^{(r)}$	regime-specific declared exposed state: X_t except (X_t, h_t) in declared-carrier regime (iii); closure boundary for every consumer read	Section 3.4
F_t	fast state: event encoding + signal channels	Section 3.2
C_t	core state: typed, segmented, rewritten each tick	Section 3.2
S_t	slow state: dispositional variables	Section 3.2
Φ	tick operator (Equation (3))	Section 3.3
h_t	recurrent carrier; declared and exposed only in regime (iii), operator-internal in regime (iv)	Section 3.3
g_t	broadcast context from workspace competition; the fourth component of X_t , not a quantity beside it	Section 3.3
W_t	working-memory summary over recent core states	Section 3.3
M_t	retrieved long-term memory content	Section 3.3
η, Δ	slow-state step size and update head (Equation (4))	Section 3.3
I_t	admissible information state at tick t (Equation (5))	Section 7.3
R_{reach}^2	reachable target fraction given I_t (Equation (6)); estimated probe-relative as $\hat{R}_{\mathcal{F}}^2$, a lower bound; reported design-time against an oracle I_t^* and again as realized reachability after training	Section 7.3
R1–R7	architectural conformance requirements: what a running system must be	Section 3.8
D1–D2	target and measurement disciplines: what a study about such a system must do; they bind reports, not membership	Section 3.8
LC1–LC5	latent-target learnability conditions	Section 7.3
M1–M9	measurement requirements	Section 9.2
H1–H17	research-program hypotheses	Section 11
S1–S4	understanding stages	Section 9.3
memory shelves	episodic, knowledge, concept, identity	Section 6.1

Table 5: Symbols and identifier families used throughout the paper.

memory have no marker yet, since B.10 reaches the meta segment only. Each is argued where it arises; the table is where they stop being scattered.

B Evidence-Marker Operationalization

Each marker below is specified by four fields. *Operationalization*: what is measured, on item-level units (M5), against noise floors established first (M1). *Instrument control*: a structurally equivalent condition in which the effect is present independently of the system under test (M4); if it fails, the verdict is *not measurable*. Some markers additionally list a *sensitivity condition*, which diagnoses the system but cannot license a null. *Failure criterion*: the condition fixed in advance (M7) under which the marker is scored negative. *Confounds*: the named alternatives that must be excluded, always including the retrieval shortcut and prompt echo where applicable. Thresholds are instantiation-specific and must be fixed in advance per study; this appendix fixes the *shape* of each test, not its numbers.

Two kinds of control, and why the distinction is enforced here. Section 9.2 separates an *instrument-positive control*—a trace in which the effect is present independently of the system under test—from a *sensitivity condition*, which only makes the effect easier for the system to produce. Only the first can license a *null*, because only the first fails for a reason that has

Family	Members	Defined in	What tests it
A1–A9	9	Section 3.1	indirectly, through R1–R7; A7 and A8 by nothing, deliberately
R1–R7	7	Section 3.8	H17a tests R4 empirically; H17b is its comparison arm
D1–D2	2	Section 3.8	D1 through LC1–LC5, D2 through M1–M9
T1–T9	9	Section 3.3	no instrument of its own; Figure 3 is the only check
LC1–LC5	5	Section 7.3	H8 (provisional); probes before and after training (D1)
M1–M9	9	Section 9.2	H12; H13 (provisional); applied to every claim (D2)
B.1–B.15	15	Section B	each carries its own failure criterion and instrument control
S1–S4	4	Section 9.3	B.4 and B.11; S4 not claimable, no transfer or composition marker
H1–H17	17	Section 11	the only family that can come out false; three marked provisional
O1–O8	8	Section 12.3	nothing—open by definition, each paired with a handle or marked open

Table 6: The identifier register. “Defined in” is a cross-reference rather than a written-out number, so it cannot go stale; the member counts and the markers and hypotheses named in the last column are checked mechanically against their definition sites rather than maintained by hand. Two entries carry no instrument by design rather than by omission: the tick order is a normative sequence that a reader checks against Figure 3, and the open problems answer to nothing, which is what makes them open. The affect paths R3 requires into integration and memory are instrumented only by the instantiation’s own intervention, since no marker for them exists yet (Section 4.3).

nothing to do with the system. The markers below are labeled accordingly, and the labels were assigned by re-reading each one against this distinction rather than by assuming the original wording was right; several were sensitivity conditions carrying the wrong name. Where a marker is a statistical detection problem—does a correlation, an ordering, or a divergence exist in the trajectories?—its instrument-positive control takes a standard form, and we name it once here instead of restating it fifteen times: a *synthetic-trace control*, in which traces carrying the effect at a known magnitude are generated directly and passed through the *identical* statistical pipeline. If the pipeline cannot recover a planted effect, no null it produces means anything. Several markers admit both kinds of control, and those report both.

B.1 Reportability. *Operationalization:* agreement between state reports (Section 3.6) and the declared exposed state they were rendered from, across scenario batteries—report contents regressed on the actual values. All report-bearing components of $\mathcal{X}_t^{(r)}$ are covered and scored separately: C_t , S_t , and g_t in every regime, plus the declared h_t field in regime (iii). The broadcast field is therefore included: a report that tracks the affect segment faithfully and states the wrong winner has failed at one field, and an aggregate would hide it. (F_t has no report field by design, Section 3.6, and is outside this marker’s scope.) *Instrument control:* synthetic-trace control—report–state pairs constructed with a planted agreement level per field and passed through the identical regression pipeline, so that a failure is attributable to the estimator rather than to the renderer. *Sensitivity condition (not a control):* reports generated from artificially fixed exposed states with known values, g_t fixed alongside the carried components—a broadcast field left free would go unmeasured while the marker reported a pass. This diagnoses the renderer, which is part of the system under test, and therefore cannot license a null. *Failure:* report–state agreement indistinguishable from shuffled-state baseline. *Confounds:* prompt echo (reports recycling probe wording); translator priors (the language model verbalizing plausible defaults regardless of state)—excluded by varying states under fixed probes.

B.2 Temporal continuity. *Operationalization:* across-session state persistence—similarity of dispositional and contextual state before and after suspension/restoration, against between-system baselines. *Instrument control:* explicit state restoration. *Failure:* post-restoration state no closer to pre-suspension state than to a fresh-start baseline. *Confounds:* external scaffolding

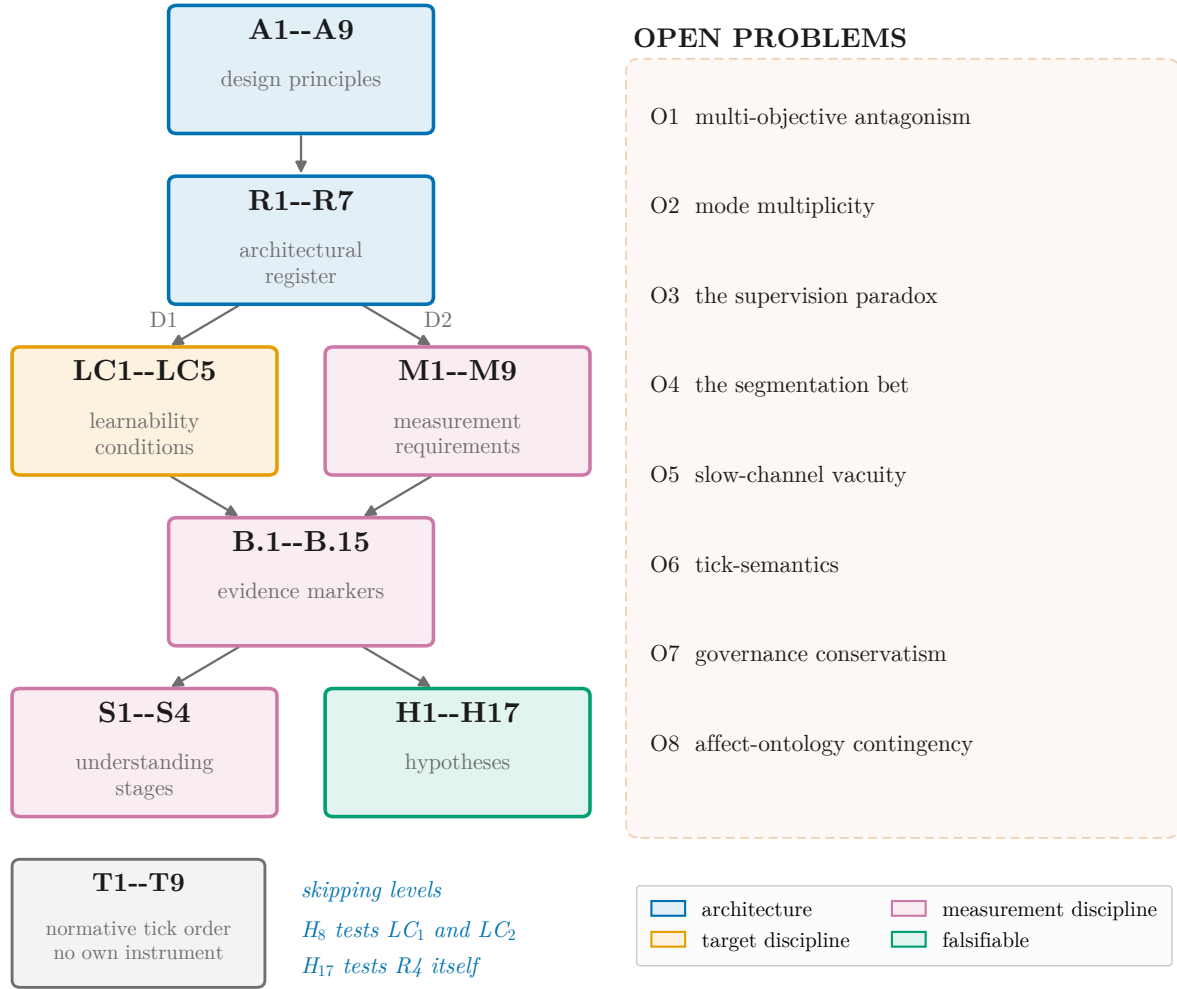


Figure 19: The identifier families and their dependence. The families are not parallel, and they are not a chain: the principles constrain the architectural register R1–R7, which decides membership in the framework’s class, while the two disciplines—D1 for targets, D2 for measurement—hang off it side by side and bind studies rather than systems, which is why they sit beside R and not inside it. Both disciplines constrain what a marker may conclude; markers in turn decide which understanding stage may be claimed and are what tests the hypotheses, the only family that can come out false. Reading upward shows which declaration justifies a dependency; reading downward shows what depends on it. Falsification is narrower: hypotheses are the only family whose members may come out false, and the named skip-level tests are shown separately. The tick-order family T1–T9 is shown without an edge because it is a normative sequence checked against Figure 3 and has no instrument of its own. The open problems on the right answer to nothing—they are difficulties the framework’s own structure predicts, each paired with a handle in Section 12.3 or marked open there.

reproducing continuity (transcript replay) rather than state persistence.

B.3 Self/world distinction. *Operationalization:* differential processing of self-referring versus world-referring inputs in the designated state channels; classification of self/world/other attribution from state trajectories. *Instrument control:* synthetic-trace control with planted self/world attribution at a known separation. *Sensitivity condition (not a control):* inputs with explicit self/world markers in the signal channels. *Failure:* attribution accuracy at chance despite carrying markers. *Confounds:* lexical shortcuts (surface pronouns) rather than structural

distinction—probed with paraphrase-controlled stimuli.

B.4 Contradiction-sensitive revision. *Operationalization:* state response to inputs contradicting previously established content—conflict-channel elevation and subsequent belief-adjacent state change, versus consistent-input controls. *Instrument control:* synthetic-trace control with a planted conflict-channel elevation of known magnitude. *Sensitivity condition (not a control):* maximally explicit contradictions of just-presented content. *Failure:* no differential conflict response with a passing control. *Confounds:* novelty response masquerading as contradiction response—controlled by novelty-matched consistent inputs.

B.5 Meta-confidence calibration. *Operationalization:* correlation between the confidence channel and realized accuracy/error across items (Section 5.4); anti-calibration explicitly reportable per M3. *Instrument control:* synthetic-trace control pairing planted confidence values with planted error rates at a known correlation. *Sensitivity condition (not a control):* items with engineered difficulty gradients—these show whether the system can express calibration, not whether the estimator can detect it. *Failure:* calibration correlation at or below zero. *Reporting:* the meta supervision regime in force (consistency, label, or the composite warm start) is reported with the marker (Section 5.4). *Confounds:* difficulty proxies leaking through input statistics; under label supervision, agreement with the generator’s confidence convention masquerading as calibration—excluded by scoring only after handover to consistency supervision.

B.6 Perturbation robustness. *Operationalization:* state-trajectory recovery after bounded injected perturbations—return time and residual displacement against unperturbed controls. *Instrument control:* synthetic trajectories with a planted recovery of known magnitude, passed through the identical analysis pipeline—recovery present by construction and independent of the system under test, which is what M4 requires. *Sensitivity conditions (not controls):* a configuration with an architecturally guaranteed restoring mechanism (explicit homeostatic decay toward baseline), which makes recovery easier for the system rather than guaranteeing it for the instrument; and perturbations beyond design bounds, where recovery is expected to degrade. Both diagnose the system; neither licenses a null. *Failure:* no systematic recovery within bounds. *Confounds:* insensitivity masquerading as robustness (a state that never moves “recovers” trivially)—excluded by verifying reactivity on the same runs.

B.7 Introspection–behavior consistency. *Operationalization:* correspondence between meta/affect channel values and the system’s subsequent action-selection statistics (e.g., low confidence preceding information-seeking tendencies). *Instrument control:* synthetic-trace control—channel values paired with subsequent action statistics at a planted coupling strength, through the identical pipeline. *Sensitivity condition (not a control):* architecturally forced couplings verified to manifest; these establish that the system *can* express the coupling, not that the estimator can see one, and a failure here is a statement about the system. *Failure:* no coupling beyond chance with controls passing. *Confounds:* common-cause input features driving both channels and behavior—handled by residualization (M6).

B.8 Workspace ignition dynamics. *Operationalization:* selection sharpness, amplification, and persistence statistics of workspace competition (Section 5.2) under competing-content loads. *Instrument control:* synthetic workspace-output traces with sharpness, amplification, and persistence values planted at known levels, passed through the same estimators without invoking the system under test; the pipeline must recover each statistic and its ordering. *Sensitivity condition (not a control):* synthetic proposal sets injected at the competition stage with a known dominant proposal, together with inputs engineered for single-winner competition. These make ignition easier for the system to express but do not establish that the statistic can detect it.

Failure: permanent soft mixing (no capacity-limited selection observable). *Confounds*: sharpness induced by annealing schedule rather than content—checked across schedule phases.

B.9 Narrative continuity. *Operationalization*: coherence of the semantic-focus trajectory across an extended interaction—topic persistence and motivated transitions versus random-walk baselines. *Instrument control*: synthetic-trace control with a planted coherence level. *Sensitivity condition (not a control)*: externally scripted topic arcs—note that these supply continuity from outside and therefore also risk the confound below. *Failure*: trajectory statistics indistinguishable from unconditioned drift. *Confounds*: continuity supplied by the interlocutor rather than the system—assessed on autonomous (input-free) tick stretches.

B.10 Causal efficacy of the meta-state. *Operationalization*: interventional test—clamping or lesioning meta channels and measuring downstream processing changes (deliberation-policy decisions, integration behavior), versus sham interventions. *Instrument control*: synthetic-trace control—intervention and sham arms with a planted downstream difference of known magnitude, through the identical contrast. *Sensitivity condition (not a control)*: clamping channels with architecturally guaranteed downstream consumers, which shows the system has a path to move and not that the contrast can detect movement. *Failure*: no downstream difference between intervention and sham. *Confounds*: intervention artifacts (off-manifold clamped values violating LC5-type constraints)—clamp within the channel’s observed operating range.

B.11 Anticipatory state shifts. *Operationalization*: the S3 signature made testable—pre-event trajectory divergence conditioned on announced outcomes: between an announcement and the event itself, state trajectories for announced-positive versus announced-negative continuations must diverge measurably *and with sign consistent with the announced valence* (affect and intention segments), relative to matched unannounced controls. *Instrument control*: synthetic-trace control with a planted pre-event divergence of known magnitude and sign. *Sensitivity condition (not a control)*: explicit announcement events of a class the appraisal channel is trained to read. *Failure*: divergence begins only at event onset. *Confounds*: the announcement text itself carrying affect the encoder projects (announcement-valence leakage)—controlled by valence-matched phrasings; persistence mimicry—excluded by echo-baseline comparison (Section 10.6).

B.12 Deliberation depth. *Operationalization*: consumed processing depth as a function of input demand, where “depth” is *two* quantities and not one. *Inter-tick depth* is counted in ticks to halt—the unit A6 defines and the only one the deliberation policy controls. *Intra-tick refinement* is counted in refinement steps within a single update (Section 10.7), which leave tick semantics untouched. The two are reported separately and never summed into a single depth count without a declared compute-normalized conversion, because the matched-compute comparison below has different answers depending on whether a refinement step is priced as a tick. Three requirements in order: a *reachability proof* first—the halting inputs (readiness, meta-confidence) must demonstrably move through a pre-specified fraction of the margin to their own halting thresholds in operation, not merely vary, before any depth claim is made (Section 8.5); then depth–demand correlation, *with demand fixed externally* by the battery’s designed difficulty ordering—the meta complexity channel is reported alongside but never used as the demand measure: under consistency supervision it is itself trained against consumed depth (Section 5.4), and under label supervision it carries the generator’s convention—either way the correlation would be circular; then the *dividend*: deeper processing must improve outcome quality over a fixed-depth control of matched average compute. *Instrument control*: synthetic-trace control pairing planted depth values with planted demand levels at a known ordering—this establishes that the depth–demand estimator can recover an ordering that is there. *Sensitivity condition (not a control)*: demand-graded input batteries with known ordering; note that a failed

reachability proof is not an instrument failure but the policy-inert verdict itself, a property of the system under test. *Failure*: two distinct modes, reported separately—depth constant across demand (policy inert), or depth varies without outcome gain (deliberation without dividend). *Confounds*: depth driven by input length rather than difficulty—length-matched batteries; post-response settling dynamics mistaken for deliberation—count only pre-response ticks.

B.13 Exposed-state sufficiency. *Operationalization*: a two-arm intervention on the typed core X_t and recurrent carrier h_t of Section 3.4, with an asymmetry that must be handled rather than glossed. Perturb X_t while holding h_t at its unperturbed value, then perturb h_t while holding X_t fixed. Where X_t is a deterministic readout of h_t , the second arm is not a physical clamp but a *null-space* perturbation: vary h_t only along directions that produce approximately zero change in X_t . Instantiations report which arm they could run and how the second was realized. Both arms measure four downstream quantities separately: next typed core, memory retrieval, halting decision, and state report.

The regime declaration fixes the marker’s object and verdict. In regimes (i) and (ii), there is no persistent carrier to compare and route closure is reported as satisfied by construction. In regime (iii), $\mathcal{X}_t^{(r)} = (X_t, h_t)$ and B.13 is a within-declared-state attribution: it reports whether X_t or the declared h_t dominates each downstream channel, but carrier dominance is not mislabeled an undeclared-route violation. In regime (iv), $\mathcal{X}_t^{(r)} = X_t$ and h_t is operator-internal; typed-core dominance on all four channels is the empirical sufficiency claim. The unconstrained-carrier arm of H17b is run only as a predeclared non-conforming comparison variant, never entered as a fifth admissible regime.

The typed core’s four components are not interchangeable here. C_t and S_t are perturbed across the tick boundary; F_t is perturbed between input translation and operator at T1; and g_t is perturbed between workspace selection and the operator. The g_t arm is reported separately, and next-state measurement includes g_{t+1} . If an instantiation cannot intervene on g_t independently, that component is *not measurable*, not omitted. Retrieval is read at the first query each intervention can causally reach: X_t interventions at M_t where they precede T4, h_t interventions at M_{t+1} or the earliest later dependent query. Both ticks are reported so an arm read too early is not mistaken for a null.

Four items are fixed before the run: on-manifold perturbation kernels matched for effect size on their own carrier; a per-downstream-channel causal estimand stated as an explicit contrast; a normalization for different carrier dimensions and scales; and a dominance margin. The marker reports the normalized $X_t:h_t$ effect ratio and verdict separately for next state, causally matched retrieval, halting, and report—never only an aggregate.

Instrument control: for each downstream quantity, paired synthetic traces plant known normalized effects for X_t and h_t and traverse the identical attribution, normalization, margin, and verdict pipeline without invoking the system under test. The suite plants ratios on both sides of the preregistered margin and must recover the correct ordering and classification for all four channels. Failed recovery forces *not measurable* for that channel; with its control passed, observed typed-core dominance licenses *positive* and observed non-dominance licenses *null*. Where no matched on-manifold kernels exist, the system result remains *not measurable* even if the synthetic estimator control passes.

Sensitivity condition (not a control): perturb a typed segment with an architecturally guaranteed downstream consumer (e.g., intention feeding the halting decision). This makes an effect easier for the system to express but does not guarantee that the estimator can recover it. *Failure*: in regime (iv), or in the H17b comparison, at least one downstream channel tracks h_t beyond the fixed margin; in regime (iii), the same outcome is reported as within-state carrier dominance rather than route non-conformance. *Confounds*: off-manifold perturbations—clamp within each carrier’s observed operating range (LC5); incomparable magnitudes—normalize before comparison; deterministic coupling—use a valid null-space arm or report *not measurable*,

never an incomparable proxy.

B.14 Non-broadcast influence. *Operationalization:* the causal footprint of losing proposals (Section 5.1). Compare subsequent state trajectories between (i) normal competition, (ii) losing specialists’ activations ablated after selection, and (iii) sham ablation—measuring whether processing that was not broadcast nonetheless shapes later proposals and states. *Instrument control:* an injected-influence condition—a known perturbation added to a losing specialist’s activation after selection, so that a downstream difference is present by construction and the ablation contrast must register it. *Sensitivity condition (not a control):* a delayed-relevance battery—inputs constructed so that a losing specialist’s content becomes relevant a few ticks later. *Failure:* no difference between ablation and sham—non-broadcast processing is then decorative, and workspace selection is a hard information bottleneck rather than a selection over ongoing processing. Within the marker’s scope both outcomes are informative and neither is assumed; outside it, only the negative one can occur, which is why the scope note below is part of the marker rather than a caveat about it. *Confounds:* ablation perturbing shared parameters rather than the specialist’s activations—intervene on activations, never weights; compute saved by ablation altering dynamics—match compute in the sham arm. *Scope:* this marker applies only to realizations in which losing specialists have a declared persistent path—the two-phase write form of Section 5.1, or any realization that declares a persistent specialist activation. In the broadcast-only form there is no such path, a null is the expected result by construction, and running the marker there measures the realization rather than the workspace; it is reported as an architectural fact, not as a verdict. Where the path exists by construction, instantiations report that fact and the broadcast-only comparison, again not the marker’s verdict as evidence.

B.15 Grounding fidelity. *Operationalization:* agreement between an emitted reply and the state report it was generated from (Section 3.6)—the fraction of assertions in the reply that the report supports, and the fraction of report content the reply misstates. Scored per reply over a battery spanning the declared coverage classes of the grounding instrument, and reported separately for *added* content (the reply asserts what the report does not contain) and *distorted* content (the reply contradicts a value the report states). *Instrument control:* reply–report pairs with violations planted by construction—an inserted numeric change, a negated relation, an added entity—which the checker must catch; the marker reports the checker’s own detection rate on these before reporting anything about the system. *Sensitivity condition (not a control):* verbose or low-constraint generation settings, which raise violation rates without establishing that the instrument sees them. *Failure—two kinds, and the marker is incomplete without both.* *Instrument failure:* detection rate on planted violations indistinguishable from chance—the grounding contract is then unverified, and results from this marker are *not measurable* rather than negative. *System failure:* with the instrument control passed, the emitted replies exceed a rate of added or distorted content fixed before the run, within the declared coverage classes. Only the second is a statement about the system; a marker carrying only the first can return “not measurable” and “passed” but never “the system’s grounding is poor,” which would leave the contract unfalsifiable on the one axis it exists to test. *Reporting:* the instrument’s declared coverage classes and its role (diagnostic or release gate) are reported with every result, since a high score under narrow coverage means less than a low score under wide coverage. *Confounds:* paraphrase counted as addition—controlled with entailment-matched rewordings that add nothing; report fields the checker cannot parse scored as absent rather than as unmeasured—the two are separated in the report.

Marker verdicts aggregate per Figure 12: *positive*, *null*, or *not measurable*, with the battery reported in full—no selective marker reporting—and no aggregate “consciousness score” is ever formed: the markers are a profile, not a sum. That last rule is an evidence policy in its own right and does not follow from M2, which would only require such a score to carry one name and

one formula; what forbids it is that no arithmetic over structural indicators yields a phenomenal claim (Section 12.1).

C Extended Positioning

This appendix refines Table 1 against three neighbor classes selected by our comparative literature audit completed on 31 August 2026.

C.1 Versus LIDA. Within that documented comparison, LIDA [33] is the closest located architectural relative: both build on Global Workspace Theory, both organize processing around a repeating cognitive cycle, both integrate multiple memory systems. The differences are substrate and posture. LIDA is a framework of coordinated symbolic-and-subsymbolic components whose behavior is designed; TASA is a differentiable substrate whose behavior is *trained*, which is why it needs the learnability theory of Section 7 that LIDA has no analogue of. And where LIDA’s affect participates as one mechanism among many, TASA commits to affect as the organizing layer of the core state. Conversely, LIDA’s procedural learning and action selection are richer than anything specified here; the frameworks are complementary along that axis.

One qualification belongs here rather than in the main text, because it removes the easiest misreading of the previous paragraph. It would be wrong to present vector-valued internal representation as the dividing line: Vector LIDA already replaces the node-and-link scheme with high-dimensional vectors over a sparse distributed memory [103]. The dividing line is narrower and should be stated as such—in that line of work the vectors are a representational choice inside a designed architecture, whereas here the typed state is what is trained and what is evaluated, which is the entire reason Section 7 exists. Anyone comparing the two should compare on that axis and not on the data structure.

C.2 Versus world-model agents. The Dreamer preprint and journal version [46, 47], treated as versions of one work rather than independent evidence, and their ancestors [44] share the recurrent-latent-state substrate and the latent-first wager. The divergence is teleological: a world model’s latent state is optimized to predict observations and value—its content is whatever control requires, and its interpretability is incidental. TASA’s core state is the product itself: typed, inspectable, affectively structured, evaluated on its own dynamics. This is why the frameworks price supervision differently: world models get well-posed targets for free (future observations, rewards), while inner-state architectures must engineer latent targets—the entire burden LC1–LC5 formalizes. A hybrid—a world-model backbone with a typed inner-state overlay—is an obvious research direction the framework’s modularity permits.

C.3 Versus LLM agents with memory scaffolds. Contemporary LLM agents attach external memory (vector stores, transcripts, summaries) to stateless models. The functional overlap with Section 6 is real—key–value retrieval, consolidation-like summarization—but the architecture differs in kind: a memory scaffold conditions *text generation*, while the memory shelves of Section 6 feed a persistent *state update*; the scaffold’s “state” is re-derived from text each turn, with no typed carriers, no consistency constraints, no timescale separation, and no parity question because there is no certified inner path to be parity-checked against. The governance machinery of Section 8 has at most a partial analogue that we have found specified for memory scaffolds—which is exactly the framework’s argument for why inner-state systems need it stated explicitly.

D Glossary

artificial inner world A persistent, inspectable, multi-timescale internal state organization whose dynamics are the primary object of study.

long-horizon aim The declared objective: functional inner state that grows measurably richer, more coherent and deeper over sustained, governed operation, with every rung of that ladder instrumented before it is claimed.

structural stance Studying architectures whose internal organization instantiates structural properties associated with conscious processing—and can be tested to—without claims about phenomenal experience.

exposed-state sufficiency Principle A9 with marker B.13: every consumer read is closed over the regime-specific declared exposed state; B.13 separately attributes influence between its typed core and any recurrent carrier.

exposure regime The declared choice among no hidden carrier, per-tick reset, declared carrier, and constrained carrier.

declared departure A system that drops or replaces a register entry (R1–R7) or a root commitment, names which and why, and states what plays its role—outside the conformance class, inside the research program. It inherits D1 and D2, which never depended on membership, and may not inherit the hypotheses unexamined, which are quantified over conforming systems. Distinct from a *comparison variant*, which exists to measure what a requirement buys.

architectural register / disciplines R1–R7, decidable by inspecting a running system, decide membership; D1–D2 bind what a *study* must check and report, apply beyond this framework’s class, and decide nothing about membership.

non-broadcast processing Losing workspace proposals shaping later processing without entering the broadcast. Realization-dependent: present by construction where specialists own a region of the update, absent by construction where proposals reach the state only through the broadcast (Section 5.1).

reference ontology The framework’s concrete worked inventory of segment contents; a declared choice whose invariant is the set of roles, not the counts.

variation evidence Evidence about how a derived quantity used as a target, gate, or instrument behaves over ordinary operation: its range, its spread, and the fraction of the decision range it visits—distinct from a constructed control, which shows only that it responds when perturbed. Its watchdog case is the reachability proof.

repair A change declared in advance as correcting a defective mechanism, with a re-adaptation budget, acceptance criterion, and revert condition fixed before the run, so that a non-degradation gate does not reject it for its transient cost.

starved interpretation Failure mode in which a correctly supervised, correctly fed-back interpretation head reads an input from which its labels are not cheaply separable; the conditioning channel then carries a near-constant vector while still consuming input width and gradient.

generator closure The case in which the state reconstructs the target generator’s rule set; caught by probe capacity rather than probe fit. It is a limit on *evidential independence*—a result on such a target cannot support a claim of emergent or persistence-dependent structure—and not by itself a degenerate learned representation.

tick One discrete update step of the internal state; the system’s native unit of processing time.

working memory A bounded, differentiable summary of the last few core states, given to the tick operator so the recurrent carrier need not hold short-horizon bookkeeping. Bounded is the requirement: an unbounded summary is a second memory.

self-action event The canonical record of an act the system just performed— type, addressee, and a pointer to the report it was grounded in—entering as an ordinary event on the next tick. The record re-enters; the generated text never does.

no backflow The rule that the translator’s text is never fed back into the core as state. The output translator is a mouthpiece, not a participant in the dynamics; at the governance layer the same rule reappears as no-self-sealing.

report grounding The requirement that the state report be the translator’s only content-bearing input, so that everything sayable is bounded by something inspectable. Non-propositional conditioning (register, person, formality, length) is permitted; anything carrying content becomes part of the declared report.

observability / sayability What an operator may watch (every channel, retrievals, checker scores) versus what the system may assert (the report, and nothing else); the first is deliberately wider.

grounding verification The requirement that an emitted reply be checked against the report, with declared coverage, an instrument control, adversarial cases, and a declared role (diagnostic or release gate); measured by B.15.

typed exposed core X_t / declared exposed state $\mathcal{X}_t^{(r)}$ X_t is the fixed-width four-part tuple of fast state, core state, slow state, and broadcast context. $\mathcal{X}_t^{(r)}$ is the regime-specific closure boundary for consumer reads: it is X_t in regimes (i), (ii), and (iv), and (X_t, h_t) in declared-carrier regime (iii). No consumer may read a channel outside $\mathcal{X}_t^{(r)}$ (A9, Figure 4).

fast / core / slow state Three of the four components of the typed exposed core: per-event encoding; the typed, per-tick central state; dispositional variables. The fourth is the broadcast context.

segment A typed, range-constrained sub-vector of a state with fixed semantics.

signal channels Typed scalar channels in the fast state carrying event-type and pragmatic cues.

tick operator The parameterized map producing the next core state (Equation (3)).

input translator / output translator The peripheral encoder into, and the peripheral language model out of, the state space.

state report The structured, human-readable extract of the current state handed to the output translator.

semantic decode The declared, separately evaluated mechanism that turns the semantic-focus vector into report content. A vector is not inspectable merely by lying in a named segment, which is why the decode is an instrument in its own right.

affect segment The dimensional–categorical affect representation (valence/arousal/dominance, basic-emotion intensities, epistemic drive).

epistemic drive The curiosity/interest variable, modeled as a motivational state distinct from basic emotions.

consistency constraints Soft constraints enforcing psychologically motivated correlations among affect variables.

affect–intention coupling The structured, learnable bias from affect variables to action-tendency logits.

appraisal context A supervised, derived interpretation of the current situation fed back as conditioning.

workspace / specialist process The competitive selection-and-broadcast mechanism, and the modules proposing content to it.

competitive selection / ignition / broadcast Sparse, capacity-limited workspace access; nonlinear amplification of winners; distribution to all consumers—and never an unconstrained soft average over all proposals, which is the form the requirement exists to exclude.

output-disjointness The principle that specialists must own disjoint output responsibility for specialization to survive.

write path The declared factorization of the tick update: broadcast-only, or two-phase with specialist-owned additive contributions. H4’s factorial cells mean different things in each, so a study reports which it ran.

meta segment / meta calibration The confidence–conflict–complexity read-out, and the requirement that it track its referents.

deliberation policy / halting criterion The per-tick continue-vs-act decision and its trigger set.

interaction loop One exchange end to end: ingest, deliberate for zero to a handful of ticks, report, translate, and close with the self-action event. Idle is a state in it, not a pause.

memory shelves Typed partitions of the unified store: episodic, knowledge, concept, identity.

consolidation The *only* route by which content moves between shelves, and—rarely and reversibly—into weights; governed at every step.

task-pressure condition Absent other pressure, memory use is learned when task information lies beyond the recurrent horizon; distance is the most reliable such pressure, not the only one.

representation learnability region The scale and objective-weight region within which a content channel is learnable.

degenerate-key guard Rejection of zero/near-zero-norm keys in cosine-similarity stores.

retrieval gate The confidence-weighted entry point through which retrieved content reaches the tick. Query from the pre-update situation, gate weight from the meta segment, so a system that is unsure retrieves more readily than one that is settled.

frozen-first The policy of fixing the encoder and training only the projection in early stages: a stable, fixed semantic geometry selected using external benchmarks, but not thereby externally calibrated, at the cost of a fixed lens the core cannot see past. Relaxed only along the staged unfreezing path.

anchor events Externally specified state targets placed at intervals in a long autoregressive rollout, so the system learns to run on its own outputs without drifting unmoored.

contrast-installed stability The hypothesis that robust affect homeostasis is learned from negative states *with recovery trajectories* rather than from calm ones—stability as a property of the training distribution’s trajectory geometry, not a regularizer added afterwards.

supervision paradox Whatever is supervised is, by construction, not emergent. Supervising the interpretive scaffold buys distinctions that may otherwise never form, and buys them at the price of their being the pipeline’s theory rather than the system’s finding.

latent target A per-tick target vector supervising a state segment.

gradient visibility LC4: a content channel must deliver a non-negligible, uncanceled gradient at the parameters it needs to shape. The loss share is the cheap proxy and not what the optimizer sees.

probe capacity The LC1 requirement that the probe family be at least as expressive as the target generator. A linear probe detects only linear triviality; getting this wrong is how generator closure passes unnoticed.

LC1–LC5 The latent-target learnability conditions: non-triviality; reachability and relative gating; persistence-discriminability; gradient visibility; on-manifold placement.

identity solution / conditional-mean collapse The failure modes of trivial and unreachable targets, respectively; *displacement* is LC5’s soft-constraint counterpart to unproducibility.

persistence baseline The predict-no-change baseline any predictive claim must beat.

displacement LC5’s soft-constraint failure mode: the target is reachable only by shifting the state distribution against the regularizer, so the cost is paid on the other supervised segments instead of showing up here.

gold-standard set Human-curated ground-truth sequences guarding against generator circularity.

echo baseline A previous-state-plus-noise baseline. By construction it excels on any stability-flavoured probe, so a probe it scores well on is measuring persistence rather than experience and has to be gated by reactivity.

supervised interpretation The principle that derived situation interpretations are effective levers only under explicit supervision.

Goodhart separation A declared stratification of metrics—development, sealed confirmatory, external—rather than a prohibition on overlap; a sealed metric may migrate to development, never back.

screen A small-scale test whose negative verdict is binding only for scale-invariant failure modes, and advisory otherwise; it may never qualify a candidate.

step-normalized schedule Optimization schedules parameterized in optimizer steps, invariant to dataset size.

staged unfreezing path The staged transition frozen encoder → adapters → co-training, each stage gated on preserved anchors and non-degraded evaluation rather than reached on a schedule.

seed / growth program The pre-growth stable system versus the governed continual-growth regime.

simulation-first The constraint that early growth operates on simulation, replayed curated experience and anchored scenarios rather than open interaction: it preserves the known input distribution against which drift is definable at all.

memory-first New knowledge lands in memory before any parametric change, and a parametric change is rare, individually verified, and reversible—a memory write does not license an automatic weight update.

no-self-sealing The system’s own outputs and single feedback partners must not dominate its learning sources.

governor / canary / drift criterion The gate roles in growth decisions: the governor may block but never green-light alone; a canary is a fixed, deterministic probe set that must not regress; drift criteria are watchdog conditions fixed in advance.

negative probes Standing tests that attempt forbidden operations and must fail.

reachability proof Operational evidence that fault-realistic operation moves a watchdog criterion’s guarded quantity through a pre-specified fraction of its threshold margin. Non-zero variance alone is not enough.

intake coverage / growth cycle The fraction of available experience consumed; one full ingestion–re-experience–consolidation pass.

train–serve parity Verified equivalence of deployed and evaluated processing paths, established by automated probes that run as standing regression checks—not by a one-time audit.

M1–M9 The measurement requirements: noise floor; aggregate uniqueness; anchor decomposability; structurally equivalent positive controls; item-level units; correlated-similarity hygiene; failure criteria fixed in advance; external anchoring; effective-configuration verification.

instrument-positive control / sensitivity condition A control in which the effect is present independently of the system under test, versus a condition that merely makes the effect easier for the system to produce; only the first can license a null (M4).

structurally equivalent positive control The form M4 requires: same probe format and same statistical pipeline as the test condition, differing only in that the effect is known to be present.

synthetic-trace control The standard instrument-positive control for a statistical detection problem: traces carrying the effect at a known magnitude, generated directly and passed through the identical pipeline. A pipeline that cannot recover a planted effect makes no null it produces mean anything.

unit of analysis The independent item—a stimulus, scenario or concept—at which statistics are run, as opposed to the repeated measurement; repetitions aggregate within items before any test runs across them (M5).

provenance / provenance graph Source, context, and confidence recorded on every write; and the dependency structure from source memories through derived shelf entries to parametric updates, which is what makes disqualification and selective reversal possible after consolidation.

noise floor The metric’s resolution at the variance level a claim generalizes over—measured as a hierarchy (M1), never as the same-configuration spread alone.

vacuous meta variables The failure mode in which meta variables decouple from their referents and drift to constants: present, typed, and empty. Its state-level analogue is slow-channel vacuity (O5).

floor effect The measurement trap in which two bounded scores both sit at their floor, so no trade-off between them can be shown—their correlation is a statement about the measurement range, not about the objectives.

not measurable The verdict class for a null result whose positive control fails.

understanding stages S1–S4 Affective reaction → pattern recognition → prospective consequence sensitivity → content understanding; the last is not currently claimable (no transfer or composition marker exists).

evidence marker An operationalized structural indicator whose failure criterion is fixed in advance; never a consciousness claim.

retrieval shortcut / prompt echo The two standing confounds of marker evaluation: lookup-produced behavior, and probe-paraphrasing reports.

E Instantiation Details

This appendix carries the full tables behind the sketch of Section 10. Everything here shares that section’s status: an abstraction of the reference implementation into ranges and alternatives, not a build report.

E.1 Objective terms

Table 7 lists the sketched recipe’s objective terms with their content and weighting logic; the structural commitments live in the term set and the two schedules, not in specific weight values.

E.2 Curriculum stages and transition gates

Table 8 tabulates the curriculum of Section 10.4 with the transition gates fixed in advance.

E.3 Training-sequence schema (abridged)

A training item, structurally (values illustrative):

```
{ scenario_family: "surprise", disposition_profile: {...explicit dims ...},
  ticks: [
    { event: { text: "A deadline moves up.", type: neutral, intensity: 0.3 },
      expected_state: {
        affect: { valence: 0.0, arousal: 0.3, dominance: 0.5,
          fear: 0.15, surprise: 0.4, curiosity: 0.6, ...},
        intention: { distribution: { explore: 0.7, reflect: 0.1, wait: 0.1, ...},
          confidence: 0.6, readiness: 0.4 },
        semantic_focus: "The plan needs rethinking under time pressure.",
        meta: { confidence: 0.5, conflict: 0.2, complexity: 0.3 }
        // warm-start labels only; see §5.4 } },
    ... ] }
```

Numeric layers convert deterministically into latent targets; the event text enters through the input translator, and the semantic-focus text is encoded by the same encoder to form the semantic segment’s target—two distinct strings, per Section 7.2. Generation rules demand gradual affect development, psychological plausibility, recovery after negative phases, and—critically for LC1-adjacent reasons—*semantic breadth* in the focus texts (situation reference, associations, abstractions, meta-comments), since monotone focus texts collapse the semantic supervision toward a few directions (Section 7.3).

E.4 Disposition profiles

Table 9 illustrates how disposition profiles parameterize generation, as a two-profile contrast over a subset of explicit dimensions.

Term	Content	Weighting logic
State supervision	Next core state vs. latent targets; hybrid squared-error + cosine (squared-error-dominant, e.g., 60:40–80:20)	scheduled up over training
Event prediction	Scenario-class (roughly ten families, up to a few tens of classes at the finest granularity) + event-type + intensity of the <i>next</i> event; raw next-embedding only as a low-weight secondary head	scheduled down over training
Affect supervision	Squared error on the 9 affect dimensions	constant, elevated (core quality)
Affect consistency	Hinge penalties on the soft correlation rules of Section 4.2	constant, small
Adaptive smoothness	Penalty on per-tick affect change beyond $\delta(I) = \delta_0 + \delta_1 I$ (event intensity I) — calm ticks strict, intense events licensed	constant, small
Intention distribution	KL divergence under softmax with ϵ -floor renormalization	constant, small
Appraisal context	Supervised scenario-family prediction feeding the conditioning channel (Section 7.4)	constant, small
Affect→intention prior	Initialization-consistency regularizer on the coupling of Section 4.3	linear decay to zero over the first 10–30% of training
Contrastive projection	Supervised contrastive shaping of the input projection over scenario families [59]	pre-training + low-weight co-training
Norm/stability	State-norm radius penalty ($R = c\sqrt{d}$); optional weak attractor pull toward the cold-start state during near-zero-intensity stretches	constant, small

Table 7: The ten objective terms of the sketched recipe. Six are *required* by the framework—state supervision, affect supervision, affect consistency, the intention distribution, the appraisal-context head, and norm/stability—in the sense that removing one drops a stated commitment. The remaining four (event prediction, adaptive smoothness, the affect→intention prior, contrastive projection) are recommended but substitutable. Exact weights are tuning surface; the structural commitments are the required term *set*, the two schedules, and per-term gradient-norm logging with automatic balancing [17, 121] as the designated upgrade.

E.5 Cold start and tick economics

Cold start. Neutral valence, low arousal, mid dominance, all emotion intensities minimal, intention distribution reflect-leaning with some wait/explore mass, meta confidence mid and conflict low, semantic focus zeroed, latent segments near-zero noise, slow state from the disposition profile. Per-sequence noise on the start state prevents origin overfitting; the data-driven alternative (training-distribution state mean) is the natural comparison to run.

Tick economics. This paper claims memory and complexity plausibility, not wall-clock cost, and the distinction matters: parameter counts bound memory analytically, whereas training time, tick latency, and evaluation throughput are measurements that depend on hardware, precision, batch size, sequence length, and implementation—none of which this paper fixes and none of which it measured. What can be stated structurally is the *cost ratio* within a tick: one operator forward pass at the sizes of Section 10.2 is roughly an order of magnitude cheaper than one sentence-encoder pass, so a deliberation of 5–10 ticks multiplies the smaller term, not the larger. Whether that translates into interactive latency, and whether repeat-run evaluation is affordable in practice, are questions for the companion implementation paper, reported with hardware,

Stage	Length (ticks)	Regime	Transition gate (fixed in advance)
1	tens	fully supervised single sequences	state-supervision validation converged; coherence above floor; no NaN/instability window
2	30–100	concatenated sequences with programmatic rest-tick transitions	coherence floor raised; mean per-tick affect change bounded; no collapse over concatenated spans
3	100–300	autoregressive rollouts with <i>soft</i> anchors (distance-weighted anchor loss), adaptive anchor density, truncated BPTT window 50–100	coherence and drift-freedom floors on rollouts; anchor distance trending down
4	500+	long rollouts, sparse anchors, optional grounded streams	anchor distance stable; long-horizon probes pass; else densify anchors rather than advance

Table 8: Curriculum stages with gate logic. Every sequence ends in programmatic recovery ticks (quiet events, exponentially decaying state target toward the cold-start state) so stillness is a learned distribution.

Explicit dimension (subset)	venturesome	deliberate-systematic
truth orientation	mid	high
harm avoidance	low	high
social orientation	mid	low-mid
risk tolerance	high	low
deliberative tendency	low-mid	high
exploration tendency	high	low

Table 9: An illustrative two-profile contrast over a subset of explicit slow-state dimensions (qualitative levels). The full profile library—its size, its archetype names, and its numeric values—is instantiation freedom; the pedagogical point is only that the *same* event stream trains *different* reactions under different slow states. **Recommendation:** begin with a *single default profile* — profile diversity multiplies the learning problem and confounds early evaluation; introduce the library once basic dynamics are stable and dispositional differentiation itself becomes the phenomenon under study.

precision, batch size, sequence length, and measured distributions.

E.6 Suite probes and score bands

The probe battery of Section 10.6, grouped by evaluated duty and tabulated with what each probe guards (the battery is illustrative coverage of the duties, not a canonical list):

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Duty	Probe	Guards	Baseline caution
affective adequacy	narrative affect arc	contextual affect through a story arc	hard-gated; echo baseline fails it (no reactivity)
	emotional complexity affect homeostasis	mixed and sequential affect return to baseline without flatness	— echo trap: pair with reactivity
differentiation & responsiveness	situation differentiation event proportionality	distinct situations → distinct trajectories response scaled to events	read out during input phases, not neutral settling guards the echo trap
stability	long-horizon coherence	no collapse/chaos over hundreds of ticks	hard-gated
	topic coherence	thematic continuity over an arc	echo scores well — pair with event proportionality
	re-stabilization	recovery after adversity	decay shape, not just endpoint
metacognition	dispositional stability	slow state steady on short horizons	vacuity risk (O5): verify the channel varies at all
	calibration	e.g., confidence–error correlation; calibration error; reactivity or resolution	<i>report components</i> (M3): a sum hides anti-calibration

Table 10: Probe battery grouped by duty, part I, with per-probe cautions.

Duty	Probe	Guards	Baseline caution
memory	round trip	write–retrieve–use	requires task pressure (H6) to be meaningful
	interference retention	retention under interference	—
decision	intention convergence	decision pressure resolves distributions	a candidate trade-off with differentiation probes (O1) — joint maxima are not to be presumed
deliberation &	depth–demand response	consumed depth varies with demand and improves outcomes at matched compute	reachability proof first (B.12): verify the halting inputs traverse a stated fraction of the margin to their thresholds, not merely that they vary; demand fixed by the battery, not by the meta channel; count only pre-response ticks; report inter-tick depth and intra-tick refinement separately, never as one number
anticipation	announced-consequence divergence	anticipatory shifts before announced events (S3, B.11)	valence-matched announcements; echo baseline must score zero here

Table 11: Probe battery grouped by duty, part II. Across both parts, each caution instantiates a requirement or open problem from the main text (O1/O5: Section 12.3). An interpretation scale of the shape *not functional* / *functional* / *good* / *excellent* (e.g., over quartile-like bands) is fixed at the suite’s first calibration and versioned with it (M2).